Rainbow subgraphs: obstructions, progress and open problems

Abstract

A subgraph of an edge-coloured graph is rainbow if its edges have distinct colours. Questions of this kind go back to Euler's thirty-six officers and to transversals in Latin squares, and they now connect graph labellings, matroid bases, group orderings, additive combinatorics, coding theory, design theory and fair division. We survey rainbow matchings, trees, paths, cycles and general spanning structures, and their analogues for digraphs and hypergraphs. For each structure we ask when a rainbow copy exists, how many disjoint copies there are and when the host decomposes into them, under proper, locally bounded and globally bounded colourings and colour-degree conditions, for graph collections and in random settings. Recent results include the Ryser–Brualdi–Stein bound $n-1$ for large Latin squares, Ringel's tree packing conjecture and Andersen's rainbow path conjecture for large n, and asymptotic forms of Rota's basis conjecture. For each topic we describe the constructions that limit what can be true and the problems that remain open. Anti-Ramsey numbers, rainbow Turán numbers and rainbow connection are not covered.

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1 Introduction

A subgraph of an edge-coloured graph is *rainbow* if no two of its edges have the same colour. The name is recent, but the questions are old. In 1723 Ozanam [230] published a card puzzle that asks for the sixteen court cards and aces to be laid out in a square so that no row or column repeats a rank or a suit. In 1782 Euler [120] asked for the same arrangement of thirty-six officers from six regiments and six ranks. Both puzzles ask for many disjoint rainbow perfect matchings at once. Most of this survey concerns three questions of this kind: when a rainbow copy of a given graph exists, how many disjoint copies there are, and when the host decomposes into them. Each time a natural statement fails, the obstructions are described before the positive results.

Latin squares are the first source of such questions. A Latin square of order n is an $n \times n$ array filled with n symbols, each appearing once in every row and column, and a transversal is a set of n cells with no repeated row, column or symbol. If rows and columns are the two sides of $K_{n,n}$ and each edge is coloured by the symbol in its cell, transversals become rainbow perfect matchings (Figure 1). Euler's officers ask for a Latin square of order 6 that decomposes into transversals. Tarry [261] showed in 1900 that none exists, and in 1959–60 Bose, Shrikhande and Parker [57, 58] showed that every order other than 2 and 6 admits one (Section 2.2). For even n the cyclic square has no transversal at all, which led Ryser in 1967 to conjecture a transversal for every odd order, and Brualdi and Stein to conjecture a partial transversal with $n-1$ cells for every order (Conjecture 2.19). The lower bounds then rose step by step: $n - \sqrt{n}$ in 1978 [68, 270], $n - O(\log^2 n)$ by Shor [248], $n - O(\log n / \log\log n)$ by Keevash, Pokrovskiy, Sudakov and Yepremyan [174], and finally $n-1$ for large n by Montgomery [214] (Theorem 2.21). Ryser's statement for odd n is still open. Random squares behave differently again: van Rees conjectured in 1990 that a random Latin square typically has no decomposition into transversals, but Bowtell and Montgomery [62] proved that it almost surely has one (Theorem 2.2).

The colouring of a Latin square is *proper*: edges at a common vertex have distinct colours. Other assumptions control colour repetition in other ways. A colouring is *locally k-bounded* if each colour appears at most k times at each vertex, and *globally k-bounded* if each colour appears at most k times in total; proper colourings are the locally 1-bounded ones. Global bounds have a long history of their own. Erdős and Spencer [118] introduced the lopsided local lemma in 1991 to show that an $n \times n$ array in which no symbol appears more than $(n-1)/16$ times has a full transversal. Hahn and Thomassen conjectured in 1986 that colour classes of size αn force a rainbow Hamilton cycle in K_n, which Albert, Frieze and Reed proved. Frieze and Krivelevich asked for the same with bounded-degree spanning trees, and Böttcher, Kohayakawa and Procacci [59] answered this for every spanning graph of maximum degree Δ, with colour classes of size $n/(51\Delta^2)$ (Theorem 5.5). Counting explains why such bounds are needed: a proper colouring of K_n may use only $n-1$ colours, too few for a rainbow spanning graph with more edges. A fourth condition asks only for many colours at each vertex. Li [193] showed that minimum *colour degree* $(n+1)/2$ forces a rainbow triangle, and the colour-degree threshold for a spanning tiling by rainbow triangles is now known to lie between $5n/7$ and $5n/6$ [201].

Trees bring in graph labellings. At the 1963 symposium in Smolenice, Ringel conjectured that K_{2n+1} decomposes into $2n+1$ copies of any tree with n edges, and in 1967 Rosa [245] introduced graceful labellings as a way to prove it. Colour the edges of a complete graph on the integers by the differences of their ends. A graceful labelling is then a rainbow copy of the tree, and a rainbow copy in the cyclic version of this colouring rotates to a decomposition (Figure 11). Montgomery, Pokrovskiy and Sudakov [219] found such rainbow copies and proved Ringel's conjecture for large n; Keevash and Staden [175] proved it independently by another method (Theorem 3.14). The graceful tree conjecture is still open. Decompositions into rainbow spanning trees have a similar history. Brualdi and Hollingsworth conjectured in 1996 that every optimal colouring of K_{2n} decomposes into rainbow spanning trees, and for many years only a constant number of disjoint ones could be found. Montgomery, Pokrovskiy and Sudakov [217] found $(1-o(1))n$ of them in every proper colouring, and Glock, Kühn, Montgomery and Osthus [137] completed the decomposition for optimal colourings (Theorem 3.7). Rota's basis conjecture of 1989 asks for the same kind of decomposition for n bases of an n-dimensional vector space (Conjecture 3.16). Its matroid version contains the case of n spanning trees (Figure 12). Montgomery and Sauermann [220] proved it asymptotically: $(1-o(1))n$ disjoint rainbow bases exist, and $(1+o(1))n$ of them cover everything (Theorem 3.18).

Paths have a shorter but similar history. A proper colouring of K_n has at least $n-1$ colours, as many as a Hamilton path has edges, and in 1980 Hahn conjectured that a rainbow Hamilton path always exists (see [63]). In 1984 Maamoun and Meyniel [210] disproved this for every n that is a power of 2 (Figure 14). Andersen [36] conjectured in 1989 that a rainbow path missing only one vertex always exists. After a long sequence of approximate results, Bowtell, Montgomery, Müyesser and Pokrovskiy [63] proved this for large n (Theorem 4.2). Orderings of group elements are rainbow paths in special colourings. In 1971 Graham [144] asked whether the elements of a subset of $\mathbb{Z}_p$ can be ordered so that all partial sums are distinct. Such an ordering is a rainbow directed path in the Cayley digraph of $\mathbb{Z}_p$, where the edge from g to $g+s$ has colour s (Conjecture 4.6, Figure 16).

Short rainbow cycles have a different origin and a different application. Behzad, Chartrand and Wall conjectured in 1970, and Caccetta and Häggkvist more generally in 1978, that digraphs with large outdegree have short directed cycles. Aharoni proposed a rainbow generalisation, and the Caccetta–Häggkvist conjecture follows from it by colouring each edge by its initial vertex (Section 4.3). In 2007 Keevash, Mubayi, Sudakov and Verstraëte [173] asked how many edges a properly coloured graph needs to contain a rainbow cycle. Bounds for this question have since been used in coding theory: Alrabiah and Guruswami [32] used rainbow cycles to prove near-tight lower bounds on the length of 3-query locally correctable binary linear codes (Section 4.4).

Several translations reach other fields. In 1961 Erdős, Ginzburg and Ziv showed that any $2t-1$ integers contain t with sum divisible by t (see [22]). Aharoni and Berger [2] conjectured in 2009 that $2^{r-1}(t-1)+1$ matchings of size t in an r-partite r-graph have a rainbow matching of size t. Alon [22] disproved this by turning zero-sum-free sequences into families of matchings (Figure 31), and Pohoata, Sauermann and Zakharov [235] showed that the right number grows like t^r (Theorem 7.1). In fair division, rainbow matchings of hypergraphs decide when maximin share allocations exist (Figure 33, Theorem 7.5), and rainbow cycles in multipartite digraphs bound the number of goods left unallocated in approximate EFX allocations (Figure 26, Theorem 6.5). In design theory, the resolutions of Steiner triple systems go back to Kirkman's schoolgirl problem of 1850, and Keevash's theory of designs includes coloured decompositions such as rainbow triangle decompositions [172].

Two settings recur in every chapter. In a *graph collection*, one edge is taken from each of several graphs on the same vertex set; colouring each graph's edges with its own colour makes this a rainbow problem (Figure 13). Aharoni conjectured that n graphs of minimum degree $n/2$ on the same n vertices have a Hamilton cycle with one edge from each, and Joos and Kim [166] proved it. So the threshold is the same as for one graph, but Montgomery, Müyesser and Pehova [216] showed that for some factors the collection threshold is higher (Section 5.5). In random settings, Frieze and McKay [126] found rainbow spanning trees in randomly coloured random graphs as soon as enough colours and edges have appeared (Theorem 3.22). Han and Yuan [152] showed in general that a rainbow structure appears in a randomly coloured random set at most a logarithmic factor after the uncoloured one (Theorem 5.14). We treat both settings within the chapters on the corresponding structures.

Pokrovskiy [239] surveyed rainbow subgraphs and their applications, Montgomery [215] surveyed transversals in Latin squares, and Sudakov [254] surveyed restricted subgraphs of edge-coloured graphs with applications to coding theory. This survey is organised by the structure that is sought, and it includes developments since these surveys, many from 2025 and 2026. We do not cover anti-Ramsey numbers, rainbow Turán numbers or rainbow connection; the question of how many edges force a rainbow cycle is included because of its use in coding theory. Within each topic, constructions that rule out the natural statement come first, then the strongest positive results, and then a discussion of what remains open and what the next target would be.

Section 2 treats matchings and transversals, and Section 3 trees, decompositions and matroid bases. Section 4 treats paths and cycles, and Section 5 general spanning structures such as factors and powers of Hamilton cycles. Sections 6 and 7 treat edge-coloured digraphs, and hypergraphs, designs and families of sets.

2 Rainbow matchings and transversals

This section is about rainbow matchings in properly coloured bipartite graphs and multigraphs. The motivating problem is to find transversals in Latin squares. Montgomery's survey [215] treats transversals

in Latin squares in more detail, and Pokrovskiy's survey [239] covers the multigraph problems.

2.1 Latin squares and three translations

A *Latin square* of order n is an $n \times n$ array filled with n symbols so that each symbol appears exactly once in each row and each column. A *partial transversal* is a set of cells which share no row, column or symbol. A *full transversal*, or simply a *transversal*, is a partial transversal with n cells. The multiplication table of a finite group is a Latin square.

A Latin square can be translated into three other objects, and each translation leads to its own generalisations. Figure 1 shows all three for the Latin square of order 5 in Figure 1a. This square has exactly three transversals, and the figure shows one of them.

Coloured bipartite graphs. Take a copy of $K_{n,n}$ whose two vertex classes are the rows and the columns of a Latin square L, and colour the edge between row r and column c with the symbol $L(r, c)$. No symbol appears twice in a row or a column, so this colouring is proper. It uses n colours, the fewest possible, so we call it *optimal*. Every optimal colouring of $K_{n,n}$ arises from a Latin square in this way. A partial transversal of L is a *rainbow matching*, that is, a matching whose edges have distinct colours, and a full transversal is a rainbow perfect matching (Figure 1b).

3-*partite hypergraphs.* Take disjoint sets R, C and S of size n for the rows, the columns and the symbols, and add the triple $\{r, c, s\}$ whenever $L(r, c) = s$. In the resulting 3-partite 3-uniform hypergraph, every pair of vertices from different parts lies in exactly one edge. A transversal of L is a perfect matching of this hypergraph (Figure 1c). This is the link between transversals and matchings in designs, such as Steiner triple systems (Section 2.12).

Coloured directed graphs. Take the complete directed graph on the vertex set $[n]$, with a loop at each vertex, and colour the edge from r to c with $L(r, c)$. A transversal consists of the cells $(r, \sigma(r))$ for a permutation σ, so it is a spanning union of vertex-disjoint directed cycles (a 1-*factor*) in which each colour appears once. When the 1-factor is a single Hamilton cycle, the transversal is called a *Hamilton transversal*. The transversal in Figure 1 is one: it gives the cycle $1 \to 4 \to 3 \to 5 \to 2 \to 1$ (Figure 1d). Of the three transversals of this square, two are Hamilton transversals. This translation relates transversals to the rainbow paths and cycles of Section 4.

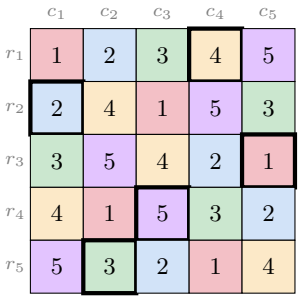
Some generalisations are natural in only one translation. In the bipartite graph, we may allow more than n colours (Section 2.8), or only require that each colour appears on at most k edges, which we call a *globally k-bounded* colouring (Section 2.9). A colouring is *locally k-bounded* if each colour appears at most k times at each vertex, so a proper colouring is a locally 1-bounded colouring. We may also replace $K_{n,n}$ by other graphs and multigraphs (Sections 2.10 and 2.11).

2.2 Decompositions

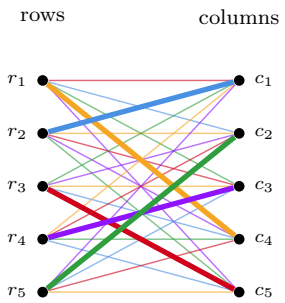
Euler's officers

The question of decomposing a Latin square into transversals is older than Latin squares as a subject. A card puzzle published by Ozanam [230] in 1723 asks for the sixteen court cards and aces of a deck (A, K, Q, J in four suits) to be laid out in a 4×4 square so that every row and every column contains each rank once and each suit once. Figure 2 shows a solution. Forgetting the suits leaves a Latin square of ranks, and forgetting the ranks leaves a Latin square of suits. The cards of one suit form a transversal of the square of ranks, so the four suits decompose it into four transversals.

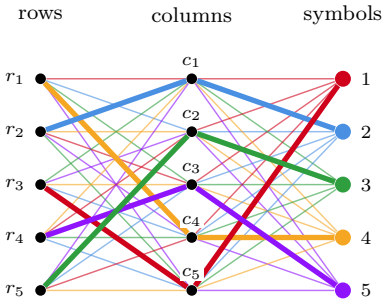
In 1782, Euler [120] asked the same question for order 6. Thirty-six officers, one of each of 6 ranks from each of 6 regiments, are to stand in a 6×6 square so that each row and each column contains one officer of each rank and one of each regiment. Euler wrote the regiments with Latin letters and the ranks with Greek letters, which is where the name *Latin square* comes from. Labelling each officer by their regiment gives a Latin square of order 6, and the officers of one rank form a transversal. So Euler asked for a Latin square of order 6 which *decomposes* into 6 disjoint transversals. Labelling by rank instead gives a second Latin square, and the two squares are *orthogonal*: when one is placed over the other, every ordered pair of symbols appears exactly once. In general, a Latin square has an orthogonal mate exactly when it decomposes into transversals. In terms of colourings, Euler asked for an optimal colouring of $K_{n,n}$ which decomposes into n rainbow perfect matchings.



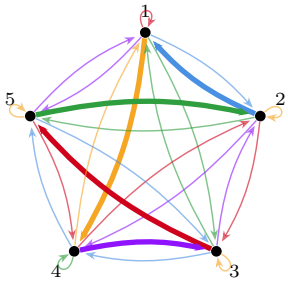
(a) A Latin square of order 5. The five cells with thick borders form a transversal.



(b) The optimally coloured $K_{5,5}$. The transversal is a rainbow perfect matching.



(c) The 3-partite hypergraph. Each triple $\{r, c, s\}$ is drawn as a path r–c–s in the colour of s. The transversal is a perfect matching.



(d) The complete directed graph with loops; the edge $r \to c$ has colour $L(r, c)$. The transversal is the rainbow Hamilton cycle $1 \to 4 \to 3 \to 5 \to 2 \to 1$.

Figure 1: The three translations of a Latin square, for one Latin square of order 5 and one of its transversals. Colours $1, \ldots, 5$ are red, blue, green, orange and purple. Faint lines show all cells of the square, and thick lines show the cells of the transversal.

Euler constructed such squares for every $n \not\equiv 2 \bmod 4$ and made the following conjecture.

Conjecture 2.1 (Euler, 1782)**.** *No Latin square of order $n \equiv 2 \bmod 4$ decomposes into transversals.*

The conjecture holds for $n = 2$, and Tarry [261] proved it for $n = 6$ in 1900. It is false for all larger n. Bose and Shrikhande [57] found counterexamples of orders 22 and 50 in 1959, and soon afterwards Bose, Shrikhande and Parker [58] found counterexamples for every $n \equiv 2 \bmod 4$ with $n \geq 10$.

Orthogonal Latin squares became important in the twentieth century through the design of experiments. In an agricultural trial with n varieties of a crop, planting them in a Latin square on an $n \times n$ field balances out differences between the rows and the columns of the field. An orthogonal pair allows a second treatment, such as n fertilisers, to be tested in the same field, and Fisher and Yates tabulated such designs for this purpose. Mutually orthogonal Latin squares are now a basic object in design theory and in coding theory. A family of k mutually orthogonal Latin squares of order n is the same as an orthogonal array of strength 2 with $k + 2$ factors. It is also the same as a code of length $k + 2$ over an alphabet of size n with n^2 codewords and minimum distance $k + 1$, and such a code meets the Singleton bound.

Euler's question asks whether *some* Latin square of order n decomposes into transversals. Many Latin squares do not. For every even n there are Latin squares with no transversal at all (Section 2.3). For every odd $n \geq 5$ there are Latin squares without a decomposition into transversals, by results of Mann, of Evans, and of Wanless and Webb [265] (see [215]). In 1990, van Rees conjectured that a uniformly random Latin square typically has no decomposition into transversals. Wanless and Webb [265] reported that computations for small orders suggest the opposite. Bowtell and Montgomery settled the question.

Theorem 2.2 (Bowtell and Montgomery [62])**.** *A uniformly random Latin square of order n decomposes into transversals with probability $1 - o(1)$.*

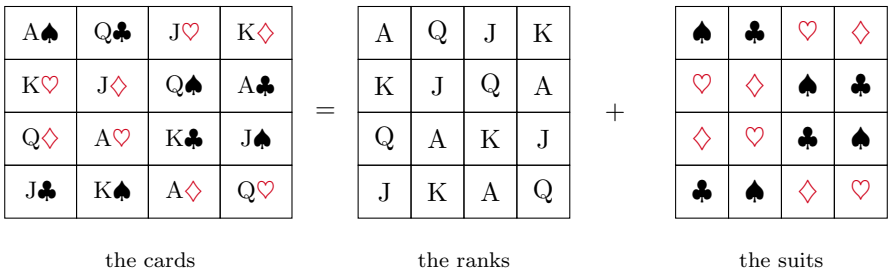


Figure 2: A solution of Ozanam's card puzzle (as drawn in [215]). The ranks and the suits form two orthogonal Latin squares of order 4: every rank–suit pair occurs once. The cards of each suit form a transversal of the square of ranks.

In particular, almost every Latin square of order $n \equiv 2 \bmod 4$ is a counterexample to Euler's conjecture. A uniformly random optimal colouring of $K_{n,n}$ with colour set $[n]$ is the same as a uniformly random Latin square. So Theorem 2.2 also settles a question in Pokrovskiy's survey [239, Conjecture 5.2], which asks whether such a colouring decomposes into rainbow perfect matchings with high probability.

Before Theorem 2.2, the following approximate result was known for all Latin squares.

Theorem 2.3 (Montgomery, Pokrovskiy and Sudakov [217])**.** *Every Latin square of order n contains $(1-o(1))n$ disjoint partial transversals, each with $(1-o(1))n$ cells.*

The proof of Theorem 2.2 first splits the random Latin square into n near-transversals, using the semi-random method as in Theorem 2.3, and then corrects all of them at once by absorption.

How many decompositions are there? Over all Latin squares of order n, the average number of decompositions into transversals is $((1+o(1))n/e^3)^{n^2}$, and Boyadzhiyska, Das and Szabó showed that no Latin square has more (see [259]). Tang and Zhao [259] ask whether a random Latin square has $((1-o(1))n/e^3)^{n^2}$ decompositions with high probability, so that Theorem 2.2 would hold with the typical count. They also ask a deterministic question in the spirit of Theorem 2.21: for large n, can the cells of every Latin square of order n other than one prescribed cell be partitioned into $n+1$ partial transversals with $n-1$ cells each? Even the existence of one such cell is open.

Room squares

Euler's problem has an analogue for complete graphs of even order. A 1-*factorisation* of K_{2n} is a partition of its edges into $2n-1$ perfect matchings. This is the same as an optimal proper colouring of K_{2n}. Two 1-factorisations are *orthogonal* if every perfect matching of one shares at most one edge with every perfect matching of the other. If we view the first 1-factorisation as a colouring, then the matchings of an orthogonal second 1-factorisation are rainbow perfect matchings. So a pair of orthogonal 1-factorisations of K_{2n} is an optimally coloured K_{2n} together with a decomposition into rainbow perfect matchings.

Such pairs are equivalent to *Room squares*, introduced by Room [244] in 1955. A Room square of side $2n-1$ is a $(2n-1)\times(2n-1)$ array whose cells are either empty or contain an unordered pair from a set of $2n$ symbols. Every pair appears in exactly one cell, and every row and every column contains each symbol exactly once. The rows give one 1-factorisation of K_{2n} and the columns give an orthogonal one (Figure 3). Room squares are used to schedule tournaments between pairs of players, such as duplicate bridge tournaments: the rows are rounds, the columns are tables, and each cell records which two partnerships meet.

Theorem 2.4 (Mullin and Wallis [222])**.** *For every $n \geq 4$ there is an optimally coloured K_{2n} which decomposes into rainbow perfect matchings. For $n \in \{2,3\}$ there is none. Equivalently, a Room square of odd side r exists if and only if $r \notin \{3,5\}$.*

Theorem 2.4 is about *some* 1-factorisation, and not every 1-factorisation has an orthogonal mate. By analogy with Theorem 2.2, one can ask whether a uniformly random 1-factorisation of K_{2n} has an orthogonal mate with high probability. We are not aware of results on this question.

		16		34	25	0∞
36	02	45		1∞		
	56	2∞	04		13	
24	3∞				06	15
01			35	26	4∞	
5∞		03	12			46
	14		6∞	05		23

Figure 3: A Room square of side 7 on the symbols $\{0, \ldots, 6, \infty\}$. Row i is the perfect matching $\{\{\infty, i\}\} \cup \{\{i-j, i+j\} : j = 1, 2, 3\}$ of K_8, with indices modulo 7, and the rows form a 1-factorisation of K_8. Each row has its own colour. Each column is a perfect matching of K_8 using one edge from each of four rows, so its four cells have four different colours: it is a rainbow perfect matching for the colouring given by the rows. The seven columns decompose K_8.

Kotzig's perfect 1-factorisation conjecture

A 1-factorisation is *perfect* if the union of any two of its perfect matchings is a Hamilton cycle.

Conjecture 2.5 (Kotzig, 1963 [183])**.** *For every $n \geq 2$, the complete graph K_{2n} has a perfect 1-factorisation.*

In the language of colourings, Kotzig's conjecture asks for an optimal colouring of K_{2n} in which every two colour classes together form a single Hamilton cycle. The union of two disjoint perfect matchings is a union of even cycles, each of length at least 4, and in a perfect 1-factorisation it is always a single cycle of length $2n$. The colourings that give most of the counterexamples in this survey are at the other end. Identify the vertices of K_{2^k} with $\mathbb{Z}_2^k$ and give the edge xy the colour $x + y$. Then the colour classes a and b form the 4-cycles x, $x + a$, $x + a + b$, $x + b$, so the union of any two colour classes consists of 4-cycles (Figure 4). This colouring has no rainbow Hamilton path (Section 4).

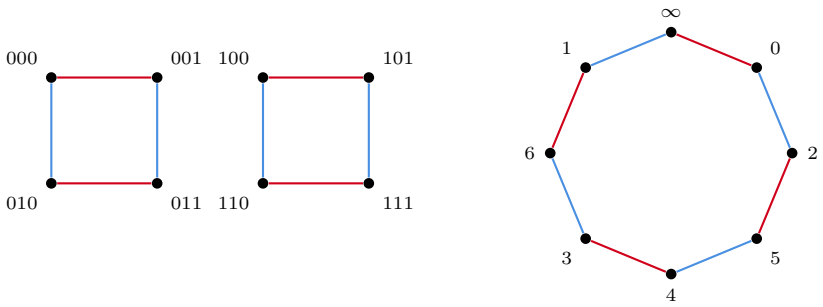


Figure 4: Left: in the colouring of K_8 by $\mathbb{Z}_2^3$, the colour classes 001 (red) and 010 (blue) form two 4-cycles. Right: the first two rows of the Room square in Figure 3, namely $\{\infty 0, 16, 25, 34\}$ (red) and $\{\infty 1, 02, 36, 45\}$ (blue), form a Hamilton cycle. The 1-factorisation given by the rows of Figure 3 is perfect: any two of its matchings form a Hamilton cycle.

Kotzig's conjecture holds when $2n - 1$ is prime, by the 1-factorisation in Figure 3, and when n is prime. It also holds for many other orders found by computer (see [133]). The smallest open case is $2n = 64$ [138, 87]. In 2026, perfect 1-factorisations of K_{92} were found [211]; 92 had been one of the open orders below 100. The analogous question for $K_{n,n}$ has also been studied by computer [19, 20]. All these constructions are explicit and algebraic. Recently, probabilistic methods have been applied to the conjecture, first in a sparse setting. A random graph is not regular, so it has no 1-factorisation at all. Glock and Sgueglia therefore asked for a fixed number of perfect matchings, any two of which form a Hamilton cycle.

Theorem 2.6 (Glock and Sgueglia [138]). *For every k there is C such that, if $p \geq C \log n/n$, then with high probability the random graph $G(2n, p)$ contains k edge-disjoint perfect matchings, any two of which form a Hamilton cycle.*

Their main result is a counting statement. Given k edge-disjoint perfect matchings of K_{2n}, a uniformly random perfect matching forms a Hamilton cycle with each of them with probability $\Theta_k(n^{-k/2})$. Cheng and Sgueglia then found almost all of a perfect 1-factorisation.

Theorem 2.7 (Cheng and Sgueglia [87]). *The complete graph K_{2n} has a 1-factorisation in which $(1 - o(1))2n$ of the perfect matchings have the property that any two of them form a Hamilton cycle.*

This is the analogue of Theorem 2.3 for Kotzig's conjecture: it controls all but a vanishing fraction of the factorisation. The exact conjecture would require the last $o(n)$ perfect matchings to be chosen as well. As in Section 2.6 below, this is the hard part.

The bipartite version has a similar status. Laufer showed that a perfect 1-factorisation of K_{n+1} gives one of $K_{n,n}$, and Wanless made the corresponding conjecture.

Conjecture 2.8 (Wanless, 1999 [262]). *The complete bipartite graph $K_{n,n}$ has a perfect 1-factorisation if and only if $n = 2$ or n is odd.*

Infinite families are known only for $n \in \{p, 2p - 1, p^2\}$ with p an odd prime [138]. Kotzig's conjecture also has a consequence outside rainbow problems. An edge-colouring is *acyclic* if it is proper and every two colour classes form a forest. Fiamčík and, independently, Alon, Sudakov and Zaks conjectured that every graph of maximum degree Δ has an acyclic edge-colouring with $\Delta + 2$ colours. Alon, Sudakov and Zaks [30] observed that a perfect 1-factorisation of K_{2n} gives acyclic edge-colourings of K_{2n} and K_{2n+1} with $2n + 1$ colours. So Conjecture 2.5 would settle the acyclic edge-colouring conjecture for complete graphs.

2.3 Latin squares without transversals

The addition table of $\mathbb{Z}_n$ with n even has no transversal. Suppose the cells $(r, \sigma(r))$, $r \in \mathbb{Z}_n$, form a transversal. Then the symbols $r + \sigma(r)$ are all the elements of $\mathbb{Z}_n$, so

$$\sum_{g \in \mathbb{Z}_n} g \;=\; \sum_{r \in \mathbb{Z}_n} \big(r + \sigma(r)\big) \;=\; 2 \sum_{g \in \mathbb{Z}_n} g,$$

and hence $\sum_g g = 0$. But for even n this sum is $n/2 \neq 0$. The same argument works for every abelian group whose elements do not sum to 0, and these are the abelian groups with exactly one element of order 2. Paige [231] showed in 1947 that the addition table of an abelian group has a transversal if and only if the group does not have exactly one element of order 2. For all finite groups, the answer is the Hall–Paige theorem (Theorem 2.11 in Section 2.4).

The same argument gives many more examples (Figure 5). Take the addition table of $\mathbb{Z}_2$ and replace each entry by a $k \times k$ Latin square, using one set S_0 of k symbols for the entries 0 and another set S_1 of k symbols for the entries 1. If k is odd, the resulting Latin square of order $2k$ has no transversal. This construction is due to Maillet (1894). Since the $k \times k$ blocks can be chosen freely, it gives many Latin squares without transversals. Cavenagh and Wanless [77] used related constructions to show that, for even n, there are at least $n^{n^{3/2}(1/2 - o(1))}$ species of Latin squares of order n without transversals. Still, by Theorem 2.2, these form a vanishing proportion of all Latin squares.

All known Latin squares without a transversal have even order. Ryser conjectured that there are none of odd order.

Conjecture 2.9 (Ryser, 1967 [246]). *Every Latin square of odd order has a transversal.*

For groups of odd order, Conjecture 2.9 also follows directly: the leading diagonal of the multiplication table is a transversal, since $g \mapsto g^2$ is a bijection when $|G|$ is odd.

Latin squares of even order can be even more restricted. Ghafari and Wanless [132] constructed, for every even $n \geq 28$ except $n = 30$, a Latin square of order n in which every two transversals share a cell. They conjecture that, for every even $n \geq 28$, there is a Latin square of order n with no two disjoint transversals in which no cell lies in every transversal, and they verify this for $n = 28$ and $32 \leq n \leq 10000$. So the failure of decompositions need not come from a single blocking cell.

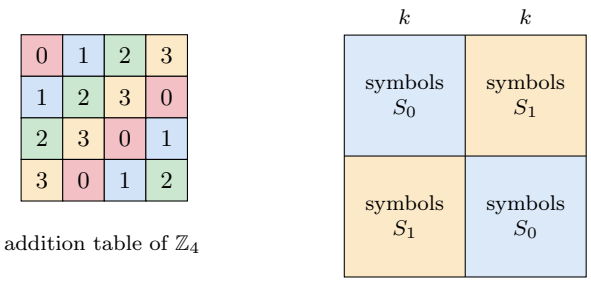


Figure 5: Two Latin squares without a transversal. Left: the addition table of $\mathbb{Z}_4$. Right: a Latin square of order $2k$ made of four $k \times k$ Latin squares, where the blue blocks use the symbol set S_0 and the orange blocks use S_1. A transversal has as many cells in the top-right block as in the bottom-left block, so it has an even number of cells in orange blocks. It also needs all k symbols of S_1, which is impossible when k is odd.

2.4 Multiplication tables of groups

Complete mappings and the Hall–Paige conjecture

By 1942, all known pairs of orthogonal Latin squares came from groups: those in the statistical tables of Fisher and Yates, and those constructed from finite fields by Bose and by Stevens. Mann [212] therefore asked which groups can be used. For a group, a single transversal is enough. If the cells $(x, \phi(x))$, $x \in G$, form a transversal of the multiplication table of G, then for every $g \in G$ the cells $(x, \phi(x)g)$ form another transversal, and these n transversals partition the table. A bijection $\theta : G \to G$ is a *complete mapping* if $x \mapsto x\theta(x)$ is also a bijection. The transversals of the multiplication table are exactly the sets $\{(x, \theta(x)) : x \in G\}$ for complete mappings θ.

Theorem 2.10 (Mann [212]; Paige [232])**.** *The multiplication table of a finite group G has an orthogonal mate if and only if G has a complete mapping.*

Mann showed that no group of order $4n+2$ has a complete mapping, so groups cannot give counterexamples to Euler's conjecture. He also found new orthogonal pairs of order 12 from groups. Paige [231] characterised the abelian groups with a complete mapping (Section 2.3), and observed that every group of odd order has one, namely $\theta(x) = x$. For general groups, Paige [232] showed that if G has a complete mapping, then the product of all elements of G, in any order, lies in the commutator subgroup. Hall and Paige showed that this condition holds exactly when the Sylow 2-subgroups of G are trivial or non-cyclic. They proved that it is sufficient for solvable groups, and for the symmetric and alternating groups, and conjectured that it is always sufficient.

Theorem 2.11 (Hall and Paige [151]; Wilcox [267], Evans [121] and Bray et al. [66])**.** *The multiplication table of a finite group G has a transversal if and only if the Sylow 2-subgroups of G are trivial or non-cyclic.*

The proof uses the classification of finite simple groups. Hall and Paige showed that if a normal subgroup N and the quotient G/N both have complete mappings, then so does G. Wilcox [267] reduced the conjecture to the simple groups and treated the groups of Lie type. Evans [121] treated the Tits group and all sporadic groups except the Janko group J_4, and Bray treated J_4 [66]. Two proofs for large groups that avoid the classification have since appeared. Eberhard, Manners and Mrazović [115] used analytic number theory to count complete mappings: a group G of order n satisfying the Hall–Paige condition has $(e^{-1/2} + o(1))\,|G^{\mathrm{ab}}|\,n!^2/n^n$ of them, where G^{ab} is the abelianisation. Müyesser and Pokrovskiy [224] gave a combinatorial proof by absorption. They showed more generally that if X, Y and Z are random-like subsets of a large group G of the same size, then the table with rows X, columns Y and symbols Z has a transversal whenever $\sum X + \sum Y = \sum Z$ in G^{ab}. This sum condition is necessary, by the argument of Section 2.3, and taking $X = Y = Z = G$ gives Theorem 2.11.

Queens and semiqueens

In 1848 Bezzel asked for eight queens on a chessboard, no two in the same row, column or diagonal. Nauck found all 92 solutions in 1850, and Pauls showed in 1874 that n non-attacking queens can be placed on an $n \times n$ board for every $n \notin \{2, 3\}$ (see [61]). A placement with one queen in each row is a permutation σ, with the queen of row j in column $\sigma(j)$, and it is non-attacking exactly when the values $\sigma(j) + j$ are distinct and the values $\sigma(j) - j$ are distinct. Pólya [242] studied placements that remain non-attacking when the board is repeated periodically to fill the plane. These are the placements on the torus, for which $\sigma(j) + j$ and $\sigma(j) - j$ are distinct modulo n (Figure 6). Pólya showed that they exist exactly when $n \equiv 1, 5 \bmod 6$.

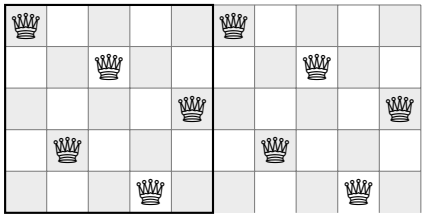


Figure 6: Five non-attacking queens on the 5×5 torus, $\sigma(j) = 2j$ on $\mathbb{Z}_5$, shown with a periodic copy to the right; the queens still do not attack each other across the boundary. Here $\sigma(j) + j = 3j$ and $\sigma(j) - j = j$ are both bijections of $\mathbb{Z}_5$.

In the language of groups, a placement of n *semiqueens* on the torus, which attack along rows, columns and one family of diagonals, is a complete mapping of $\mathbb{Z}_n$, that is, a transversal of the addition table of $\mathbb{Z}_n$. A toroidal placement of n queens is a *strong complete mapping* of $\mathbb{Z}_n$, for which $x \mapsto x + \sigma(x)$ and $x \mapsto -x + \sigma(x)$ are both bijections. The counting questions for queens were solved by the methods used for Latin squares.

Theorem 2.12 (Eberhard, Manners and Mrazović [114])**.** *For odd n, the addition table of $\mathbb{Z}_n$ has $(e^{-1/2} + o(1))\, n\, n!^2/n^n$ transversals.*

This proved a conjecture of Vardi. For queens, Luria [208] showed that the number $T(n)$ of toroidal placements is at most $((1 + o(1))ne^{-3})^n$, and Bowtell and Keevash [61] showed that this is tight for $n \equiv 1, 5 \bmod 6$, using a random greedy process followed by absorption. For the ordinary board, Simkin [249] showed that the number of placements is $((1 \pm o(1))ne^{-\alpha})^n$ with $\alpha = 1.942 \pm 0.003$. Glock, Munhá Correia and Sudakov [135] answered a question of Nauck on completing a partial placement, using rainbow matchings: for large n, any $n/60$ non-attacking queens can be completed to n, while some placements of about $n/4$ queens cannot. Determining the completion threshold asymptotically is open. Glock, Munhá Correia and Sudakov show that their $n/4$ example has no fractional completion either, and conjecture that the threshold equals its fractional relaxation asymptotically, which splits the problem into two parts.

Subsquares: Snevily's conjecture

What happens if we keep only some rows and columns of a group table? Snevily [252] raised this question for $\mathbb{Z}_n$ in 1999. One of his motivations was a statement about finite fields: if A and B are subsets of the same size k of a finite field of odd characteristic, then the elements of B can be ordered as $b_{\pi(1)}, \ldots, b_{\pi(k)}$ so that the sums $a_i + b_{\pi(i)}$ are distinct.

Conjecture 2.13 (Snevily, 1999 [252])**.** *Let G be an abelian group of odd order and $A, B \subseteq G$ with $|A| = |B| = k$. Then the $k \times k$ subsquare of the addition table with rows A and columns B has a transversal.*

For groups of even order, the subsquare may be the table of a subgroup with no transversal, such as $\{0, 2\} \times \{0, 2\}$ in $\mathbb{Z}_4$, and Snevily conjectured that in $\mathbb{Z}_n$ these and their translates are the only exceptions. Alon [21] proved Conjecture 2.13 for $\mathbb{Z}_p$ using the Combinatorial Nullstellensatz, Dasgupta, Károlyi, Serra and Szegedy [105] for all cyclic groups of odd order, and Arsovski [38] for all abelian groups of odd

order. Müyesser and Pokrovskiy [224] determined, for large subsquares of arbitrary groups, exactly when a transversal exists. Goddyn and Halasz [140] showed that every group table has a partial transversal with $n-1$ cells, which is the Ryser–Brualdi–Stein conjecture for group tables. Müyesser and Pokrovskiy ask whether the Fourier-analytic counting of Eberhard, Manners and Mrazović can be adapted to count transversals of subsquares, or sequencings of groups (Section 4.2) [224].

Cycle types

A transversal of a group table can also be read as a permutation. A bijection $\psi : G \to G$ is an *orthomorphism* if $x \mapsto x^{-1}\psi(x)$ is also a bijection, and orthomorphisms correspond to transversals as complete mappings do. Orthomorphisms with prescribed cycle types are useful in constructions. Ringel's solution of the Heawood map colouring problem used orthomorphisms consisting of a single cycle, and orthomorphisms whose cycles all have length 6 give cyclic Steiner triple systems (see [223]).

Conjecture 2.14 (Friedlander, Gordon and Tannenbaum, 1981 [125])**.** *Let G be an abelian group of order n satisfying the Hall–Paige condition, and let $k \geq 2$ divide $n-1$. Then some orthomorphism of G fixes the identity and permutes the other elements in disjoint cycles of length k.*

Friedlander, Gordon and Tannenbaum verified the conjecture for groups of order at most 15 and for abelian p-groups with $p \geq 3$. Even the case $k = 3$ for cyclic groups remained open for forty years. Müyesser proved the conjecture for large groups, using the absorption methods described in Section 2.5.

Theorem 2.15 (Müyesser [223])**.** *The Friedlander–Gordon–Tannenbaum conjecture holds for all sufficiently large groups.*

The method gives many other cycle types as well. Müyesser conjectured that the same holds for all large groups satisfying the Hall–Paige condition, abelian or not, and asked which cycle types an orthomorphism of an abelian group can have [223]. The case $k = 3$ also has a version for arbitrary Latin squares: he conjectured that every proper colouring of K_n with n colours has a rainbow set of vertex-disjoint triangles covering all but a bounded number of vertices.

2.5 Full transversals in random Latin squares

For general Latin squares, the progress on Conjecture 2.9 has come from random Latin squares. These are hard to study, because changing one cell of a Latin square forces changes in many other cells. Kwan compared a random Latin square with the outcome of a random triangle removal process.

Theorem 2.16 (Kwan [189])**.** *With high probability, a uniformly random Latin square of order n has $((1-o(1))n/e^2)^n$ transversals.*

Gould and Kelly [142] used switchings in random Latin rectangles, which are less rigid than Latin squares, and transferred the results to random Latin squares using bounds on permanents, following McKay and Wanless.

Theorem 2.17 (Gould and Kelly [142])**.** *With high probability, a uniformly random Latin square has a Hamilton transversal.*

This proves, for almost all Latin squares, a conjecture of Gyárfás and Sárközy: every Latin square of order n should have a partial transversal with $n-2$ cells which contains no cycle in the directed translation. Bowtell, Montgomery, Müyesser and Pokrovskiy [63] later proved this conjecture for all large n, as a by-product of their proof of Andersen's conjecture (Section 4.1). Eberhard, Manners and Mrazović used a version of the circle method from analytic number theory to count transversals precisely.

Theorem 2.18 (Eberhard, Manners and Mrazović [116])**.** *With high probability, a uniformly random Latin square of order n has $(e^{-1/2} + o(1))\, n!^2/n^n$ transversals.*

Their result holds for every Latin square satisfying a spectral quasirandomness condition. This includes random Latin squares with high probability and the multiplication tables of quasirandom groups. In an

earlier paper [115], they proved an asymptotic formula of the same kind for the number of transversals in the multiplication table of every large group satisfying the Hall–Paige condition. This gave a proof of Theorem 2.11 for large groups which does not use the classification of finite simple groups. Müyesser and Pokrovskiy [224] later gave a combinatorial proof for large groups, based on absorption (Section 2.4).

2.6 Partial transversals

Brualdi conjectured that every Latin square has a partial transversal with $n-1$ cells, and Stein made a stronger conjecture (Section 2.9). Together with Ryser's conjecture, this is called the Ryser–Brualdi–Stein conjecture.

Conjecture 2.19 (Ryser–Brualdi–Stein; sources from 1967, 1975, 1991 [246, 70, 253])**.** *Every Latin square of order n has a partial transversal with $n-1$ cells, and a full transversal if n is odd.*

For even n, the examples of Section 2.3 show that $n-1$ cannot be replaced by n. In terms of colourings, the first part of Conjecture 2.19 says that every optimal colouring of $K_{n,n}$ has a rainbow matching with $n-1$ edges.

Any maximal partial transversal has at least $n/2$ cells. Koksma improved this to $2n/3$ in 1969 and Drake to $3n/4$ in 1977. In 1978, Brouwer, de Vries and Wieringa [68] and Woolbright [270] independently proved that there is always a partial transversal with $n-\sqrt{n}$ cells. Shor [248] improved this to $n-O(\log^2 n)$ in 1982. His proof had an error, which Hatami and Shor [153] corrected in 2008.

Theorem 2.20 (Keevash, Pokrovskiy, Sudakov and Yepremyan [174])**.** *Every Latin square of order n has a partial transversal with $n-O(\log n/\log\log n)$ cells.*

The proof starts from a random partial transversal with $n-o(n)$ cells and enlarges it one cell at a time, by exchanging edges along alternating paths in the coloured $K_{n,n}$. Montgomery [215] explains why $\log n/\log\log n$ is the limit of this method: with fewer unused colours, there are too few alternating paths to reach a typical pair of vertices. To go further, one has to use the structure of the particular Latin square. Montgomery did this and removed the error term entirely.

Theorem 2.21 (Montgomery [214])**.** *For every sufficiently large n, every Latin square of order n has a partial transversal with $n-1$ cells.*

So, for large n, Ryser's conjecture is the only part of Conjecture 2.19 that is still open. The proof of Theorem 2.21 uses absorption. Since a general Latin square does not come from a group, Montgomery finds approximate algebraic structure in the colouring and uses it to build the absorber. The method needs the two cells that are missing from the final transversal, and Montgomery [215] explains why a full transversal for large odd n needs new ideas.

Two refinements of Theorem 2.21 appeared recently. Tang and Zhao [259] showed that every large Latin square of order n has $((1\pm n^{-c})\,n/e^2)^n$ partial transversals with $n-1$ cells, for some constant $c>0$. Anastos, Fabian, Müyesser and Szabó [34] proposed a version of the Ryser–Brualdi–Stein conjecture with prescribed numbers of edges of each colour. For a bipartite multigraph on $2n$ vertices which is the union of three perfect matchings M_1, M_2, M_3, they conjectured that for all $a_1, a_2, a_3 \geq 0$ with $a_1+a_2+a_3 = n-1$ there is a matching M with $|M\cap M_i| = a_i$ for each i. They proved this with $n-2$ in place of $n-1$, and Boyadzhiyska, Christoph and Szabó [64] proved the conjecture. More generally, Anastos, Fabian, Müyesser and Szabó proposed a multiplicity version of the whole Ryser–Brualdi–Stein conjecture.

Conjecture 2.22 (Anastos, Fabian, Müyesser and Szabó, 2022 [34])**.** *Let $K_{n,n}$ be decomposed into perfect matchings $M_1,\dots,M_n$, and let $a_1,\dots,a_n \geq 0$ be integers with $a_1+\cdots+a_n = n-1$. Then there is a matching M with $|M\cap M_i| = a_i$ for every i.*

Taking $a_i = 1$ for $i<n$ and $a_n = 0$ gives the first part of Conjecture 2.19, which Theorem 2.21 proves for large n. The theorem of Boyadzhiyska, Christoph and Szabó covers all sequences with at most three non-zero entries. Already four non-zero entries seem to require new ideas, and a first target is to find matchings with $n-f(k)$ edges when k entries are non-zero [64]. No counterexample can come from the addition table of an abelian group, by an old result of Hall [34].

2.7 Counting transversals

The maximum number of transversals in a Latin square of order n is $((1+o(1))n/e^2)^n$. The upper bound is due to Taranenko [260], and Glebov and Luria [134] gave a simpler proof and a matching construction. By Theorems 2.16 and 2.18, a random Latin square has this many transversals, up to the $o(1)$ term.

For odd n, Ryser's conjecture says that the minimum number of transversals is positive. Wanless [263] asked for upper bounds on this minimum. Dai, Divoux and Kelly [104] showed that some Latin squares of odd order have exponentially fewer transversals than a random one, and Luria found a simpler construction with fewer transversals.

Theorem 2.23 (Luria [209])**.** *For every $n \equiv 3 \bmod 6$, there is a Latin square of order n with at most $((1+o(1))\,2n/(3e^2))^n$ transversals.*

The construction is a small change to a group table. Write $n = 3q$, index rows, columns and symbols by $\mathbb{Z}_3 \times \mathbb{Z}_q$, and put the symbol $(i+j,\ a+b+\mathbf{1}_{i=j=0})$ in the cell $((i,a),(j,b))$. This is the addition table of $\mathbb{Z}_3 \times \mathbb{Z}_q$, except that one of the nine $q \times q$ blocks is shifted by one. Summing the symbols along a transversal, as in Section 2.3, shows that every transversal either lies inside the three diagonal blocks or avoids them.

These constructions bound the minimum from above. No lower bound is known, since even positivity for odd n is Ryser's conjecture. Ghafari and Wanless asked whether a Latin square of odd order can have exactly one transversal, which they consider unlikely, and Dai, Divoux and Kelly believe that every Latin square of large odd order n has at least $(n/C)^n$ transversals [104]. Tang and Zhao ask more generally for the smallest positive number of transversals [259].

2.8 Proper colourings with more colours

A *generalised Latin square* of order n is an $n \times n$ array in which no symbol appears twice in a row or column, but which may use more than n symbols. These are the proper colourings of $K_{n,n}$. Theorems 2.20 and 2.21 hold for all proper colourings of $K_{n,n}$ [174, 214]. Best, Pula and Wanless [55] conjectured that every proper colouring of $K_{n-1,n}$ has a rainbow matching with $n-1$ edges, and Montgomery [215] notes that the methods of [214] prove this for large n.

More colours should make a rainbow perfect matching easier to find. Montgomery, Pokrovskiy and Sudakov [217] showed that if at most $(1-o(1))n$ colours appear more than $(1-o(1))n$ times in a proper colouring of $K_{n,n}$, then there are $(1-o(1))n$ disjoint rainbow perfect matchings. Akbari and Alipour asked how many colours force one rainbow perfect matching.

Conjecture 2.24 (Akbari and Alipour, 2004 [15])**.** *For large n, every proper colouring of $K_{n,n}$ with at least $n^2/2$ colours has a rainbow perfect matching.*

Barát and Nagy [46] proved this with $3n^2/4$ colours, and Best, Hendrey, Wanless, Wilson and Wood [54] with $(2-\sqrt{2})n^2$ colours.

Theorem 2.25 (Keevash and Yepremyan [177]; Montgomery, Pokrovskiy and Sudakov [217])**.** *There is $\varepsilon > 0$ such that, for large n, every proper colouring of $K_{n,n}$ with at least $n^{2-\varepsilon}$ colours has a rainbow perfect matching. In particular, Conjecture 2.24 holds.*

Keevash and Yepremyan obtained $\varepsilon = 1/200$, and the methods of [174] reduce the number of colours to $O(n \log n/\log\log n)$.

2.9 Globally bounded colourings

In 1975, Stein [253] made several conjectures in which the condition on rows and columns is replaced by a condition on the total number of times each symbol appears. An *equi-n-square* is an $n \times n$ array filled with n symbols, each appearing exactly n times. These correspond to the colourings of $K_{n,n}$ with n colours, each on exactly n edges.

Conjecture 2.26 (Stein, 1975 [253])**.** *Every equi-n-square has a partial transversal with $n-1$ cells.*

Every Latin square is an equi-n-square, so this contains Brualdi's conjecture. If no symbol appears too often, there is a full transversal. Erdős and Spencer proved this using a new version of the Lovász Local Lemma, the lopsided Local Lemma, which they introduced for this purpose.

Theorem 2.27 (Erdős and Spencer [118]). *Every $n \times n$ array in which no symbol appears more than $(n-1)/16$ times has a full transversal.*

Bissacot, Fernández, Procacci and Scoppola [56] improved $1/16$ to $27/256$. Alon, Spencer and Tetali [29] showed that, if n is a power of 2 and no symbol appears more than εn times, then the array decomposes into transversals.

For equi-n-squares, Stein showed that there is a partial transversal with about $(1-1/e)n$ cells. Aharoni, Berger, Kotlar and Ziv [5] improved this to $2n/3$ using a topological version of Hall's theorem. Anastos and Morris [35] found partial transversals with $(3/4-o(1))n$ cells by a polynomial-time algorithm. Stein's conjecture is false.

Theorem 2.28 (Pokrovskiy and Sudakov [240]). *For large n, there are equi-n-squares with no partial transversal with more than $n - \frac{1}{42}\ln n$ cells.*

Pokrovskiy [239] asked whether every equi-n-square has a partial transversal with $n - o(n)$ cells. The answer is yes.

Theorem 2.29 (Chakraborti, Christoph, Hunter, Montgomery and Petrov [78]). *Every equi-n-square has $n - n^{1-\Omega(1)}$ disjoint partial transversals, each with $n - n^{1-\Omega(1)}$ cells. There are equi-n-squares with no partial transversal with more than $n - \Omega(\sqrt{n})$ cells.*

So Stein's conjecture fails by a polynomial amount, but it is true up to a polynomially smaller error. The positive part also gives almost a decomposition into near-transversals, in the spirit of Theorem 2.3. The behaviour of equi-n-squares is therefore different from that of Latin squares, where Theorem 2.21 gives $n-1$ cells.

Stein's conjecture asks for $n-1$ cells when each colour appears exactly n times. If each colour appears slightly fewer times, a full transversal may exist. Erdős and Spencer needed every colour to appear at most about $n/16$ times, and Pokrovskiy and Sudakov [240] asked how far this can be pushed.

Problem 2.30 (Pokrovskiy and Sudakov, 2017 [240]). Does every colouring of $K_{n,n}$ in which each colour appears at most $(1-o(1))n$ times have a rainbow perfect matching?

With an extra local condition the answer is yes, even in a strong form. Kim, Kühn, Kupavskii and Osthus [178] showed that if each colour appears at most $(1-o(1))n$ times in total and at most $o(n/\log^2 n)$ times at each vertex, then $K_{n,n}$ has $(1-o(1))n$ edge-disjoint rainbow perfect matchings. Without a local condition, Problem 2.30 is open.

For equi-n-squares, the gap left by Theorem 2.29 is between a polynomial error $n^{1-\Omega(1)}$ and the deficit of order $\sqrt{n}$ in the construction. Chakraborti, Christoph, Hunter, Montgomery and Petrov suggest that their construction is optimal up to the constant.

Conjecture 2.31 (Chakraborti, Christoph, Hunter, Montgomery and Petrov, 2024 [78]). *There is a constant C such that every equi-n-square has a partial transversal with at least $n - C\sqrt{n}$ cells.*

One of Stein's conjectures from 1975 survives the counterexamples [215]. It weakens the Latin condition on one side only.

Conjecture 2.32 (Stein, 1975 [253]). *Every colouring of $K_{n,n-1}$ in which each vertex of the larger class meets at most one edge of each colour has a rainbow matching with $n-1$ edges.*

This contains the Best–Pula–Wanless conjecture of Section 2.8, and Montgomery [215] describes it as widely open.

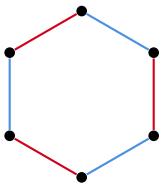
For locally and globally bounded colourings of complete multipartite graphs and hypergraphs, see Kamčev, Sudakov and Volec [167]. Globally bounded colourings of K_n and rainbow Hamilton cycles are discussed in Section 4.

2.10 Multigraphs and families of matchings

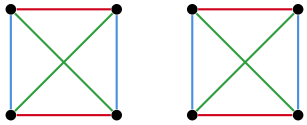
The colour classes of a proper colouring are matchings. So we may forget the host graph and consider a family $F_1, \dots, F_m$ of matchings, which may share edges. A *rainbow matching* is a matching with at most one edge from each F_i, and it is *full* if it has exactly one edge from each F_i.

Conjecture 2.33 (Aharoni and Berger, 2009 [2]). *Any n matchings of size $n+1$ in a bipartite multigraph have a full rainbow matching.*

Conjecture 2.33 implies Brualdi's conjecture. Delete one symbol from a Latin square of order n. The other $n-1$ symbols give $n-1$ perfect matchings of size n in $K_{n,n}$, and a full rainbow matching of them is a partial transversal with $n-1$ cells. The size $n+1$ cannot be replaced by n (Figure 7).



M_1 (red) and M_2 (blue); two copies of M_1, one of M_2



two copies of K_4, each properly coloured with the same three colours

Figure 7: Left: three matchings of size 3 in a bipartite graph with no full rainbow matching, since the only perfect matchings of the 6-cycle are M_1 and M_2. In general, $n-1$ copies of M_1 and one copy of M_2 in a $2n$-cycle show that size n is not enough in Conjecture 2.33. Right: three matchings of size 4 with no full rainbow matching, since two disjoint edges in a copy of K_4 have the same colour. This is why Conjecture 2.37 needs size $n+2$.

If each matching has $2n$ edges, a greedy argument gives a full rainbow matching. Aharoni, Charbit and Howard [7] improved $2n$ to $7n/4$, Kotlar and Ziv [182] to $5n/3$, Pokrovskiy [236] to $(\varphi + o(1))n$ where φ is the golden ratio, Clemens and Ehrenmüller [89] to $(3/2 + o(1))n$, and Aharoni, Kotlar and Ziv [12] to $3n/2 + 1$. In a different direction, Barát, Gyárfás and Sárközy [45] showed that n matchings of size $n+1$ in a bipartite multigraph have a rainbow matching of size $n - \sqrt{n}$.

Theorem 2.34 (Pokrovskiy [237]). *Any n matchings of size $n + o(n)$ in a bipartite multigraph have a full rainbow matching.*

Correia, Pokrovskiy and Sudakov [94] gave a short proof with $o(n)$ replaced by $n^{3/4}$. Their sampling trick turns results which miss a few of the matchings into results which use all of them, and they applied it to several of the problems below.

How many matchings of size n force a rainbow matching of size n? For bipartite graphs, Drisko answered this.

Theorem 2.35 (Drisko [112]). *Any $2n-1$ matchings of size n in a bipartite graph have a rainbow matching of size n.*

The two perfect matchings of a $2n$-cycle, each taken $n-1$ times, show that $2n-2$ matchings are not enough. Drisko's theorem extends to the intersection of two matroids (Section 3.6).

Conjecture 2.36 (Aharoni, Berger, Chudnovsky, Howard and Seymour, 2016 [3]). *Any $2n$ matchings of size n in a graph have a rainbow matching of size n. If n is odd, $2n-1$ matchings suffice.*

They proved this with $3n-2$ matchings, and Aharoni, Briggs, Kim and Kim [6] improved this to $3n-3$.

Without bipartiteness, the analogue of Conjecture 2.33 needs matchings of size $n+2$ (Figure 7).

Conjecture 2.37 (Gao, Ramadurai, Wanless and Wormald, 2017 [130]). *Any n matchings of size $n+2$ in a multigraph have a full rainbow matching.*

Gao, Ramadurai, Wanless and Wormald proved an asymptotic version for multigraphs with small edge multiplicities. Keevash and Yepremyan [176] found rainbow matchings of size $n - O(1)$ when the multiplicities are $o(n)$ and the matchings have size $n + o(n)$. For arbitrary multigraphs, Aharoni, Berger, Chudnovsky and Zerbib [4] showed that n matchings of size n have a rainbow matching of size $2n/3 - 1$.

Theorem 2.38 (Correia, Pokrovskiy and Sudakov [94]). *For large n, any n matchings of size at least $n + 20n^{1-1/16}$ in a multigraph have a full rainbow matching.*

Chakraborti and Loh [81] allowed colour classes with maximum degree $o(n)$ and multiplicities $o(n)$. As an application, they showed that every proper colouring of a graph with $2n + o(n)$ colours, each on at least n edges, has a rainbow matching of size n.

Grinblat [147] studied a variant that came from algebras of sets. Let $v(n)$ be the least v such that every n-edge-coloured multigraph, in which each colour class is a vertex-disjoint union of cliques covering at least v vertices, has a full rainbow matching. Taking $n - 1$ disjoint triangles, each in all n colours, shows that $v(n) \geq 3n - 2$.

Theorem 2.39 (Correia and Sudakov [95]). *For large n, $v(n) = 3n - 2$. This proves a conjecture of Grinblat.*

Earlier bounds were $10n/3+o(n)$ by Grinblat, $16n/5+O(1)$ by Nivasch and Omri [227], and $3n+O(\sqrt{n})$ by Clemens, Ehrenmüller and Pokrovskiy [90]. For simple graphs, Munhá Correia and Yepremyan [96] showed that $2n + o(n)$ vertices suffice.

Between these two extremes, let $v_m(n)$ be the least number of vertices that each colour class must cover when every edge has multiplicity at most m. Then $v_n(n) = 3n - 2$ and $v_1(n) = 2n + o(n)$, and Munhá Correia, Pokrovskiy and Sudakov showed $v_m(n) \leq 2n + 2m + O(n/(\log n)^{1/4})$, which is asymptotically tight when m is a small constant fraction of n [94, 95]. The case $m = 1$ is closely tied to transversals. Add $c/2$ disjoint stars with n edges each, one of each colour, to a properly n-coloured $K_{n,n}$. Every colour class is now a union of cliques covering $2n + c$ vertices, so a bound $v_1(n) \leq 2n + c$ would give a rainbow matching of size n and hence a partial transversal with $n - c/2$ cells in every Latin square [95]. Munhá Correia and Yepremyan ask whether $2n + C \log^2 n$ vertices suffice [96].

For the conjectures of Aharoni and Berger and of Gao, Ramadurai, Wanless and Wormald, the known error terms are polynomial. For Latin squares the error is $O(\log n/ \log \log n)$ in Theorem 2.20, or one cell in Theorem 2.21. Munhá Correia, Pokrovskiy and Sudakov point out that sub-polynomial errors would need new ideas, since their sampling trick requires polynomial errors [94], and Montgomery [215] notes that the absorption methods behind Theorem 2.21 do not seem to apply to Conjecture 2.33.

Rainbow matchings in families of sets, such as the Aharoni–Howard conjecture and the rainbow version of the Erdős matching conjecture, are discussed in Section 7.

2.11 General host graphs

The *colour degree* of a vertex is the number of distinct colours on the edges at it, and $\delta^c(G)$ is the minimum colour degree of G.

Theorem 2.40 (Kostochka and Yancey [181]). *Every edge-coloured graph G has a rainbow matching of size $\lceil \delta^c(G)/2 \rceil$.*

This was conjectured by Li and Wang [194], and LeSaulnier, Stocker, Wenger and West [190] had proved the bound $\lfloor \delta^c(G)/2 \rfloor$. A properly coloured complete graph on $\delta^c(G) + 1$ vertices shows that it is tight. If G has many vertices, there is a rainbow matching of size $\delta^c(G)$: Kostochka, Pfender and Yancey [180] proved this when $|V(G)| \geq 4.25\, \delta^c(G)^2$.

For proper colourings, Wang asked whether every properly coloured graph with minimum degree δ and at least $f(\delta)$ vertices has a rainbow matching of size δ. Diemunsch, Ferrara, Moffatt, Pfender and Wenger [108] showed that $f(\delta) = 6.5\delta$ works, and later work [199, 150, 200] reduced this to $4\delta-4$ for $\delta \geq 4$. A properly coloured $K_{2\delta-1}$ shows that $f(\delta) \geq 2\delta$. Closing the gap between 2δ and $4\delta - 4$ is open. Other conditions, such as average colour degree or the number of edges plus the number of colours, are studied in [185, 274, 204, 205]. For example, Zhou [274] conjectures that, apart from one coloured graph on four vertices, every edge-coloured graph with average colour degree at least $2k - 1$ has a rainbow matching of size k. Colour-degree conditions for rainbow cycles and tilings are discussed in Sections 4 and 5.

2.12 Steiner triple systems

A *Steiner triple system* of order n is a 3-uniform hypergraph on n vertices in which every pair of vertices lies in exactly one edge. They exist exactly when $n \equiv 1, 3 \bmod 6$. By the hypergraph translation of Section 2.1, a Latin square is a 3-partite analogue of a Steiner triple system, and a transversal is a perfect matching. Brouwer made the analogue of Brualdi's conjecture.

Conjecture 2.41 (Brouwer, 1981 [67]). *Every Steiner triple system of order n has a matching with at least $(n-4)/3$ edges.*

The Fano plane shows that the bound is tight for $n = 7$ (Figure 8), and constructions of Wilson and of Bryant and Horsley show that it is tight for infinitely many n (see [215]).

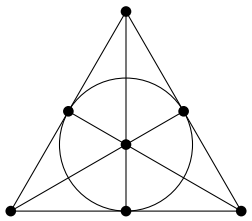


Figure 8: The Fano plane, a Steiner triple system of order 7 whose seven triples are the three sides, the three medians and the circle. Any two triples meet, so its largest matching has one edge, and $1 = (7-4)/3$.

Brouwer proved a bound of $n/3 - O(n^{2/3})$, and Alon, Kim and Spencer [26] improved it to $n/3 - O(n^{1/2}\log^{3/2} n)$. Keevash, Pokrovskiy, Sudakov and Yepremyan [174] turned the problem into a rainbow matching problem. They split the vertices at random into three sets A, B and C, and join $a \in A$ to $b \in B$ by an edge of colour $c \in C$ whenever abc is a triple. This gives a properly coloured bipartite graph whose rainbow matchings are matchings of the Steiner triple system, and they obtained $n/3 - O(\log n/\log\log n)$.

Theorem 2.42 (Montgomery [214]). *For large n, every Steiner triple system of order n has a matching with $\lfloor n/3 \rfloor - 1$ edges. In particular, Conjecture 2.41 holds for large n.*

As for Latin squares, the bound cannot be improved in general, because of the constructions above. Tang and Zhao [259] showed that the number of such matchings is $((1 \pm n^{-c})\, n/(2e^2))^{\lfloor n/3 \rfloor}$, so these near-perfect matchings are not rare. For random Steiner triple systems much more is true, as we now discuss.

A Steiner triple system is *resolvable* if its triples split into perfect matchings. The existence of resolvable systems is Kirkman's schoolgirl problem from 1850: fifteen schoolgirls walk out in five rows of three on seven days, and no two of them should walk in the same row twice. Ray-Chaudhuri and Wilson [243] showed that resolvable Steiner triple systems exist for every $n \equiv 3 \bmod 6$. For random Steiner triple systems, Kwan [189] showed that almost every Steiner triple system of order $n \equiv 3 \bmod 6$ has a perfect matching, and Ferber and Kwan [124] found $(1/2 - o(1))n$ disjoint perfect matchings, which cover almost all triples.

Conjecture 2.43 (Ferber and Kwan, 2019 [124]). *Almost every Steiner triple system of order $n \equiv 3 \bmod 6$ is resolvable.*

Theorem 2.2 is the 3-partite version of Conjecture 2.43, because a Latin square decomposes into transversals exactly when its 3-partite hypergraph is resolvable. Bowtell and Montgomery [62] note that some tools in their proof come from counting perfect matchings in bipartite graphs and have no known analogue for Steiner triple systems. There is also a counting version. The number of resolutions of a Steiner triple system of order n is at most $((1+o(1))n/(2e^3))^{n^2/6}$, and Tang and Zhao [259] ask whether a random Steiner triple system has this many with high probability, which would strengthen Conjecture 2.43.

3 Rainbow spanning trees and bases

This section is about rainbow spanning trees and rainbow copies of given trees. In a properly coloured complete graph, a rainbow spanning tree always exists: the star at any vertex is one. So the main questions

are how many edge-disjoint rainbow spanning trees there are, and which trees have rainbow copies. The second question is closely tied to graph labelling and to Ringel's tree packing conjecture.

3.1 Rainbow spanning trees in general graphs

In an arbitrary edge-coloured graph, Suzuki characterised when a rainbow spanning tree exists. Deleting all edges of r colours from a connected graph must leave at most $r+1$ components, since a rainbow spanning tree loses at most r edges. This necessary condition is also sufficient.

Theorem 3.1 (Suzuki [256]). *An edge-coloured connected graph G on n vertices has a rainbow spanning tree if and only if, for every $1 \le r \le n-2$, deleting the edges of any r colours leaves a graph with at most $r+1$ components.*

A rainbow spanning tree is a common basis of two matroids on the edge set: the graphic matroid of G, and the partition matroid in which a set is independent if it has at most one edge of each colour. So Theorem 3.1 is a case of Edmonds' matroid intersection theorem, and a rainbow spanning tree can be found in polynomial time. We return to matroids in Section 3.6.

When the colouring has fewer than $n-1$ colours, a spanning tree cannot be rainbow, and the natural object is a forest using each colour exactly once. Mosallam and Zaslavsky [221] studied these total rainbow forests. Using a necessary and sufficient condition for their existence, they showed that there are very many minimal edge-coloured graphs without one. They ask for a description of all minimal obstructions, which they expect to be very difficult, and, more modestly, for a general construction that contains their known families [221]. Instead of one spanning tree one can also ask for few vertex-disjoint rainbow trees covering all vertices; for complete graphs coloured with a given number of colours, the minimum number needed was determined by Jin and Li [164].

An older result of this kind has a nice consequence. Graham and Pollak [145] studied a problem about addressing in telephone networks. In a loop-switching network, a message should be routed using addresses from which the distance between any two nodes can be read off, and this led them to ask how few complete bipartite graphs are needed to partition the edges of a graph. They showed that K_n needs at least $n-1$ of them. Alon, Brualdi and Shader proved a rainbow statement that implies this.

Theorem 3.2 (Alon, Brualdi and Shader [23]). *If the edges of K_n are coloured so that every colour class is a complete bipartite graph, then there is a rainbow spanning tree.*

A rainbow spanning tree uses $n-1$ different colours. So a partition of $E(K_n)$ into complete bipartite graphs, viewed as a colouring, has at least $n-1$ parts, which is the Graham–Pollak theorem (Figure 9).

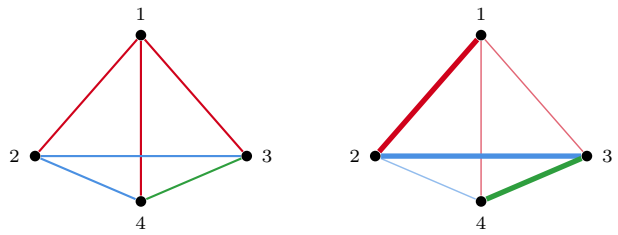


Figure 9: Left: K_4 partitioned into three complete bipartite graphs, the star at 1 (red), the star from 2 to $\{3,4\}$ (blue) and the edge 34 (green). Right: a rainbow spanning tree. By Theorem 3.2, any partition of $E(K_n)$ into complete bipartite graphs has a rainbow spanning tree, so it has at least $n-1$ parts.

3.2 Decomposing properly coloured complete graphs

A spanning tree of K_{2n} has $2n-1$ edges, and K_{2n} has $n(2n-1)$ edges. So K_{2n} contains at most n edge-disjoint spanning trees, and n of them decompose it. For optimal colourings of K_{2n}, which are 1-factorisations, Brualdi and Hollingsworth conjectured that such a decomposition into rainbow trees exists.

Conjecture 3.3 (Brualdi and Hollingsworth, 1996 [69]). *For every $n \geq 3$, every optimal colouring of K_{2n} decomposes into n rainbow spanning trees.*

The condition $n \geq 3$ is needed. The optimal colouring of K_4 is unique up to isomorphism, and its only rainbow spanning trees are the four stars. Any two stars in K_4 share an edge, so there are no two edge-disjoint rainbow spanning trees (Figure 10).

Conjecture 3.3 asks for a decomposition into rainbow trees of any shapes. Constantine asked whether the shapes can be made the same. This links the problem to the classical question of decomposing a complete graph into copies of one fixed graph, which we meet again in Ringel's conjecture (Section 3.5).

Conjecture 3.4 (Constantine, 2002 [92]). *For every $n \geq 3$, every optimal colouring of K_{2n} decomposes into n isomorphic rainbow spanning trees.*

The common tree cannot always be a Hamilton path. For n a power of 2, the colouring of K_{2n} by $\mathbb{Z}_2^k$ in Section 2.2 has no rainbow Hamilton path at all, by an argument of Maamoun and Meyniel (Section 4). So the shape of the trees has to depend on the colouring, or be chosen cleverly.

Both conjectures are about optimal colourings, in which every colour class is a perfect matching. Kaneko, Kano and Suzuki asked what happens for an arbitrary proper colouring. Then there may be many more colours, which should make rainbow trees easier to find, but the colour classes can have very different sizes. They conjectured that the same number of trees still exists.

Conjecture 3.5 (Kaneko, Kano and Suzuki, 2002 [168]). *For every $n \geq 6$, every properly coloured K_n contains $\lfloor n/2 \rfloor$ edge-disjoint rainbow spanning trees.*

For even n, the trees in Conjecture 3.5 decompose K_n. In particular, a colour with $n/2$ edges must appear in every tree, so there is no room for error. Suzuki [257] proposed a version without any condition on the colouring. For an arbitrary colouring of K_{2n}, a rainbow tree may not exist, so he asked for a decomposition into n spanning trees each of which has roughly the same colour distribution as K_{2n} itself, and he proved that a single such tree exists.

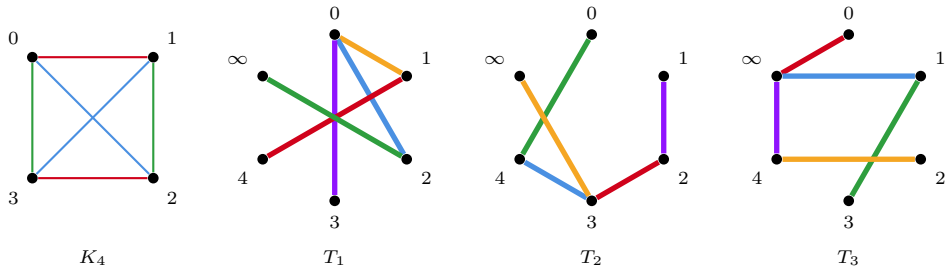


Figure 10: Left: the optimal colouring of K_4. Its rainbow spanning trees are the four stars, and any two of them share an edge. Right: three edge-disjoint rainbow spanning trees T_1, T_2, T_3 which decompose K_6, for the optimal colouring of K_6 on $\mathbb{Z}_5 \cup \{\infty\}$ in which colour i is $\{\infty i\} \cup \{\{i-j, i+j\} : j = 1, 2\}$. Colours $0, \ldots, 4$ are red, blue, green, orange and purple.

For a long time only a constant number of trees was known. Brualdi and Hollingsworth found 2, Krussel, Marshall and Verrall found 3, and Kaneko, Kano and Suzuki also found 3 for all proper colourings. Fu, Lo, Perry and Rodger [127] gave an explicit construction of $\lfloor \sqrt{6n+9}/3 \rfloor$ trees in an optimally coloured K_{2n}. Horn [155] gave the first linear bound, εn for some $\varepsilon > 0$, for optimal colourings. For all proper colourings, Balogh, Liu and Montgomery [43] and, independently, Pokrovskiy and Sudakov [241] found linearly many trees. Pokrovskiy and Sudakov found $10^{-6}n$ edge-disjoint rainbow spanning trees which are moreover isomorphic: they are all copies of one spider, a star with some of its edges subdivided once.

Theorem 3.6 (Montgomery, Pokrovskiy and Sudakov [217]). *Every properly coloured K_n contains $(1 - o(1))n/2$ edge-disjoint rainbow spanning trees.*

This proves Conjectures 3.3 and 3.5 asymptotically. The remaining difficulty is the last $o(n)$ trees. For even n, an exact decomposition must use every edge of K_n, so there is no slack at all, and a near-decomposition cannot simply be completed. For optimal colourings, Glock, Kühn, Montgomery and Osthus overcame this using absorption, and they obtained isomorphic trees as well.

Theorem 3.7 (Glock, Kühn, Montgomery and Osthus [137]). *For every large n, every optimal colouring of K_{2n} decomposes into n isomorphic rainbow spanning trees.*

Thus Conjectures 3.3 and 3.4 hold for large n, while Conjecture 3.5 is open. The method of Theorem 3.7 uses the fact that all colour classes of an optimal colouring are perfect matchings. For an arbitrary proper colouring the colour classes may have very different sizes, and this is where new ideas are needed. One natural intermediate step is to replace properness by a bound on the size of each colour class, which is the subject of the next subsection.

3.3 Bounded colourings and rainbow forests

The condition that the colouring is proper can be weakened to a bound on the size of each colour class. In a colouring of K_n in which each colour has at most $n/2$ edges, Akbari and Alipour [16] found 2 edge-disjoint rainbow spanning trees, and Carraher, Hartke and Horn [76] found $\lfloor n/(1000 \log n) \rfloor$. Horn and Nelsen [156] extended this to general host graphs, under a condition involving the second-smallest eigenvalue of the normalised Laplacian. On the other hand, Pokrovskiy and Sudakov [241] showed that some bounded colourings of K_n have no rainbow spanning spider at all, so an analogue of Constantine's conjecture for bounded colourings would need a different family of trees.

Glock, Kühn, Montgomery and Osthus [137] asked how many edge-disjoint rainbow spanning trees can be guaranteed when each colour of K_n has at most $n/2$ edges, and in particular whether linearly many can be found. They expected that a full decomposition is not possible in this generality.

A different generalisation keeps properness but changes the host graph. Montgomery, Pokrovskiy and Sudakov [217] conjectured a generalisation of Theorem 3.6 from complete graphs to all properly coloured graphs: a properly coloured graph with km edges, in which each colour has at most m edges, should contain $(1 - o(1))m$ edge-disjoint rainbow forests with k edges each. Xu showed that this holds exactly when $k \leq 2m$.

Theorem 3.8 (Xu [271]). *For every $\varepsilon > 0$ and large m, every properly coloured graph with km edges, where $1 \leq k \leq 2m$ and every colour class has at most m edges, contains $(1 - \varepsilon)m$ edge-disjoint rainbow forests with k edges each. For every $k > 2m$ there are such graphs with no rainbow forest with k edges.*

To see why $k \leq 2m$ is needed, take an optimal colouring of K_{2m+1}. It has $2m + 1$ colours, each with m edges, so it has km edges for $k = 2m + 1$. But a forest in K_{2m+1} has at most $2m$ edges. For K_n with n even, taking $m = n/2$ and $k = n - 1 \leq 2m$ in Theorem 3.8 recovers Theorem 3.6, since a forest with $n - 1$ edges in K_n is a spanning tree. So Theorem 3.8 extends the asymptotic Brualdi–Hollingsworth and Kaneko–Kano–Suzuki results from complete graphs to all properly coloured graphs, in the largest possible range.

3.4 Rainbow copies of a given tree

Which trees have a rainbow copy in every colouring of a given type? For locally bounded colourings of complete graphs, Montgomery, Pokrovskiy and Sudakov gave an essentially optimal answer.

Theorem 3.9 (Montgomery, Pokrovskiy and Sudakov [218]). *Every locally k-bounded colouring of K_n contains a rainbow copy of every tree with at most $(1 - o(1))n/k$ vertices.*

The bound is tight up to the $o(1)$ term. Merging the colour classes of an optimal colouring of K_n in groups of k gives a locally k-bounded colouring with about n/k colours, and a rainbow tree in it has fewer than about n/k edges. Theorem 3.9 is the main tool for the labelling problems in Section 3.5.

For proper colourings, the case $k = 1$, Theorem 3.9 gives every tree on $n - o(n)$ vertices, and the error cannot be removed entirely: some proper colourings of K_n have no rainbow Hamilton path (Section 4.1). Montgomery, Pokrovskiy and Sudakov conjectured that a bounded error suffices.

Conjecture 3.10 (Montgomery, Pokrovskiy and Sudakov, 2018 [218]). *There is a constant C such that every properly coloured K_n contains a rainbow copy of every tree on $n - C$ vertices.*

For paths this is a weak form of Andersen's conjecture, now proved for large n (Theorem 4.2). For general trees the right value of C is unclear [239], and the conjecture would improve the applications to labelling problems below.

Theorem 3.9 is about complete host graphs. For sparse host graphs the question is different, since the host graph itself may have few trees. The hypercube Q_n is a natural first case. It is n-regular, so a proper colouring of it may use only n colours, and then a rainbow tree has at most n edges. Crawford, Sankar, Schildkraut and Spiro showed that every tree of this size appears.

Theorem 3.11 (Crawford, Sankar, Schildkraut and Spiro [101])**.** *Every proper colouring of the hypercube Q_n contains a rainbow copy of every tree with at most n edges.*

Colouring each edge of Q_n by the coordinate in which its endpoints differ gives a proper colouring with n colours, so n edges is best possible. In this colouring there is no rainbow cycle at all, since every coordinate changes an even number of times around a cycle.

The hypercube suggests a question about all host graphs. Crawford, Sankar, Schildkraut and Spiro [101] have asked whether there is a constant C such that, for every graph of minimum degree d, every proper colouring contains a rainbow copy of every tree with at most $d - C$ edges. For paths in d-regular graphs this would give Schrijver's problem (Section 4.1) up to the constant C. Their proof works for every graph in which, for every edge xy, some matching separates x from y; in such graphs trees with d edges are already guaranteed.

3.5 Graceful labellings and Ringel's conjecture

Several old problems on trees ask for a rainbow copy of a tree in a specific colouring of a complete graph.

Graceful labellings

A tree T with n edges is *graceful* if its vertices can be labelled with distinct numbers from $\{0, 1, \dots, n\}$ so that the differences $|a - b|$ along its edges are $1, \dots, n$, each exactly once.

Conjecture 3.12 (Graceful tree conjecture; Ringel, Kotzig and Rosa, 1960s)**.** *Every tree is graceful.*

Rosa [245] introduced these labellings, under the name β-valuations, as a tool for Ringel's tree packing problem below: a graceful labelling of a tree T with n edges gives a decomposition of K_{2n+1} into $2n + 1$ copies of T.

Colour the edge ab of the complete graph on $\{0, 1, \dots, n\}$ by $|a - b|$. Then a graceful labelling of T is exactly a rainbow copy of T in this colouring (Figure 11). Each colour d has $n + 1 - d$ edges, so the colouring is far from proper, and a rainbow copy of T must use every colour.

Two asymptotic versions are known. Adamaszek, Allen, Grosu and Hladký [1] showed that every tree on n vertices with maximum degree $O(n/\log n)$ can be labelled injectively by $\{1, \dots, (1+\varepsilon)n\}$ with distinct edge differences. Letzter, Pokrovskiy and Williams [191] showed that every large tree on n vertices has a bijective labelling by $\{1, \dots, n\}$ with at least $(1-\varepsilon)n$ distinct edge differences. Li [195] determined the total number of graceful labellings of trees with N edges up to a factor $\exp(O(\log^4 N))$. This proves an estimate that Knuth's *The Art of Computer Programming* states conditionally on an empirical observation.

Ringel's conjecture

A typical decomposition problem asks whether the edges of a graph can be partitioned into copies of another graph. Ringel posed the following problem at the symposium on graph theory in Smolenice in 1963. The complete graph K_{2n+1} has $n(2n + 1)$ edges, so it could in principle be split into $2n + 1$ copies of any tree with n edges, and Ringel conjectured that this is always possible.

Conjecture 3.13 (Ringel, 1963)**.** *For every tree T with n edges, the complete graph K_{2n+1} decomposes into $2n + 1$ copies of T.*

Ringel's conjecture also has a rainbow form. Identify the vertices of K_{2n+1} with $\mathbb{Z}_{2n+1}$ and colour the edge ab by its *near distance* $\min\{|a - b|, 2n + 1 - |a - b|\}$, viewing a, b as integers in $[0, 2n]$. Each of the n

colours has $2n+1$ edges, and the rotation $x \mapsto x+1$ permutes the edges of each colour cyclically. So if T has a rainbow copy, then this copy and its $2n$ rotations decompose K_{2n+1} (Figure 11). A graceful labelling gives such a rainbow copy, so the graceful tree conjecture implies Ringel's conjecture. The near-distance colouring is locally 2-bounded, and Theorem 3.9 gives a rainbow copy of every tree with about $n/2$ edges in it. To embed a tree with n edges, Montgomery, Pokrovskiy and Sudakov developed the method much further.

Theorem 3.14 (Montgomery, Pokrovskiy and Sudakov [219]; Keevash and Staden [175])**.** *Ringel's conjecture holds for all large n.*

Keevash and Staden proved more generally that every quasirandom graph with N vertices and nN edges decomposes into N copies of any fixed tree with n edges. The proof of Montgomery, Pokrovskiy and Sudakov finds a rainbow copy of T in the near-distance colouring, as described above, but not a graceful labelling. So the graceful tree conjecture (Conjecture 3.12) remains open, and the results of [1, 191] are its best approximations. The two approximations relax different things. Adamaszek, Allen, Grosu and Hladký keep the differences distinct but use slightly more labels and need a degree bound. Letzter, Pokrovskiy and Williams use exactly the right labels but allow a few repeated differences. Letzter, Pokrovskiy and Williams propose as the next step an injective labelling of every tree on n vertices by $\{1, \dots, (1+o(1))n\}$ with distinct differences, without a degree condition [191].

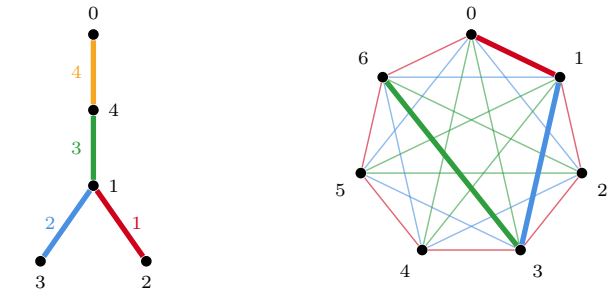


Figure 11: Left: a graceful labelling of a tree with 4 edges; edges are coloured by the difference of their labels. Right: K_7 on $\mathbb{Z}_7$ with edges coloured by near distance 1 (red), 2 (blue) and 3 (green). The path $0, 1, 3, 6$ is rainbow, and its seven rotations decompose K_7, as Ringel's conjecture predicts for the path with 3 edges.

Harmonious labellings

Graham and Sloane [146] introduced harmonious labellings in 1980, in work on additive bases. They conjectured that every tree T with n edges has a labelling $\phi : V(T) \to \mathbb{Z}_n$, using exactly one label twice, for which the sums $\phi(x) + \phi(y)$ along the edges are distinct. Let $K_{\mathbb{Z}_n}$ be the complete graph on $\mathbb{Z}_n$ in which the edge xy has colour $x+y$. This colouring is proper, since $x+y = x+z$ only when $y = z$. A rainbow copy of an n-vertex tree in $K_{\mathbb{Z}_n}$ gives a harmonious labelling of the tree with one more edge obtained by adding a leaf, so it is enough to show that $K_{\mathbb{Z}_n}$ contains a rainbow copy of every tree on n vertices [225].

Theorem 3.15 (Müyesser and Pokrovskiy [225])**.** *For every Δ there is n_0 such that, for $n \geq n_0$, the coloured complete graph $K_{\mathbb{Z}_n}$ contains a rainbow copy of every tree on n vertices with maximum degree at most Δ.*

This confirms the conjecture of Graham and Sloane for large bounded-degree trees. Müyesser and Pokrovskiy in fact determine, for every abelian group G of order n and every large bounded-degree tree T, when the analogous colouring K_G contains a rainbow copy of T. The answer is not always yes. For example, if $G = \mathbb{Z}_2^k$ and T is a path, then K_G has no rainbow copy of T, by an argument of Maamoun and Meyniel which we discuss in Section 4. Labellings by groups are discussed further in Section 2.4.

For trees of unbounded degree the Graham–Sloane conjecture is open. Hovey's cordial labelling conjecture generalises it: for every k, the vertices of every tree can be labelled by $\mathbb{Z}_k$ so that the labels on vertices, and the sums of the labels along edges, are each as evenly distributed as possible. Taking k to be the number of edges gives the harmonious labelling conjecture. Müyesser and Pokrovskiy note that their methods give Hovey's conjecture for bounded-degree trees when k is large compared to the maximum degree [225].

3.6 Bases of matroids: Rota's basis conjecture

A spanning tree of a graph is a basis of its graphic matroid, and many of the questions in this section make sense for bases of any matroid. The most famous of them comes from linear algebra.

Conjecture 3.16 (Rota, 1989; see Huang and Rota [160])**.** *Let $B_1, \dots, B_n$ be bases of an n-dimensional vector space. Then the n^2 vectors can be split into n disjoint bases, each containing exactly one vector from each B_i.*

Rota made the same conjecture for matroids of rank n. Equivalently, the vectors can be written in an $n \times n$ array whose i-th row consists of the vectors of B_i and whose every column is a basis (Figure 12). Colour the vectors of B_i with colour i. A basis meeting every B_i once is then a rainbow basis, and the conjecture asks for a decomposition into rainbow bases. One rainbow basis always exists: if every choice of $v_i \in B_i$ gave $\det(v_1, \dots, v_n) = 0$, then, as the determinant is linear in each argument and each B_i spans the space, the determinant would vanish identically. Rota asks for n rainbow bases at once, using every vector exactly once.

B_1	100	010	001
B_2	110	011	101
B_3	111	100	110

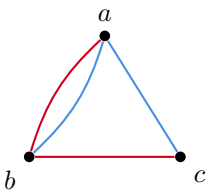


Figure 12: Left: Rota's conjecture for three bases of $\mathbb{R}^3$, with vectors written as coordinate strings. The rows are the given bases B_1, B_2, B_3, and each column has determinant 1, so each column is a basis; 6 of the 36 ways to order rows 2 and 3 work. Right: the graphic case with $n = 2$. The edges of a red and a blue spanning tree of a multigraph on $\{a, b, c\}$ split into the rainbow spanning trees $\{ab_{\text{red}}, ac\}$ and $\{bc, ab_{\text{blue}}\}$.

If all the B_i are the standard basis $\{e_1, \dots, e_n\}$, then a column is a basis exactly when its indices are distinct, so an array as in Conjecture 3.16 is a Latin square. In this sense the conjecture asks for a Latin square in which "distinct" is replaced by "linearly independent". If the vectors are $e_i - e_j$ in $\mathbb{R}^{n+1}$, then a set of them is linearly independent exactly when the corresponding edges ij form a forest. In this graphic case the conjecture says that if the edges of a multigraph on $n+1$ vertices are coloured with n colours and each colour class is a spanning tree, then the edges decompose into n rainbow spanning trees, which is a question of the same kind as Conjecture 3.3.

The conjecture came from invariant theory. For vectors $v_1, \dots, v_n$ the bracket $[v_1 \cdots v_n] = \det(v_1, \dots, v_n)$ is unchanged when all the v_i are moved by the same linear map of determinant 1. Brackets are the basic invariants of a configuration of vectors, and an array as in Conjecture 3.16 is a non-zero product of n brackets, one for each column. Huang and Rota [160] found implications between the basis conjecture, a conjecture on the supersymmetric bracket algebra, and a conjecture of Alon and Tarsi on Latin squares, which came from graph colouring.

The Alon–Tarsi conjecture

Dinitz asked in 1979 whether, given an $m \times m$ array of m-element sets, one can always choose an element from each set so that the chosen elements are distinct in every row and every column. If all the sets are $\{1, \dots, m\}$, a choice is a Latin square, so Dinitz asked whether Latin squares survive when each cell has its own list of m symbols. In graph terms, this asks whether the line graph of $K_{m,m}$ is m-choosable. Alon

and Tarsi [31] gave an algebraic criterion for choosability. Applied to Dinitz's problem, it requires the following count to be non-zero. The *sign* of a Latin square is the product of the signs of its n rows and n columns, read as permutations, and $l(n)$ is the number of even Latin squares of order n minus the number of odd ones. For odd $n > 1$, swapping two rows is a sign-reversing involution, so $l(n) = 0$.

Conjecture 3.17 (Alon and Tarsi, 1992 [31]). *For every even n, the numbers of even and odd Latin squares of order n are different.*

Galvin [128] solved Dinitz's problem for all m by a different, combinatorial argument, but Conjecture 3.17 remains open. It holds when $n = p + 1$ (Drisko [111]) and when $n = p - 1$ (Glynn [139]) for an odd prime p. Huang and Rota showed, and Onn [229] gave a short proof using a colourful determinantal identity, that if $l(n) \neq 0$ in a field $\mathbb{F}$ then Conjecture 3.16 holds for $\mathbb{F}^n$. So Rota's conjecture holds for vector spaces over fields of characteristic 0 of dimension $p \pm 1$.

Progress

Rota's conjecture is known for some classes of matroids, including paving matroids (Geelen and Humphries), strongly base orderable matroids (Wild) and matroids of rank at most 4 (Cheung); see [220]. For random bases of $\mathbb{F}_q^n$ it holds with high probability [247]. For general matroids, the question is how many disjoint rainbow bases can be found. Geelen and Webb found $\lceil \sqrt{n-1} \rceil$, and Dong and Geelen [110] found $\Omega(n/\log n)$. Bucić, Kwan, Pokrovskiy and Sudakov [72] found $(1/2 - o(1))n$, and Pokrovskiy [238] found $n - o(n)$ disjoint rainbow independent sets of size $n - o(n)$.

Theorem 3.18 (Montgomery and Sauermann [220]). *Let $B_1, \dots, B_n$ be bases of a matroid of rank n. Then there are $(1 - o(1))n$ disjoint bases, each containing exactly one element of each B_i. Moreover, the n^2 elements can be covered by $(1 + o(1))n$ such bases.*

The covering bound improves the earlier bounds $2n$ of Aharoni and Berger and $2n - 2$ from the Polymath 12 project (see [220]). Both parts of Theorem 3.18 are tight up to the $o(1)$ terms, so the asymptotic form of Rota's conjecture is settled. The exact conjecture is open, even for graphic matroids, that is, for the rainbow spanning tree problem in Figure 12. Sauermann's theorem for random bases suggests a random version of the graphic case. Let $F_1, \dots, F_{n-1}$ be independent uniformly random spanning trees of K_n. Sauermann [247] asks whether, with high probability, their edges can be partitioned into $n - 1$ spanning trees, each containing exactly one edge of each F_i. For random bases of a vector space over a finite field, this is Sauermann's theorem.

Kahn proposed a common generalisation of Rota's conjecture and Dinitz's problem, in which every cell of the array has its own basis to choose from, just as every cell in Dinitz's problem has its own list of symbols.

Conjecture 3.19 (Kahn, 1994 [160]). *Let $B_{i,j}$, $1 \leq i, j \leq n$, be bases of a matroid of rank n. Then there are elements $b_{i,j} \in B_{i,j}$ such that every row $\{b_{i,1}, \dots, b_{i,n}\}$ and every column $\{b_{1,j}, \dots, b_{n,j}\}$ is a basis.*

Taking $B_{i,j} = B_i$ for all j gives Rota's conjecture. For the uniform matroid of rank n, whose bases are all n-element sets, Conjecture 3.19 is exactly Dinitz's problem, so this case holds by Galvin's theorem. Kahn's conjecture is open in general.

Matroid intersection and algorithms

Many results of Section 2 extend from bipartite graphs to the intersection of two matroids, since matchings of a bipartite graph are the common independent sets of two partition matroids. For example, Drisko's theorem (Theorem 2.35) holds for sets that are independent in two matroids M and N, by a result of Kotlar and Ziv (see [53]). Brualdi's conjecture also has a natural analogue in this setting.

Conjecture 3.20 (Aharoni and Berger, 2009 [2]). *Let M and N be matroids on the same ground set, and let $F_1, \dots, F_n$ be disjoint sets of size n, each independent in both M and N. Then there is a set of size $n - 1$, independent in both M and N, containing at most one element of each F_i.*

For two partition matroids, it says that n disjoint matchings of size n in a bipartite graph have a rainbow matching of size $n-1$, which contains Brualdi's conjecture. Latin squares without transversals show that $n-1$ cannot be raised to n in general. Aharoni, Kotlar and Ziv [11] found such a set of size $n-\sqrt{n}$, following the method of Woolbright and of Brouwer, de Vries and Wieringa. Bérczi, Király, Yamaguchi and Yokoi [74] observed that no counterexample to the full version, with size n, is known when the common bases are spanning arborescences. An *arborescence* is a directed spanning tree in which every vertex except a root has exactly one incoming edge. The arborescences of a digraph are the common bases of the graphic matroid of the underlying graph and the partition matroid that allows one incoming edge at each vertex. This led them to the rainbow arborescence conjecture, which we state and discuss in Section 6.4.

A single rainbow spanning tree can be found in polynomial time by matroid intersection (Section 3.1). Packing rainbow bases is much harder. Bérczi and Schwarcz showed that deciding whether a matroid can be partitioned into rainbow bases, for a given colouring of its elements, is intractable, and Hörsch, Kaiser and Kriesell [161] showed that it remains hard for graphic matroids. Bérczi, Csáji and Király [73] studied the complexity of packing rainbow spanning trees in more detail.

Two packing and covering questions for matroid intersection remain. For a matroid M let $\gamma(M)$ be the largest number of disjoint bases, and for two matroids on the same ground set let $\gamma(M_1, M_2)$ be the largest number of disjoint common bases. Every common basis is a basis of both, so $\gamma(M_1, M_2) \le \min\{\gamma(M_1), \gamma(M_2)\}$, but no lower bound in terms of $\gamma(M_1)$ and $\gamma(M_2)$ is known, even when the ground set is a union of bases of both matroids [73]. On the covering side, Aharoni and Berger conjectured that if the ground set can be covered by k independent sets of M_1 and by $\ell \ge k$ independent sets of M_2, then it can be covered by $\max\{k+1, \ell\}$ common independent sets. Hörsch, Kaiser and Kriesell [161] point out a consequence for rainbow bases. If the ground set of M is a union of k disjoint bases and is partitioned into classes of size at most $k-1$, then it should split into k rainbow bases, each meeting every class at most once. Already classes of size at most 2, with $k \ge 3$, are open.

3.7 Graph collections

A *graph collection* is a sequence $\mathbf{G} = (G_1, \dots, G_m)$ of graphs on a common vertex set; the graphs need not be distinct. A graph H with m edges is a *transversal* of $\mathbf{G}$ if its edges can be matched bijectively to $G_1, \dots, G_m$ so that each edge lies in its graph. This is a rainbow problem: give every edge of G_i the colour i, so that an edge lying in several graphs becomes several parallel edges of different colours. A transversal is then a rainbow subgraph using every colour (Figure 13).

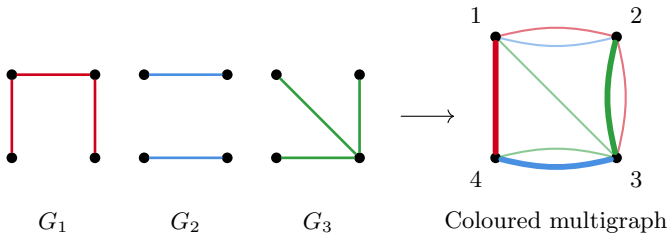


Figure 13: Left: a collection of three graphs on four vertices. Right: the same collection as an edge-coloured multigraph, in which G_i supplies the edges of colour i. The thick edges form a Hamilton path 1 4 3 2 with one edge from each graph, that is, a transversal.

For a single spanning tree, collections bring nothing new. Given $n-1$ graphs on a common set of n vertices, a transversal spanning tree is a rainbow spanning tree of the coloured multigraph. The matroid-intersection argument behind Theorem 3.1 applies to multigraphs without change, so one can decide in polynomial time whether a transversal spanning tree exists.

Prescribed trees are a different matter. If all G_i are equal to G, a transversal copy of a tree T is just a copy of T in G, so a degree condition on each G_i must at least force T in a single graph. Komlós, Sárközy and Szemerédi showed that minimum degree $(1/2+\alpha)n$ forces every n-vertex tree of maximum degree at most $cn/\log n$. Montgomery, Müyesser and Pehova [216] proved the transversal version.

Theorem 3.21 (Montgomery, Müyesser and Pehova [216]). *For each $\alpha > 0$ there are $c > 0$ and n_0 such that the following holds for all $n \geq n_0$. If each of $n-1$ graphs on a common set of n vertices has minimum degree at least $(1/2+\alpha)n$, then the collection contains a transversal copy of every n-vertex tree with maximum degree at most $cn/\log n$.*

Taking all graphs equal recovers the theorem of Komlós, Sárközy and Szemerédi, so neither condition can be weakened beyond the value of c. When the uncoloured degree threshold for a family of graphs also suffices for collections, the family is called *colour-blind* [149]. Theorem 3.21 says that bounded-degree spanning trees are colour-blind. Factors need not be (Section 5.5).

3.8 Random colourings

Now let the colours be random. Add the edges of K_n one by one in a uniformly random order, and give each edge a colour chosen independently and uniformly from a set of at least $n-1$ colours. Two conditions are necessary for a rainbow spanning tree: the graph must be connected, and at least $n-1$ colours must have appeared. Frieze and McKay [126] showed that nothing more is needed.

Theorem 3.22 (Frieze and McKay [126]). *In the randomly coloured random graph process with at least $n-1$ colours, with high probability the first graph that contains a rainbow spanning tree is the first graph that is connected and uses at least $n-1$ colours.*

With exactly $n-1$ colours, the colours are the bottleneck. The graph becomes connected after about $\frac{1}{2}n\log n$ edges, but by the coupon collector problem all $n-1$ colours appear only after about $n\log n$ edges. So in $G(n,p)$ coloured with $n-1$ colours, the threshold for a rainbow spanning tree is $2\log n/n$, twice the threshold for connectivity [93]. Bal, Bennett, Frieze and Prałat [41] let each edge choose $k \geq 2$ colours, of which a rainbow tree may use any one. The same hitting-time statement holds; for $k = 2$ both necessary conditions appear at about the same time. Similar results hold for the random k-out graph with $k \geq 2$ [42], and for random subgraphs of dense regular graphs [65]. In the latter, Bradshaw shows that the factor 2 in the edge probability $(2+\varepsilon)\log n/d$ cannot be relaxed in general.

A prescribed tree is harder to find. Krivelevich, Kwan and Sudakov showed that adding a random graph $G(n, C/n)$ to any n-vertex graph of minimum degree δn creates a copy of every bounded-degree spanning tree. Aigner-Horev, Hefetz and Lahiri [13] proved a rainbow version with edge probability $\omega(1)/n$ and $(1+\alpha)n$ colours. Katsamaktsis, Letzter and Sgueglia [171] removed both losses.

Theorem 3.23 (Katsamaktsis, Letzter and Sgueglia [171]). *For every $\delta \in (0,1)$ and integer $d \geq 2$ there is $C > 0$ such that the following holds. Let T be an n-vertex tree with maximum degree at most d, and let G_0 be an n-vertex graph with $\delta(G_0) \geq \delta n$. If $G_0 \cup G(n, C/n)$ is coloured uniformly with $n-1$ colours, then with high probability it contains a rainbow copy of T.*

Both the number of colours and, up to the constant C, the edge probability are optimal. With $(1+\varepsilon)e(H)$ colours instead of exactly $e(H)$, a coupling argument of McDiarmid turns many uncoloured results of this kind into rainbow ones; the exact palette needs different methods. The same paper shows that $G(n, C/n)$ coloured with n colours contains a rainbow copy of any given bounded-degree tree on $(1-\varepsilon)n$ vertices. In $G(n,p)$ alone, rainbow copies of prescribed spanning trees with exactly $n-1$ colours follow from the rainbow threshold theorem of Section 5.8, up to a logarithmic factor in p. Katsamaktsis, Letzter and Sgueglia ask for a universal version of Theorem 3.23: does the coloured randomly perturbed graph contain rainbow copies of all bounded-degree spanning trees at once [171]?

Sparser random graphs have no spanning tree at all, and the question becomes how large a rainbow tree is. Let $G(n,p)$ be coloured uniformly with $c = \Theta(n)$ colours. Cooley, Do, Erde and Missethan [93] showed that just above the phase transition, when $p = (1+\varepsilon)/n$ with $\varepsilon \to 0$ and $\varepsilon^3 n \to \infty$, the largest rainbow tree has order $(2+o(1))\varepsilon n$. This is the order of the giant component, so the giant contains an almost spanning rainbow tree. For fixed small ε and $c = \alpha n$, they conjectured that the largest rainbow tree has order $(1 \pm O(\varepsilon))2\varepsilon n$, again matching the largest tree. Bell and Frieze [49] proved this up to a logarithmic factor in the error term: the largest rainbow tree has order $(1 \pm O(\varepsilon\log(1/\varepsilon)))2\varepsilon n$.

Further above the phase transition less is known. For fixed $\alpha > 0$ and $d > 1$, Cooley, Do, Erde and Missethan ask for the order of the largest rainbow tree in $G(n, d/n)$ coloured with αn colours. They ask

whether, for small α, it is asymptotically the number of colours appearing in the giant component and, for large α, the order of the giant itself. They also ask whether the order of the largest rainbow tree is concentrated around its mean [93]. Bradshaw [65] suggests a matroid version of these questions: when does a random subset of the ground set of a matroid, with random colours, contain a rainbow basis? His edge-replacement method for forests has a natural analogue in matroids.

4 Rainbow paths and cycles

For matchings and spanning trees, the main questions in properly coloured complete graphs were about decompositions. For paths and cycles, even the existence of one long rainbow path is hard, because an algebraic obstruction appears already for Hamilton paths. This section starts with long rainbow paths in complete graphs and their connection to orderings of groups, and then turns to short rainbow cycles, to the number of edges that forces a rainbow cycle, and to rainbow Hamilton cycles in dense graphs.

4.1 Long rainbow paths in properly coloured complete graphs

Hahn's question and the Maamoun–Meyniel colouring

A properly coloured K_n uses at least $n-1$ colours, which is exactly the number of edges of a Hamilton path. Hahn asked whether a Hamilton path can always be made rainbow. His conjecture was in fact more general: if $n = 2k \geq 6$ and the edges of K_n are coloured so that each colour is used at most k times, then there should be a rainbow Hamilton path. Maamoun and Meyniel [210] showed in 1984 that this fails whenever n is a power of 2, already for proper colourings. Identify the vertices of K_{2^k} with $\mathbb{Z}_2^k$ and give the edge xy the colour $x+y$. This is an optimal colouring, and its colours are the 2^k-1 non-zero elements of $\mathbb{Z}_2^k$. If $v_1, v_2, \ldots, v_{2^k}$ were a rainbow Hamilton path, its edges would use every non-zero element once, so

$$v_1 + v_{2^k} \;=\; \sum_{i=1}^{2^k-1} (v_i + v_{i+1}) \;=\; \sum_{x \in \mathbb{Z}_2^k \setminus \{0\}} x \;=\; 0$$

for $k \geq 2$, and so $v_1 = v_{2^k}$, which is impossible (Figure 14). This is the same zero-sum argument that rules out transversals in the addition table of $\mathbb{Z}_{2m}$ (Section 2.3).

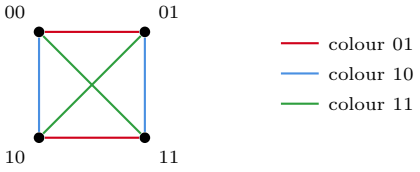


Figure 14: The colouring of K_4 by $\mathbb{Z}_2^2$, where xy has colour $x+y$. A Hamilton path has three edges, so a rainbow one would use each colour once, and the sum of its colours would be $01+10+11 = 00$. But this sum equals the sum of the two endpoints, which are distinct. Hence there is no rainbow Hamilton path. The same holds for K_{2^k} for every $k \geq 2$.

Andersen's conjecture

After this counterexample, the natural question is how close to Hamilton a rainbow path can be. Andersen [36] proved in 1989 that every properly coloured K_n has a Hamilton cycle with at least $n - \sqrt{2n}$ distinct colours, and conjectured that one vertex less than a Hamilton path is always possible.

Conjecture 4.1 (Andersen, 1989 [36])**.** *Every properly coloured K_n contains a rainbow path with $n-1$ vertices.*

By the Maamoun–Meyniel colouring this would be best possible. Progress came in two directions: long rainbow paths and cycles, and Hamilton cycles with many colours. Gyárfás and Mhalla found rainbow

paths on $(2n+1)/3$ vertices, Gebauer and Mousset [131] and Chen and Li [84] on $(3/4-o(1))n$ vertices, and Alon, Pokrovskiy and Sudakov [28] found a rainbow cycle on $n - O(n^{3/4})$ vertices. This was the first asymptotic version of Conjecture 4.1. Their main tool was the observation that the edges of a random set of colours in a properly coloured K_n behave much like a random graph. Balogh and Molla [44] improved the error to $O(\sqrt{n}\log n)$, and also found a Hamilton cycle with $n - O(\log^2 n)$ colours, improving Andersen's original bound. For random colourings, Gould, Kelly, Kühn and Osthus [143] showed that almost every optimal colouring of K_n has a rainbow Hamilton path and a rainbow cycle using all the colours. Recently, the conjecture was proved for all large n.

Theorem 4.2 (Bowtell, Montgomery, Müyesser and Pokrovskiy [63])**.** *For every sufficiently large n, every properly coloured K_n contains a rainbow path with $n-1$ vertices.*

The same paper proves a directed version, which settles a conjecture of Gyárfás and Sárközy on Latin squares for large n (Section 6.1). It also gives a rainbow Hamilton path whenever the colouring uses at least n colours, so Hahn's conjecture holds for large odd n, and a Hamilton cycle with at least $n-2$ colours, as Akbari, Etesami, Mahini and Mahmoody [17] had conjectured.

For cycles the picture is less complete. Bowtell, Montgomery, Müyesser and Pokrovskiy [63] find a rainbow cycle through all but a constant number C of vertices, but they do not determine C. Akbari, Etesami, Mahini and Mahmoody asked for the exact value.

Problem 4.3 (Akbari, Etesami, Mahini and Mahmoody, 2007 [17])**.** Does every 1-factorisation of K_n contain a rainbow cycle through all but two vertices?

The problem is the cycle analogue of Andersen's conjecture: removing an edge from such a cycle leaves a rainbow path on $n-2$ vertices, one fewer than Theorem 4.2 provides. Gould, Kelly, Kühn and Osthus [143] showed that almost all 1-factorisations of K_n have a rainbow Hamilton path, so colourings like that of Maamoun and Meyniel are rare. A different extension replaces the path by a tree: Montgomery, Pokrovskiy and Sudakov conjectured that every tree on $n-C$ vertices has a rainbow copy in every proper colouring of K_n, for some constant C [239].

Globally bounded colourings

Hahn's conjecture also leads in a second direction, where the colouring need not be proper but each colour may be used only a few times. Here the question is how strong the global bound must be to guarantee a rainbow Hamilton cycle. Hahn and Thomassen proved that a bound of $O(n^{1/3})$ suffices, Albert, Frieze and Reed improved this to $n/64$, and Böttcher, Kohayakawa and Procacci to $n/38$ (see [239]). Hahn's original bound is not enough. Pokrovskiy and Sudakov [240] adapted their counterexample to Stein's conjecture (Section 2.9) to colourings of K_n with fewer than $n/2$ edges of each colour in which every rainbow path misses at least $(\ln n)/42$ vertices.

So the threshold for a rainbow Hamilton cycle lies between $n/38$ and $n/2$, and two questions remain. The first is where it lies. The counterexamples need colour classes of size close to $n/2$, and Pokrovskiy [239] suggests that the threshold is asymptotically $n/2$: for every fixed $\varepsilon > 0$, a globally $(1/2-\varepsilon)n$-bounded colouring of a large K_n might always contain a rainbow Hamilton cycle. This would move the constant from $1/38$ to the largest value that the counterexamples allow. The second question is what remains true at the bound $n/2$ itself. Pokrovskiy suggests that a rainbow cycle of length $n - o(n)$ still exists. Its bipartite analogue has been answered: for equi-n-squares, partial transversals with $n - n^{1-\Omega(1)}$ cells always exist, and a deficit of order $\sqrt{n}$ can be forced (Theorem 2.29). For cycles in K_n no such bound is known.

Schrijver's problem

Andersen's conjecture concerns complete graphs, but the same counting applies to every regular graph. A properly d-coloured d-regular graph has d colours, so a rainbow path has at most d edges. Schrijver asked whether one can always find a rainbow path with $d-1$ edges, and verified this for $d \le 10$. For $d = n-1$ and K_n, this is Andersen's conjecture.

Theorem 4.4 (Bucić, Frederickson, Müyesser, Pokrovskiy and Yepremyan [71])**.** *Every properly coloured d-regular graph contains a rainbow path with $(1-o(1))d$ edges.*

Previously, the best general bound was about $2d/3$, and Babu, Chandran and Rajendraprasad had found rainbow paths with $d - o(d)$ edges in graphs without 4-cycles. Theorem 4.4 works for host graphs that may be very sparse, where the random-subgraph method of Alon, Pokrovskiy and Sudakov is not available.

Regularity is essential to Theorem 4.4. If one only asks that every vertex sees many colours, much less is known. Here a natural target is again one less than the trivial bound.

Conjecture 4.5 (Chen and Li, 2005 [83])**.** *Every edge-coloured graph in which every vertex is incident with edges of at least d colours contains a rainbow path with $d - 1$ edges.*

Rainbow paths with $2d/3 + 1$ edges are known in general, and with $d - 1$ edges for $d \leq 7$. For proper colourings of graphs with minimum degree d, Johnston, Palmer and Sarkar [165] found rainbow paths with $2d/3$ edges. Bucić, Frederickson, Müyesser, Pokrovskiy and Yepremyan [71] point out that even $d - o(d)$ edges, in either setting, would be new; their methods need the host to be close to regular.

The main motivation for Theorem 4.4 was a problem about groups, which we now describe.

4.2 Sequencings and Graham's rearrangement conjecture

Round dances and carry-over effects

In his *Récréations mathématiques*, Lucas described the following problem, crediting its solution to Walecki. Children dance in a ring, holding hands. How should the rounds be arranged so that every child holds hands with every other child exactly once? A round with $2n + 1$ children is a Hamilton cycle on $2n + 1$ vertices, so the question asks for a decomposition of K_{2n+1} into n Hamilton cycles. Walecki's solution places one child in the centre and rotates a zigzag path through the others (Figure 15).

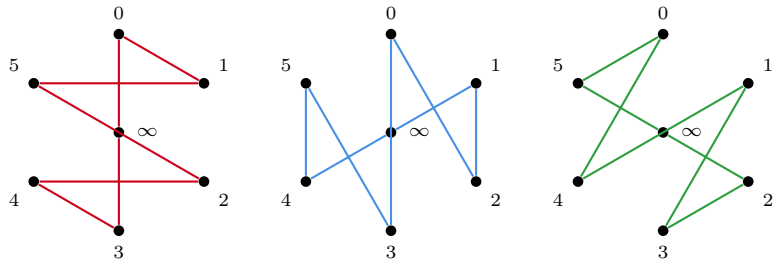


Figure 15: Walecki's decomposition of K_7 into three Hamilton cycles, $\infty, 0, 1, 5, 2, 4, 3$ and its two rotations. Each cycle is one round of the dance for seven children.

A related problem arose in the design of experiments. Williams [268] considered experiments in which each subject receives several treatments in succession, for example different diets given to cows in successive periods, where each treatment may leave a residual effect on the next one. To balance these effects, every ordered pair of distinct treatments should occur exactly once as consecutive treatments of some subject. A Latin square is *row-complete* if every ordered pair (x, y) with $x \neq y$ appears exactly once in adjacent cells of a row. Reading each row as a directed path, a row-complete Latin square of order n is the same as a decomposition of the complete directed graph on n vertices into n Hamilton paths.

Gordon's question

Gordon [141] found row-complete Latin squares from groups. A *sequencing* of a group G of order n is an ordering $b_1, \ldots, b_n$ of its elements such that the partial products $a_i = b_1 b_2 \cdots b_i$ are all distinct. Gordon showed that then the square with entries $a_i^{-1} a_j$ is row-complete and column-complete, and asked which groups have a sequencing. For abelian groups he showed that this happens exactly when the group has a unique element of order 2. For non-abelian groups the question is open in general. Keedwell conjectured that the dihedral groups of orders 6 and 8 and the quaternion group are the only non-abelian groups without a sequencing, and this has been verified for all orders up to 255 by computer [228].

Ringel considered a variant in his solution of the Heawood map colouring problem. An *R-sequencing* is an ordering of the elements with $b_1 \cdots b_n$ equal to the identity and the first $n-1$ partial products distinct. An R-sequencing gives a complete mapping, and Friedlander, Gordon and Miller conjectured that an abelian group has an R-sequencing exactly when it satisfies the Hall–Paige condition of Section 2.4. This was proved by Alspach, Kreher and Pastine [228].

Graham's conjecture

Graham [144] asked the same question for subsets of cyclic groups of prime order, in a 1971 paper on sums of integers taken from a fixed sequence. Later papers of Graham suggest that the problem was also motivated by juggling patterns [71].

Conjecture 4.6 (Graham, 1971 [144])**.** *Let p be a prime and $S \subseteq \mathbb{Z}_p \setminus \{0\}$. Then the elements of S can be ordered as $s_1, \ldots, s_k$ so that the partial sums s_1, $s_1 + s_2$, $\ldots$, $s_1 + \cdots + s_k$ are distinct.*

The same question was raised independently in design theory. Archdeacon introduced Heffter arrays to construct cycle decompositions and surface embeddings, and Archdeacon, Dinitz, Mattern and Stinson proposed a version of Conjecture 4.6 for all cyclic groups. Alspach, studying cycle decompositions of complete graphs, conjectured that if the elements of S do not sum to 0, then the partial sums can be made distinct and non-zero [71].

The translation into rainbow paths is as follows. For a group G and $S \subseteq G \setminus \{e\}$, the *Cayley digraph* has an edge from g to gs for each $g \in G$ and $s \in S$. Colour this edge with s. The colouring is proper in the directed sense: each vertex has one outgoing and one incoming edge of each colour. If $s_1, \ldots, s_k$ is an ordering with distinct partial products, then the partial products s_1, $s_1 s_2$, $\ldots$, $s_1 \cdots s_k$ form a directed path whose edges have the distinct colours $s_2, \ldots, s_k$. Conversely, a rainbow directed path with $k-1$ edges misses exactly one colour $c \in S$, and translating it to start at c gives an ordering with distinct partial products. So Graham's question asks for a rainbow directed path using all but one colour in the Cayley digraph of $\mathbb{Z}_p$ (Figure 16).

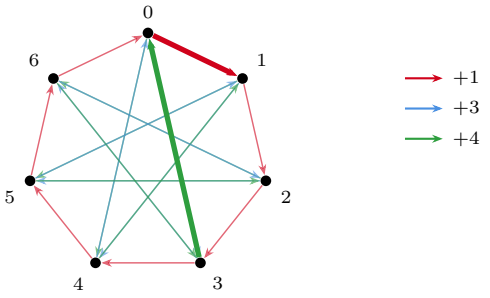


Figure 16: The Cayley digraph of $\mathbb{Z}_7$ with $S = \{1, 3, 4\}$, edges coloured by the generator. The ordering $3, 4, 1$ has partial sums $3, 0, 1$, which are distinct, and gives the rainbow path $3 \to 0 \to 1$ (bold) with colours 4 and 1. The ordering $1, 3, 4$ has partial sums $1, 4, 1$ and fails. Four of the six orderings work.

Progress on Conjecture 4.6 came separately for small and for large sets. For small sets, the polynomial method gives $|S| \leq 12$, and rectification arguments of Sawin and of Kravitz [184] give $|S| \leq \log p / \log\log p$. Bedert and Kravitz [48] went beyond this barrier, to $|S| \leq e^{c(\log p)^{1/4}}$. For large sets, the random Hall–Paige theorem of Müyesser and Pokrovskiy [224] gives the conjecture when $|S| \geq (1 - o(1))p$. The rainbow-path translation brought progress for arbitrary groups. Bucić, Frederickson, Müyesser, Pokrovskiy and Yepremyan [71] showed that every subset of every group has an ordering in which all but $o(|S|)$ partial products are distinct, using a directed version of Theorem 4.4. Bedert, Bucić, Kravitz, Montgomery and Müyesser [47] showed that every subset of size at least $|G|^{1-c}$ of any group G has an ordering with distinct partial products, and that the same holds in $\mathbb{F}_2^n$ for all subsets of size at least a constant. The remaining middle range was covered by Pham and Sauermann.

Theorem 4.7 (Pham and Sauermann [234], with [48, 47])**.** *Graham's rearrangement conjecture holds for all sufficiently large primes p.*

Pham and Sauermann proved the conjecture for $C_\alpha \le |S| \le p^{1-\alpha}$, and together with the results for small and large sets this covers every S when p is large. The thresholds are not explicit, so the conjecture is still open for small primes.

Beyond cyclic groups of prime order, Alspach and Liversidge conjectured that every subset of every finite abelian group, not containing 0, has an ordering with distinct partial sums. For cyclic groups, Costa, Della Fiore and coauthors [98] proved this for several families of composite orders. Very recently, Kuang and Wang settled the conjecture for all large sets.

Theorem 4.8 (Kuang and Wang [187]). *There is an absolute constant n_0 such that, for every finite abelian group G, every set $S \subseteq G \setminus \{0\}$ with $|S| \ge n_0$ has an ordering with distinct partial sums.*

The key new ingredient is additive dimension: if S contains a dissociated subset of size of order $(\log|S|)^2/\log\log|S|$, then S has an ordering in which no proper interval of consecutive terms sums to 0. This is within a factor $O(\log|S|/\log\log|S|)$ of the smallest additive dimension that a set of this size can have. So for abelian groups, only sets of bounded size remain.

The directed Schrijver problem

For non-abelian groups the question is open, and the rainbow-path language suggests a purely graph-theoretic version. In the Cayley digraph of G with respect to S, every vertex has exactly one outgoing and one incoming edge of each of the $|S|$ colours. So it is a d-regular digraph with $d = |S|$, properly coloured with d colours, and an ordering of S with distinct partial products is a rainbow directed path with $d-1$ edges. Bucić, Frederickson, Müyesser, Pokrovskiy and Yepremyan asked whether such a path exists in every digraph of this kind, not only in Cayley digraphs.

Problem 4.9 (Directed Schrijver problem; Bucić, Frederickson, Müyesser, Pokrovskiy and Yepremyan, 2025 [71]). Let D be a d-regular digraph, in which every vertex has indegree and outdegree d, whose edges are coloured with d colours so that each vertex has one incoming and one outgoing edge of each colour. Does D contain a rainbow directed path with $d-1$ edges?

A positive answer would show that every subset not containing the identity of every finite group, abelian or not, has an ordering with distinct partial products. That would contain Graham's conjecture, Theorem 4.8 and the corresponding questions for non-abelian groups. It would also contain the undirected Schrijver problem, since replacing each edge of a properly coloured d-regular graph by two opposite directed edges of the same colour gives a digraph as in Problem 4.9. For complete symmetric digraphs, an asymptotic answer follows from work of Benzing, Pokrovskiy and Sudakov [51], and Bucić, Frederickson, Müyesser, Pokrovskiy and Yepremyan answered it asymptotically when d is linear in the number of vertices. Wilson removed the density condition.

Theorem 4.10 (Wilson [269]). *Every d-regular digraph, properly coloured with d colours as in Problem 4.9, contains a rainbow directed path with $d - o(d)$ edges.*

Applied to Cayley digraphs, Theorem 4.10 shows that every subset S of every group, not containing the identity, contains a subset of size $|S| - o(|S|)$ which has an ordering with distinct partial products. The exact form of Problem 4.9 is open, as is the undirected Schrijver problem.

Two directions remain open beyond the exact problem. The first is algebraic. For abelian groups, Theorem 4.8 leaves only sets of bounded size. For non-abelian groups, valid orderings are known for sets of size at least $|G|^{1-c}$ in every finite group [47], and for small sets in dihedral and related semidirect products [97]. In the dihedral group of order $2p$, Costa, Della Fiore, Feng and Liu point out that only an intermediate range of set sizes is missing [98].

Problem 4.11 (Bedert, Bucić, Kravitz, Montgomery and Müyesser, 2025 [47]). Does every subset of every finite group, not containing the identity, have an ordering with distinct partial products?

Bedert, Bucić, Kravitz, Montgomery and Müyesser consider an affirmative answer feasible. Problem 4.9 would give it.

The second direction is to weaken regularity in Theorem 4.10. Bucić et al. [71] asked whether every properly coloured digraph with minimum outdegree d contains a rainbow path with $d - o(d)$ edges. A

simple greedy argument gives $d/2$, and for general digraphs nothing better is known. Wilson's proof works when every vertex has outdegree at least d and indegree $O(d)$, and he gives an example showing that removing the bound on the indegrees requires new ideas [269]. For Eulerian digraphs, where in- and outdegrees agree, Bucić et al. also ask for the longest rainbow path in terms of the average outdegree.

4.3 Short rainbow cycles

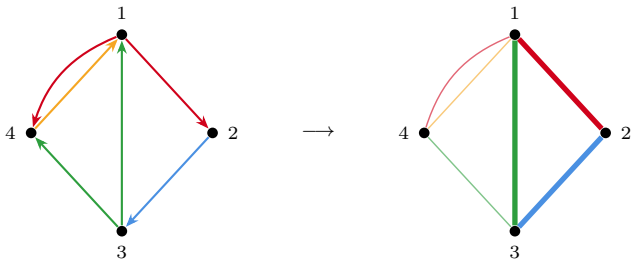
So far we have looked for long rainbow paths. The opposite question asks for a short rainbow cycle. Its motivation comes from directed graphs. Behzad, Chartrand and Wall conjectured in 1970 that a regular digraph with outdegree r has a directed cycle of length at most $\lceil n/r \rceil$, and Caccetta and Häggkvist extended this to all digraphs.

Conjecture 4.12 (Caccetta and Häggkvist, 1978)**.** *Every digraph on n vertices with minimum outdegree at least r contains a directed cycle of length at most $\lceil n/r \rceil$.*

Conjecture 4.12 is one of the best-known problems on digraphs, and it is open even for $r = n/3$, that is, for directed triangles. Shen proved that a cycle of length at most $\lceil n/r \rceil + 73$ exists. Aharoni proposed a rainbow generalisation.

Conjecture 4.13 (Aharoni, 2017)**.** *Let G be a simple graph on n vertices whose edges are coloured with n colours, each used at least r times. Then G contains a rainbow cycle of length at most $\lceil n/r \rceil$.*

To see that Conjecture 4.13 implies Conjecture 4.12, take a digraph D with minimum outdegree r. Replace each edge from u to v by an undirected edge uv of colour u (Figure 17). Every vertex gives a colour class of size at least r. In a rainbow cycle, each vertex u is the tail of at most one of its edges, and since there are as many vertices as edges on the cycle, every vertex is the tail of exactly one. So the edges are oriented consistently around the cycle, which is therefore a directed cycle of D.



a digraph, edges coloured by their tail the coloured graph; a rainbow triangle

Figure 17: The reduction from the Caccetta–Häggkvist conjecture to Aharoni's conjecture. Each edge $u \to v$ becomes an undirected edge of colour u (vertex 1 red, 2 blue, 3 green, 4 orange). The rainbow triangle $1, 2, 3$ comes from the directed triangle $1 \to 2 \to 3 \to 1$.

The case $r = 1$ of Conjecture 4.13 is easy. DeVos, Drescher, Funk, González Hermosillo de la Maza, Guo, Huynh, Mohar and Montejano [107] proved the case $r = 2$, and Aharoni, DeVos and Holzman [8] studied the case of rainbow triangles. Hompe and Huynh [154] proved that for every fixed r the conjecture holds up to an additive constant $\alpha_r = O(r^5 \log^2 r)$, in analogy with Shen's theorem. Clinch, Goerner, Huynh and Illingworth [91] survey the conjecture and its variants. If many colour classes are not stars, rainbow cycles become much shorter: Guo [148] showed, extending earlier work of Aharoni, Berger, Chudnovsky, Guo and Zerbib, that a small fraction of non-star colour classes already forces a rainbow cycle of length $O(\log n)$. In the reduction from digraphs every colour class is a star, so these results do not apply to the Caccetta–Häggkvist conjecture.

The smallest open case is $r = 3$. The Caccetta–Häggkvist conjecture is known for $r = 3$, and also for $r = 4$ and $r = 5$, but Aharoni's conjecture is not. For $r = 3$, the theorem of Hompe and Huynh gives a rainbow cycle of length at most $n/3 + \alpha_3$, improving the earlier bound $4n/9 + 7$ [91]; what is missing is the additive constant. DeVos et al. [107] proposed proving Conjecture 4.13 for $r \in \{3, 4, 5\}$ as a concrete target, where the uncoloured statement is already available.

The conjecture can also be stated for matroids. A graph on n vertices has a graphic matroid of rank $n-1$, and cycles are its circuits. DeVos et al. conjectured that the case $r = 2$ holds for all regular matroids. Thus, if the elements of a simple regular matroid of rank $n-1$ are coloured with n colours, each used at least twice, there should be a rainbow circuit of size at most $\lceil n/2 \rceil$. McGuinness [213] proved this in 2026. The same statement fails for binary matroids [91]. Whether regular matroids also satisfy the analogue of Conjecture 4.13 for every r is the natural next question; it would contain Aharoni's conjecture.

4.4 How many edges force a rainbow cycle?

A graph with n vertices and n edges has a cycle. In 2007, Keevash, Mubayi, Sudakov and Verstraëte [173] asked how many edges a properly coloured graph on n vertices must have to contain a rainbow cycle. The hypercube shows that a linear number is not enough. Colour each edge of Q_k by the coordinate in which its endpoints differ. Around any cycle, each coordinate changes an even number of times, so every colour appears an even number of times and no cycle is rainbow (Figure 18). Since Q_k has $n = 2^k$ vertices and average degree $k = \log_2 n$, the answer is at least of order $n \log n$.

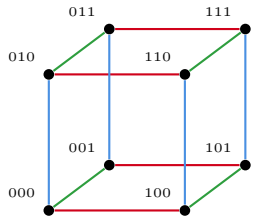


Figure 18: The cube Q_3 with each edge coloured by the coordinate in which its endpoints differ. This colouring is proper, and every cycle uses each colour an even number of times, so no cycle is rainbow.

The problem became important because of a connection to coding theory, which we now describe in more detail; see Sudakov's survey [254] for a fuller account.

Even covers and sparse linear equations

An *even cover* in a hypergraph is a non-empty set of edges that covers every vertex an even number of times. In a graph, a smallest even cover is a shortest cycle. For hypergraphs, the size of a smallest even cover therefore plays the role of the girth. Give each vertex v a variable x_v over $\mathbb{F}_2$, and give each edge E the equation $\sum_{v \in E} x_v = b_E$. The edges in an even cover correspond exactly to a non-empty collection of equations whose left-hand sides sum to zero. Thus a short even cover is a short linear dependency among the rows of the incidence matrix.

Sparse linear equations also define low-density parity-check codes, where each equation checks a small number of codeword positions. In a k-XOR instance, an even cover certifies inconsistency when the corresponding right-hand sides sum to 1. Feige's 2008 conjecture on short even covers was motivated by certificates of unsatisfiability for random 3-SAT formulas; the connection to SAT requires further conditions. The conjecture asks how many edges force a small even cover in a uniform hypergraph. Guruswami, Kothari and Manohar proved it up to logarithmic factors; Hsieh, Kothari and Mohanty then improved these factors. Hsieh, Kothari, Mohanty, Munhá Correia and Sudakov [157] gave a combinatorial proof and further improvements for odd uniformity.

The translation uses a properly coloured auxiliary graph. Let H be a k-uniform hypergraph with k even, and let $\ell \geq k/2$. The *Kikuchi graph* has as vertices the ℓ-element subsets of $V(H)$. Join S and T by an edge of colour $E \in E(H)$ when their symmetric difference is E (Figure 19). This is a proper colouring: for each S and E there is at most one possible neighbour $T = S \triangle E$. A cycle in this graph has a useful interpretation in H. The hyperedges whose colours occur an odd number of times form an even cover, provided this collection is non-empty. In particular, every rainbow cycle gives an even cover of the same size. A cycle with a colour occurring exactly once also gives an even cover of at most its length.

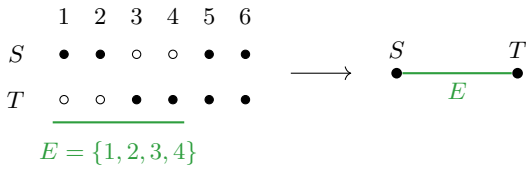


Figure 19: An edge of the Kikuchi graph with $k = 4$ and $\ell = 4$. A filled dot in a row means that the vertex belongs to that set. The sets $S = \{1, 2, 5, 6\}$ and $T = \{3, 4, 5, 6\}$ differ on $E = \{1, 2, 3, 4\}$. If E is an edge of H, the corresponding Kikuchi vertices are joined by an edge of colour E.

Locally decodable codes

A binary code $C : \{0,1\}^m \to \{0,1\}^n$ is *q-query locally decodable* if a randomised decoder can recover any chosen message bit by reading at most q codeword positions. The recovery probability must exceed $1/2$ by a fixed positive constant, even after a fixed positive fraction of the positions have been corrupted. Such codes arise in private information retrieval, where a user wants to retrieve a bit of a database without revealing which bit is requested. The basic question is how small the codeword length n can be in terms of m. For $q = 2$, the Hadamard code has $n = 2^m$, and exponential length is necessary. For $q = 3$, the known constructions have subexponential length, while the lower bounds are polynomial.

The connection to Kikuchi graphs led to an improvement over the earlier quadratic lower bounds.

Theorem 4.14 (Alrabiah, Guruswami, Kothari and Manohar [33])**.** *For fixed positive error tolerance and decoding advantage, every binary 3-query locally decodable code with m message bits has length $n = \widetilde{\Omega}(m^3)$.*

Here $\widetilde{\Omega}$ suppresses factors polynomial in $\log m$. The theorem applies to nonlinear codes as well. Hsieh, Kothari, Mohanty, Munhá Correia and Sudakov [157] obtained a combinatorial proof of the near-cubic bound for binary *linear* codes using coloured Kikuchi graphs.

A *locally correctable* code allows recovery of any codeword bit, rather than only a message bit. For binary linear codes this stronger requirement gives a much larger lower bound on the length. Alrabiah and Guruswami [32] proved that a binary linear 3-query locally correctable code of length n has dimension $O(\log^2 n \cdot \log\log n)$, for fixed positive error tolerance. Quadratic Reed–Muller codes have dimension of order $\log^2 n$, so this is nearly tight. Their argument uses rainbow cycles in properly coloured graphs that encode linear dependencies of the code.

Rainbow cycles

For rainbow cycles themselves, after several earlier bounds, Tomon proved that average degree $(\log n)^{2+o(1)}$ suffices, and Janzer and Sudakov and, independently, Kim, Lee, Liu and Tran improved this to $O(\log^2 n)$. The following theorem is tight up to the $o(1)$ term, by the hypercube.

Theorem 4.15 (Alon, Bucić, Sauermann, Zakharov and Zamir [24])**.** *Every properly coloured graph on n vertices with average degree at least $(\log n)^{1+o(1)}$ contains a rainbow cycle.*

Alon, Bucić, Sauermann, Zakharov and Zamir also observed a connection between this problem and questions in additive number theory, which allowed them to extend results on those questions from abelian to non-abelian groups. For the coding applications, what is needed is a short cycle in which at least one colour appears exactly once. This is weaker than being rainbow, and Sudakov's survey [254] gives a short proof of the bound on the number of edges that forces it.

The hypercube shows that average degree $\log_2 n$ is not enough, and Keevash, Mubayi, Sudakov and Verstraëte conjectured that this is the truth up to a constant factor.

Conjecture 4.16 (Keevash, Mubayi, Sudakov and Verstraëte, 2007 [173])**.** *There is a constant C such that every properly coloured graph on n vertices with at least $Cn\log n$ edges contains a rainbow cycle.*

The theorem of Alon, Bucić, Sauermann, Zakharov and Zamir gives average degree $C\log n\log\log n$, so only the factor $\log\log n$ is missing. They explain why their argument stops there. It splits the colours

at random into two halves and looks for rainbow walks in each half from a common vertex. There are properly coloured expanders, built from Hamming graphs, with average degree $\log n(\log\log n)^{1-o(1)}$ in which a random half of the colours leaves only small components. As a first step towards Conjecture 4.16, they ask whether the colours of every properly coloured robust sublinear expander with average degree $C\log n$ can be split into two classes so that both resulting subgraphs are spanning and connected [24].

On the coding side, the combinatorial proofs leave two gaps. Feige's conjecture asks for an even cover of size $O(\ell\log n)$ in every k-uniform hypergraph on n vertices with $Cn(n/\ell)^{k/2-1}$ edges, and the known results still lose logarithmic factors [157]. For 3-query locally decodable codes, the lower bound of Theorem 4.14 is nearly cubic, while the shortest known codes have subexponential length. Hsieh, Kothari, Mohanty, Munhá Correia and Sudakov call a super-cubic lower bound an outstanding open question. For linear locally correctable codes the lower bound of Alrabiah and Guruswami is already exponential, but their proof uses the extra structure that local correction provides.

4.5 Colour-degree conditions

In an arbitrary edge-coloured graph, the analogue of minimum degree is the minimum colour degree $\delta^c(G)$, the least number of colours at a vertex. Li [193], and independently Li, Ning, Xu and Zhang [192], proved a rainbow version of Mantel's theorem: every edge-coloured graph on n vertices with $\delta^c(G)\geq(n+1)/2$ contains a rainbow triangle. This is tight: a properly coloured complete bipartite graph with parts of size $n/2$ has colour degree $n/2$ and no triangle at all. Ning and Tian [226] recently determined the minimum number of rainbow triangles under this condition, answering a question of Li, Ning, Shi and Zhang. Colour-degree conditions for many vertex-disjoint rainbow triangles are discussed in Section 5.6. Without triangles, longer rainbow cycles appear under weaker conditions. Czygrinow and Parvatikar [103] showed that every edge-coloured triangle-free graph on n vertices with $\delta^c(G)\geq(n+7)/5$ contains a rainbow 4-cycle, which is best possible. Rainbow cycles of other fixed lengths, and their relation to cycles in oriented graphs, are discussed in Section 6.2.

4.6 Rainbow Hamilton cycles in dense graphs

Dirac's theorem says that every graph on $n\geq 3$ vertices with minimum degree at least $n/2$ has a Hamilton cycle. In a proper colouring, a rainbow Hamilton cycle needs n colours, and the examples of Section 4.1 show that a global bound on the colour classes is needed even in K_n. Coulson and Perarnau proved a rainbow Dirac theorem under such a bound.

Theorem 4.17 (Coulson and Perarnau [100])**.** *There is $\mu>0$ such that, for large n, every graph on n vertices with minimum degree at least $n/2$ contains a rainbow Hamilton cycle in every colouring in which each colour is used at most μn times.*

They also showed that the statement fails for $\mu>1/8$, and the optimal constant is not known. For near-rainbow Hamilton cycles the threshold $1/8$ is exact. Peng and Yan [233] showed that every properly coloured graph with minimum degree $(1/2+\varepsilon)n$ in which each colour is used at most $n/8$ times has a Hamilton cycle with $n-o(n)$ colours, and that $1/8$ cannot be increased. Without a global bound, Cheng, Huang, Liu, Wang and Yan [85] showed that a properly coloured graph with $\delta(G)\geq cn$, for $1/2<c\leq 1$, has a Hamilton cycle with $cn-O(n^{1/2})$ colours, which is best possible up to the error term. For $c=1$ this recovers Andersen's bound. Ding, Hu and Jiang [109] found the exact degree condition for rainbow Hamilton cycles in strongly edge-coloured graphs, in which every colour class is an induced matching: minimum degree $(n+1)/2$ suffices for large n.

The constant $1/8$ appears twice above: it is the limit of Theorem 4.17 for arbitrary colourings, and the exact threshold for near-rainbow Hamilton cycles in proper colourings. Peng and Yan conjectured that for proper colourings it is also the exact threshold for rainbow Hamilton cycles.

Conjecture 4.18 (Peng and Yan, 2024 [233])**.** *Every proper colouring of a graph on n vertices with minimum degree at least $n/2$, in which each colour is used at most $n/8$ times, contains a rainbow Hamilton cycle.*

In Theorem 4.17 the constant μ is far below $1/8$, so the conjecture asks for the sharp constant, and for proper colourings only. Without a global bound, Cheng, Huang, Liu, Wang and Yan [85] believe that

their bound $cn - O(n^{1/2})$ on the number of colours already holds at $c = 1/2$, that is, for Dirac graphs themselves, as Peng and Yan had also conjectured. Their theorem gives one colourful Hamilton cycle. Graphs with minimum degree $(1/2 + \varepsilon)n$ contain $n/8$ edge-disjoint Hamilton cycles. Motivated by this, they conjecture that for $c > 1/2$ every proper colouring contains at least $n/8$ Hamilton cycles, each using $cn - o(n)$ colours. They also propose a colourful version of the Komlós–Sárközy–Szemerédi theorem for bounded-degree spanning trees.

Rainbow versions of classical partition and decomposition problems have also been studied. Lehel conjectured that the vertices of every 2-edge-coloured complete graph can be partitioned into two monochromatic cycles, and Bessy and Thomassé proved this in 2010. Araújo, Liu, Martins, Mendonça, Moreira, Santos and Yan [37] proved a rainbow analogue: for large n, the vertices of every properly coloured K_n can be partitioned into two rainbow cycles. Liu, Xu and Yang [198] determined asymptotically how many rainbow paths are needed to cover the edges of a properly coloured dense graph. In the other direction, one may prescribe a few edges and ask for a Hamilton decomposition that separates them. Wu conjectured in 1999 that for every subgraph H of K_{2n+1} with n edges, the complete graph K_{2n+1} has a decomposition into Hamilton cycles in which the edges of H lie in different cycles. If the edges of H are given different colours and the edges of K_{2n+1} outside H are uncoloured, the cycles must each receive at most one colour. Javadi and Miralaei [163] proved Wu's conjecture. They conjecture more generally that if the edges of H are coloured so that each colour class is a linear forest, then the colouring extends to all of K_{2n+1} with n colours, each forming a Hamilton cycle.

The rainbow version of Lehel's conjecture also leaves two questions. The two rainbow cycles may share colours, and this cannot always be avoided; Araújo et al. [37] ask how few colours they need to share. They also suggest partitions into rainbow powers of cycles. The r-th power of a cycle has r edges per vertex, and K_n can be properly coloured with n colours, so a rainbow r-th power of a cycle has at most about n/r vertices. A bounded number of vertex-disjoint rainbow r-th powers of cycles might still cover K_n. As a first step they ask for one such power on $n/r - o(n)$ vertices in every properly coloured K_n.

Rainbow Hamilton cycles in graph collections, where each edge must come from a different graph of a family, and in randomly coloured random graphs are discussed in Sections 5.5 and 5.8.

5 General rainbow spanning structures

In 1986 Hahn and Thomassen conjectured that there is a constant $\alpha > 0$ such that every colouring of K_n in which each colour is used at most αn times contains a rainbow Hamilton cycle. Albert, Frieze and Reed proved this (Section 4.1). Frieze and Krivelevich then conjectured in 2008 that the same holds for every spanning tree of bounded maximum degree (see [255]). Both questions are cases of a single one: which spanning graphs H have a rainbow copy in every colouring of a given host with small colour classes? This chapter is about that question and its variants.

The first restriction comes from counting. A rainbow copy of H needs $e(H)$ colours. A proper colouring of K_n may use only $n - 1$ colours, so properness alone cannot give a rainbow spanning graph with more than $n - 1$ edges. We therefore bound the sizes of the colour classes, or weaken what we ask of H.

The simplest spanning targets are factors. For a fixed graph F, an *F-factor* is a collection of vertex-disjoint copies of F covering all vertices. In a *rainbow F-factor* all edges have distinct colours. In a *tiling by rainbow copies of F* each copy is rainbow, but a colour may be used once in each of several copies. Figure 20 shows the difference for triangles.



(a) Rainbow triangles (b) Rainbow factor

Figure 20: Two triangle-factors. Left: each triangle is rainbow, but the colours recur in the other component. Right: all six colours are distinct. Edge labels identify the colours.

5.1 Proper colourings

Let $6 \mid n$ and colour K_n by a 1-factorisation. This proper colouring has $n-1$ colours, while a triangle-factor has n edges, so no triangle-factor is rainbow. Every triangle in it is rainbow, however, so the obstruction says nothing about tilings. Yuster [273] showed that for tilings properness is enough, under the minimum-degree condition that Alon and Yuster proved for uncoloured F-factors.

Theorem 5.1 (Yuster [273]). *For every fixed graph F with at least one edge there is a constant $C = C(F)$ with the following property. If $v(F)$ divides n and G is an n-vertex graph with*

$$\delta(G) \geq \left(1 - \frac{1}{\chi(F)}\right) n + C,$$

then every proper edge-colouring of G contains a spanning tiling by rainbow copies of F.

For triangles there is nothing to colour-check, since every triangle is rainbow in a proper colouring. For K_4 it is different: the two disjoint edges of a 4-cycle in K_4 may share a colour.

Can one ask for a rainbow factor if one more colour is available? Müyesser [223] arrived at this question from the cycle types of orthomorphisms. The Friedlander–Gordon–Tannenbaum conjecture (Section 2.4) asks for an orthomorphism of a group whose non-identity elements lie in cycles of length k. After proving it for large groups, Müyesser proposed a version of the case $k = 3$ for arbitrary proper colourings, just as Ryser's conjecture (Conjecture 2.9) is the version of the Hall–Paige theorem for arbitrary Latin squares.

Conjecture 5.2 (Müyesser, 2023 [223]). *There is an absolute constant C such that every complete graph K_n properly coloured with n colours contains a rainbow union of vertex-disjoint triangles covering all but at most C vertices.*

The Rödl nibble gives a rainbow triangle packing leaving at most $n^{1-\varepsilon}$ vertices uncovered, for some fixed $\varepsilon > 0$. Müyesser suggests a polylogarithmic number of uncovered vertices as the next step. For partial transversals of Latin squares the analogous progress is complete: from $n - O(\log n / \log\log n)$ cells (Theorem 2.20) to $n-1$ cells (Theorem 2.21). He knows no example ruling out $C = 2$.

5.2 Locally bounded colourings

From now on we bound the colour classes. A colouring is *globally k-bounded* if every colour is used at most k times, and *locally k-bounded* if every colour is used at most k times at each vertex. A rainbow copy of H in a host G needs $e(H)$ colours, so a globally bounded colouring of G can be forced to contain one only if the bound is at most $e(G)/e(H)$. Ehard, Glock and Joos [117] showed that a bound just below this suffices for every bounded-degree spanning H, if each colour also has bounded degree and the host is quasirandom.

For disjoint vertex sets S, T, write $e_G(S,T)$ for the number of edges between them. An n-vertex graph G is (ε, d)*-quasirandom* if every vertex has degree $(d \pm \varepsilon)n$ and

$$e_G(S,T) = (d \pm \varepsilon)|S||T|$$

whenever S, T are disjoint and $|S|, |T| \geq \varepsilon n$. The complete graph is $(\varepsilon, 1)$-quasirandom, and the random graph $G(n,d)$ is (ε, d)-quasirandom with high probability.

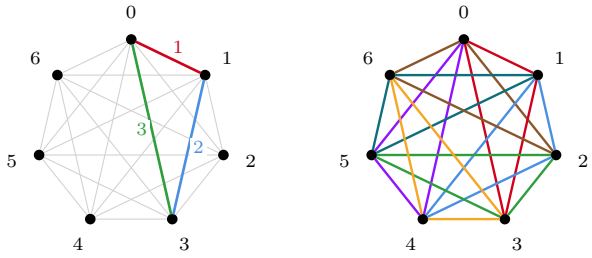
Theorem 5.3 (Ehard, Glock and Joos [117]). *For all $d \in (0,1]$, $\gamma \in (0,1)$ and integers $\Delta, \Lambda \geq 1$, there are $\varepsilon > 0$ and n_0 such that the following holds for $n \geq n_0$. Let G be an (ε, d)-quasirandom graph on n vertices and let H have n vertices, at least one edge and maximum degree at most Δ. Every locally Λ-bounded, globally $(1-\gamma)e(G)/e(H)$-bounded colouring of G contains a rainbow copy of H.*

In K_n, a triangle-factor may therefore have colour classes of size about $(1-\gamma)n/2$, and a Δ-regular spanning graph colour classes of size about $(1-\gamma)n/\Delta$. Up to the factor $1-\gamma$, these are the largest classes for which a rainbow copy can be forced. The proof uses a rainbow version of the blow-up lemma, extending one of Glock and Joos [136].

For triangle-factors the local bound need not be constant. Kim, Kühn, Kupavskii and Osthus [178] allow up to $\eta n / \log^4 n$ edges of a colour at each vertex; we state their theorem in Section 5.7.

Rainbow copies also give decompositions, as for Ringel's conjecture in Section 3.5. Let a group Γ act on the vertices of G, and colour each edge by its orbit under Γ. If every orbit has $|\Gamma|$ edges, then the images of a rainbow copy of H under Γ are edge-disjoint [117]. For K_{2n+1} and the rotations of $\mathbb{Z}_{2n+1}$, the orbits are the near-distance classes. A rainbow copy of a graph with n edges therefore gives a *cyclic* decomposition of K_{2n+1}, formed by the $2n+1$ rotations of one copy (Figure 21). Kotzig conjectured in 1973 that every tree with n edges has a cyclic decomposition of K_{2n+1} [117]. The proof of Theorem 3.14 finds a rainbow copy in the near-distance colouring, so it gives this for large n. The divisibility conditions behind Kotzig's conjecture do not use that T is a tree, and Ehard, Glock and Joos proposed the following.

Conjecture 5.4 (Ehard, Glock and Joos, 2019 [117])**.** *For every integer $\Delta \geq 1$ there is n_0 such that, for all $n \geq n_0$, every graph H with n edges and maximum degree at most Δ has a cyclic decomposition of K_{2n+1} into copies of H.*



(a) A rainbow triangle (b) Its seven rotations

Figure 21: (a) In K_7 on $\mathbb{Z}_7$, the triangle $0, 1, 3$ is rainbow in the near-distance colouring; its edges have near distances 1, 2 and 3. (b) Its seven rotations $\{i, i+1, i+3\}$, one colour each, partition the 21 edges of K_7; the copies form the Fano plane. A computer check confirms the partition.

Their evidence is an approximate version. For every Δ there is $\varepsilon > 0$ such that, for large n, every graph with at most n vertices, maximum degree at most Δ and at most $(1-\varepsilon)n/2$ edges has a cyclic packing in K_n. That is, the n rotations of one copy are edge-disjoint. This is Theorem 5.3 applied to the orbit colouring, which is locally 2-bounded and has colour classes of size n. In K_{2n+1} it handles graphs with $(1-\varepsilon)n$ edges, so Conjecture 5.4 asks for the last εn edges, with no slack left. For trees this last step needed the methods behind Theorem 3.14; for bounded-degree graphs with cycles it is open.

5.3 Golomb rulers and Sidon sets

Section 3.5 coloured the edge ab of the complete graph on $\{0, 1, \dots, N\}$ by $|a-b|$. A graceful labelling of a tree is a rainbow spanning tree in this colouring. Rainbow cliques in the same colouring are a classical object as well.

A *Golomb ruler* is a set of integer marks such that no two pairs of marks are the same distance apart. For example, the marks $0, 1, 4, 6$ measure each of the distances $1, \dots, 6$ exactly once. A set of marks in $\{0, \dots, N\}$ is a Golomb ruler exactly when it spans a rainbow clique in the difference colouring (Figure 22). The problem of the shortest ruler with k marks is therefore to find the least N for which this colouring contains a rainbow K_k.

In additive combinatorics these sets are called *Sidon sets*: a set of integers is Sidon if all sums $a+b$ with $a \leq b$ are distinct. Since $a+b = c+d$ is the same as $a-c = d-b$, a set is Sidon exactly when its positive differences are distinct. Erdős and Turán [119] showed that a Sidon subset of $\{0, \dots, N\}$ has at most $(1+o(1))\sqrt{N}$ elements, and Singer's construction [250] gives Sidon sets of this size. So the largest rainbow clique in the difference colouring of K_{N+1} has order $(1+o(1))\sqrt{N}$.

How does this compare with arbitrary colourings? The difference colouring is not proper, but it is locally 2-bounded: at the vertex a, the colour d appears only on the edges to $a-d$ and $a+d$. Babai [39] asked how large a rainbow clique every proper colouring of K_N contains. Alon, Lefmann and Rödl [27] showed that the answer is of order $(N \log N)^{1/3}$. Alon, Jiang, Miller and Pritikin [25] extended this to locally m-bounded colourings. They showed that every locally m-bounded colouring of K_N contains a

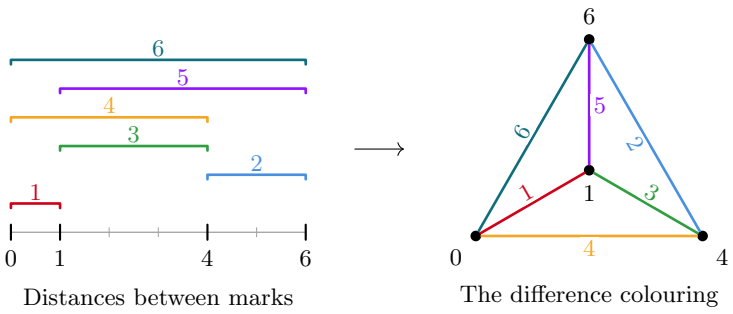


Figure 22: The marks $0, 1, 4, 6$ form a Golomb ruler. Left: the six distances, coloured and labelled by length. Right: the same six distances colour the edges of a rainbow K_4; vertex labels are the marks.

rainbow K_t if $N \geq Cmt^3/\log t$, and that this is tight up to the constant C. For fixed m the guaranteed clique again has order $(N \log N)^{1/3}$.

The difference colouring is therefore far from the worst case. Its additive structure gives rainbow cliques of order $\sqrt{N}$, much more than the $(N \log N)^{1/3}$ guaranteed by local boundedness alone.

5.4 Globally bounded colourings

We return to the Frieze–Krivelevich question. It asks for a rainbow H under a global bound alone, with no condition at the vertices. Sudakov and Volec [255] showed that the bound must be smaller than n/Δ^2 by a constant factor. For every even Δ and every n divisible by $\Delta/2 + 1$, they gave an n-vertex graph H of maximum degree Δ and a globally $16n/\Delta^2$-bounded colouring of K_n with no rainbow copy of H. Böttcher, Kohayakawa and Procacci [59] had already proved that this order is enough. Their theorem settled the Frieze–Krivelevich conjecture for all spanning graphs of bounded degree.

Theorem 5.5 (Böttcher, Kohayakawa and Procacci [59])**.** *Let H have n vertices and maximum degree at most Δ, where $\Delta \geq 1$. Every globally $n/(51\Delta^2)$-bounded colouring of K_n contains a rainbow copy of H.*

The bound is explicit in Δ, and Δ may grow with n. All edges of a colour may meet a single vertex. The proof applies the local lemma to a uniformly random bijection from $V(H)$ to $V(K_n)$.

Maximum degree is not the only parameter that works. A *cherry* is a path with two edges, and a graph of maximum degree Δ has at most $n\Delta^2/2$ of them. Sudakov and Volec [255] bounded the colour classes in terms of the number of cherries instead.

Theorem 5.6 (Sudakov and Volec [255])**.** *If H is an n-vertex graph with at most r cherries, then every globally $n/(1512\, r^{3/4})$-bounded colouring of K_n contains a rainbow copy of H.*

This allows H to have vertices of large degree, as long as there are few of them. When H has $O(n^{4/3})$ cherries, colour classes of bounded size suffice. The number $n^{4/3}$ cannot be increased: Sudakov and Volec constructed globally 9-bounded colourings of K_n and trees with $\Theta(n^{4/3})$ cherries that have no rainbow copy.

In a complete host any vertex can be used anywhere. If the host only has large minimum degree, some spanning graphs may not embed at all. Here one asks whether the uncoloured degree threshold still suffices once the colour classes are small. For a fixed graph F with at least one edge, let δ_F be the minimum-degree threshold for an F-factor. More precisely, δ_F is the infimum of the numbers a such that, for every $\varepsilon > 0$, minimum degree at least $(a + \varepsilon)n$ forces an F-factor for all sufficiently large n divisible by $v(F)$. Coulson, Keevash, Perarnau and Yepremyan [99] proved that δ_F also gives a rainbow F-factor.

Theorem 5.7 (Coulson, Keevash, Perarnau and Yepremyan [99])**.** *For every fixed graph F with at least one edge and every $\varepsilon > 0$, there are $\mu > 0$ and n_0 such that the following holds. If $n \geq n_0$, $v(F) \mid n$ and $\delta(G) \geq (\delta_F + \varepsilon)n$, then every globally μn-bounded colouring of G contains a rainbow F-factor.*

For K_r-factors this is the Hajnal–Szemerédi threshold $(1-1/r)n$. For other F the threshold δ_F can be below the Alon–Yuster bound of Theorem 5.1. The theorem is the graph case of a result on hypergraphs (Section 7).

Beyond factors, the bandwidth theorem of Böttcher, Schacht and Taraz embeds every spanning graph of bounded degree and small bandwidth at a degree threshold depending on its chromatic number. The *bandwidth* of H is the minimum, over orderings $\pi : V(H) \to [|V(H)|]$, of $\max_{uv\in E(H)} |\pi(u) - \pi(v)|$. For example, an F-factor has bandwidth at most $v(F) - 1$, and the r-th power of a Hamilton cycle has bandwidth at most $2r$. Glock and Joos [136] proved a rainbow bandwidth theorem.

Theorem 5.8 (Glock and Joos [136])**.** *For all $\varepsilon > 0$ and integers $\Delta \geq 1$, $r \geq 2$, there are $\beta, \mu > 0$ and n_0 such that the following holds for $n \geq n_0$. Let H be an r-colourable graph on n vertices with maximum degree at most Δ and bandwidth at most βn. If G has n vertices and minimum degree at least $(1-1/r+\varepsilon)n$, then every globally μn-bounded colouring of G contains a rainbow copy of H.*

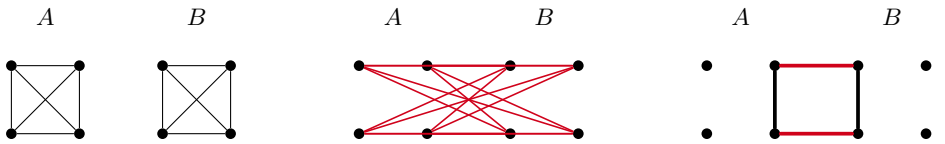
This covers connected spanning graphs such as powers of Hamilton cycles. For factors, Theorem 5.7 is stronger whenever $\delta_F < 1 - 1/\chi(F)$.

In both theorems a global bound μn leaves the uncoloured degree threshold unchanged. Coulson, Keevash, Perarnau and Yepremyan [99] suggested that this should hold asymptotically for every Dirac-type problem. Section 7.7 discusses this for hypergraphs, where tight Hamilton cycles show that a global bound alone is not always enough.

5.5 Graph collections

Graph collections and their translation into coloured multigraphs were introduced in Section 3.7. As noted there, a degree condition on each G_i can force a transversal copy of H only if it forces H in a single graph. Aharoni conjectured that for Hamilton cycles nothing more is needed. Joos and Kim [166] proved this: if $n \geq 3$ and each of n graphs on a common n-vertex set has minimum degree at least $n/2$, then the collection has a transversal Hamilton cycle. So Hamilton cycles are colour-blind, as are bounded-degree spanning trees (Theorem 3.21).

Factors need not be colour-blind. Montgomery, Müyesser and Pehova [216] observed the following obstruction. Let F have no bridge, that is, no edge whose removal disconnects its component. Split the vertex set into halves A and B. Let all graphs but one be the union of a clique on A and a clique on B, and let the last graph be the complete bipartite graph between A and B. A transversal F-factor must use exactly one edge between A and B. The copy of F containing that edge meets both A and B, so it must cross between them at least twice, since a single crossing edge would be a bridge. Each graph in this collection has minimum degree about $n/2$, but there is no transversal F-factor (Figure 23).



(a) $G_1 = \cdots = G_7$ (b) $G_8 = K_{A,B}$ (c) Two edges from G_8

Figure 23: The obstruction of Montgomery, Müyesser and Pehova for $F = C_4$ and $n = 8$. The eight graphs consist of seven copies of the graph in (a) and the graph in (b). In (c), a 4-cycle meeting both A and B uses two edges between them, and both must come from G_8. A computer check confirms that none of the 315 C_4-factors of K_8 is a transversal.

They showed that this is the only additional obstruction. Let $\delta_F^{\mathrm{T}} = \max\{\delta_F, 1/2\}$ if F has no bridge, and $\delta_F^{\mathrm{T}} = \delta_F$ otherwise.

Theorem 5.9 (Montgomery, Müyesser and Pehova [216])**.** *Let $\varepsilon > 0$ and let F be a graph with r vertices and t edges. For all sufficiently large n, every collection of tn graphs on a common set of rn vertices, each with minimum degree at least $(\delta_F^{\mathrm{T}} + \varepsilon)rn$, contains a transversal F-factor.*

Compare this with Theorem 5.7. There, small colour classes leave the threshold δ_F unchanged. Here a single graph of the collection may contain all edges between two halves, and for bridgeless F with $\delta_F < 1/2$ the threshold rises to $1/2$.

For connected spanning graphs, Gupta, Hamann, Müyesser, Parczyk and Sgueglia [149] gave a general sufficient condition for colour-blindness. One of its applications is a transversal version of the Pósa–Seymour conjecture, proved for large graphs by Komlós, Sárközy and Szemerédi.

Theorem 5.10 (Gupta, Hamann, Müyesser, Parczyk and Sgueglia [149])**.** *Let $r \geq 2$ and $\varepsilon > 0$. For all sufficiently large n, every collection of rn graphs on a common set of n vertices, each with minimum degree at least $(r/(r+1)+\varepsilon)n$, contains a transversal r-th power of a Hamilton cycle.*

The degree condition is the uncoloured threshold for the r-th power of a Hamilton cycle, up to the error εn, so this family is colour-blind. Their framework also covers Hamilton cycles in hypergraph collections (Section 7).

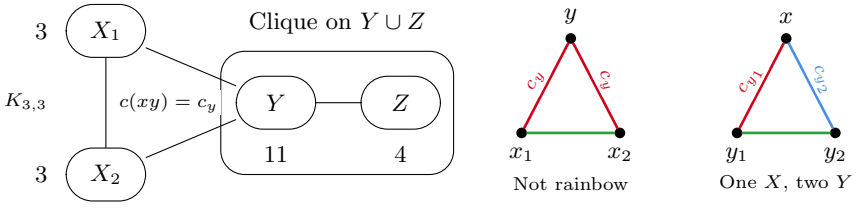
The error term εn is the remaining gap. For Hamilton cycles the theorem of Joos and Kim needs no error term, so this family is colour-blind exactly. Gupta, Hamann, Müyesser, Parczyk and Sgueglia ask whether minimum degree $rn/(r+1)$ already gives a transversal r-th power of a Hamilton cycle, as it gives the uncoloured power for large n by the theorem of Komlós, Sárközy and Szemerédi [149]. More generally, they suggest proving exact versions of their applications through extremal collections, in which essentially every graph is close to an extremal graph for the uncoloured problem. They do not expect a single general theorem here, since the extremal constructions differ from one structure to the next.

5.6 Colour-degree conditions

A different way to restrict a colouring is to require many colours at every vertex, while leaving the sizes of the colour classes unrestricted. The *colour degree* of a vertex is the number of colours on its edges, and $\delta^c(G)$ is the minimum colour degree. Even $\delta^c(K_n) = n-1$ does not give a rainbow triangle-factor, by the 1-factorisation example, so here we ask for tilings by rainbow triangles.

The Corrádi–Hajnal theorem gives a triangle-factor when $3 \mid n$ and $\delta(G) \geq 2n/3$. Its colour-degree analogue is false. Lo and Williams [201] constructed coloured graphs with minimum colour degree $5n/7-1$ and no tiling by rainbow triangles, for infinitely many n divisible by 3. By the rainbow Mantel theorem (Section 4.5), colour degree $(n+1)/2$ gives one rainbow triangle. Hu, Li and Yang proposed that $\delta^c(G) \geq (n+k)/2$ should give k vertex-disjoint rainbow triangles whenever $n \geq 3k$. The Lo–Williams construction disproves this for suitable $k \geq 5n/17$.

The construction is shown in Figure 24. Partition the vertices into X_1, X_2, Y, Z of sizes $3t, 3t, 12t-1, 3t+1$, and write $X = X_1 \cup X_2$. The graphs between X_1 and X_2 and within $Y \cup Z$ are complete, and all their edges receive distinct colours. Join X completely to Y, and give xy a new colour c_y that depends only on y. There are no edges between X and Z. A triangle with two vertices in X uses two edges of the same colour c_y, so every rainbow triangle meeting X has two vertices in Y. Covering the $6t$ vertices of X would need $12t$ vertices of Y, one more than there are. At $t = 1$, the graph has 21 vertices and minimum colour degree 14.



(a) The 21-vertex graph (b) Triangles meeting X

Figure 24: The Lo–Williams construction for $t = 1$. Left: the vertex sets and complete pairs; $Y \cup Z$ induces a clique. The edges in $G[X]$ and $G[Y \cup Z]$ all have distinct colours. Right: edges from X to a fixed y have colour c_y, excluding triangles with two vertices in X. A rainbow triangle through X needs two vertices of Y, but $|Y| = 11 < 2|X|$.

On the positive side, Lo and Williams proved that a larger colour degree suffices.

Theorem 5.11 (Lo and Williams [201]). *For every $\varepsilon > 0$ there is n_0 such that every edge-coloured graph G on $n \geq n_0$ vertices with $3 \mid n$ and*

$$\delta^c(G) \geq (5/6 + \varepsilon)n$$

has a spanning tiling by rainbow triangles.

The colour-degree threshold for a tiling by rainbow triangles thus lies between $5/7$ and $5/6$. Lo and Williams also gave sufficient conditions for tilings by rainbow K_r for every fixed $r \geq 4$. If only k disjoint rainbow triangles are wanted, Liu, Ning and Tian [197] showed that $\delta^c(G) \geq (n+k)/2$ suffices when $n \geq 42.5k + 48$. This is the Hu–Li–Yang bound in the range where it holds. Between the two ranges, for $17k/5 < n < 42.5k + 48$, it is not known whether colour degree $(n+k)/2$ gives k disjoint rainbow triangles, or what the right threshold is when it does not.

Lo and Williams [201] conjectured that their construction is optimal, and gave the corresponding bound for all clique sizes.

Conjecture 5.12 (Lo and Williams, 2024 [201]). *For every integer $r \geq 3$ and all sufficiently large n divisible by r, every edge-coloured graph G on n vertices with*

$$\delta^c(G) \geq \left(1 - \frac{r-1}{r^2-2}\right) n$$

contains a spanning tiling by rainbow copies of K_r.

For $r = 3$ the bound is $5n/7$, one more than the colour degree of the construction. Lo and Williams show that for $r = 3$ the absorbing step of their proof already works at colour degree $(2/3 + \gamma)n$. What is missing is a tiling by rainbow triangles covering almost all vertices under the conjectured condition. Their argument for this step gives $0.832n$, and they expect that new ideas are needed to go further.

5.7 Approximate decompositions

Once one rainbow factor exists, one can ask how many edge-disjoint rainbow factors there are. An exact decomposition of K_n into F-factors needs $v(F) \mid n$ and exactly

$$t = \frac{v(F)(n-1)}{2e(F)}$$

factors, so t must be an integer. If the factors are rainbow, each colour is used at most once in each of them, so every colour class has at most t edges.

An *α-decomposition* is a collection of edge-disjoint subgraphs covering at least $(1-\alpha)e(G)$ edges. Kim, Kühn, Kupavskii and Osthus [178] showed that a global bound just below t, together with a local bound that grows almost linearly, gives an α-decomposition into rainbow F-factors. The permitted local bound depends on the following parameter. For a graph F with at least one edge, put

$$a(F) = \max\Big\{\Delta(F),\ \max_{uv \in E(F)} (d_F(u) + d_F(v) - 2),\ \max_{uvw \text{ a path in } F} (d_F(u) + d_F(v) + d_F(w) - 4)\Big\},$$

where an empty maximum is taken to be 0.

Theorem 5.13 (Kim, Kühn, Kupavskii and Osthus [178]). *Fix a graph F with f vertices and $h \geq 1$ edges, and let $0 < \alpha < 1$. There are $\eta > 0$ and n_0 such that, whenever $n \geq n_0$ and $f \mid n$, every colouring of K_n which is globally $(1-\alpha)fn/(2h)$-bounded and locally $\eta n / \log^{2a(F)} n$-bounded has an α-decomposition into rainbow F-factors.*

For $F = K_3$ we have $a(F) = 2$. The theorem then gives at least $(1-\alpha)(n-1)/2$ edge-disjoint rainbow triangle-factors in any globally $(1-\alpha)n/2$-bounded, locally $\eta n / \log^4 n$-bounded colouring of K_n. Since the local bound tends to infinity, the theorem applies to every proper colouring of K_n whose colour classes have at most $(1-\alpha)n/2$ edges. Kim, Kühn, Kupavskii and Osthus prove the result more generally for

quasirandom hosts. The exact decomposition problems for matchings and trees (Sections 2 and 3) ask for no edges to be left over.

Kim, Kühn, Kupavskii and Osthus point to two directions beyond Theorem 5.13 [178]. The first is to decompose into other rainbow spanning structures, such as regular spanning graphs of bounded degree. The second is to improve the error term or to obtain exact decompositions. Among spanning structures, an exact rainbow decomposition is known for spanning trees in optimal colourings (Theorem 3.7).

5.8 Random colourings

So far the colourings have been adversarial. If the colours are random, the counting lower bound of $e(H)$ colours may be close to sufficient. In a random graph with uniformly random colours, how much does the rainbow condition change the threshold for a copy of H? Bell, Frieze and Marbach [50] posed this question in the framework of Frankston, Kahn, Narayanan and Park, which relates thresholds to expectation thresholds. Han and Yuan [152] answered it.

To state their result, let X be a finite set and $\mathcal{F}$ an increasing family of subsets of X. Let ℓ be the largest size of a minimal member of $\mathcal{F}$. Keep each element of X independently with probability p, and give each kept element a uniformly random colour from $[k]$, where $k \geq \ell$. The *k-colour rainbow threshold* is the value of p at which, with probability $1/2$, the kept set contains a minimal member of $\mathcal{F}$ whose elements have distinct colours. For $X = E(K_n)$ and $\mathcal{F}$ the graphs containing a copy of H, this is the threshold for a rainbow copy of H in $G(n,p)$ coloured with $k \geq e(H)$ colours.

Theorem 5.14 (Han and Yuan [152])**.** *There is a constant K such that, for every finite X, increasing family $\mathcal{F} \subseteq 2^X$ and integer $k \geq \ell$, the k-colour rainbow threshold of $\mathcal{F}$ is at most $K\, q_f(\mathcal{F}) \log \ell$, where $q_f(\mathcal{F})$ is the fractional expectation threshold.*

The fractional expectation threshold is a lower bound for the threshold, defined through fractional covers of $\mathcal{F}$ by up-sets; in particular it is at most the ordinary threshold. So even with exactly ℓ colours, the rainbow threshold is at most a factor $O(\log \ell)$ above the uncoloured threshold. In particular, for every graph H, the random graph $G(n,p)$ coloured with exactly $e(H)$ colours contains a rainbow copy of H once p exceeds the threshold for an uncoloured copy by a factor $O(\log e(H))$. Bell, Frieze and Marbach had proved this under an additional spread assumption. Han and Yuan also prove a rainbow version of Spiro's refinement, which gives rainbow thresholds for powers of Hamilton cycles in $G(n,p)$. Aigner-Horev, Hefetz, Person and Trushkin [14] used Theorem 5.14 to find rainbow perfect matchings, Hamilton cycles, bounded-degree spanning trees and clique factors with an exact palette in random subgraphs of pseudorandom graphs.

Theorem 5.14 loses a logarithm compared with the uncoloured theory. Park and Pham proved (see [152]) that the threshold is at most $K\, q(\mathcal{F}) \log \ell$, where $q(\mathcal{F}) \leq q_f(\mathcal{F})$ is the expectation threshold, while Theorem 5.14 gives only $K\, q(\mathcal{F}) \log^2 \ell$ in these terms. Han and Yuan [152] note two ways to remove the extra factor. One is a transversal version of their argument, which they could not prove. The other is a conjecture of Talagrand that $q(\mathcal{F}) \geq q_f(\mathcal{F})/K$ for an absolute constant K and every increasing $\mathcal{F}$. The second gap concerns structures for which the uncoloured logarithm can be reduced but not removed, such as K_r-factors. There, a coupling of Riordan reduces the uncoloured problem to perfect matchings in random r-uniform hypergraphs. The coupling acts on r-sets of vertices, not on coloured edges, and Han and Yuan do not see how to transfer it. So their method does not give the right rainbow threshold for K_r-factors.

6 Edge-coloured digraphs

A digraph may have two edges between a pair of vertices, but then their directions are opposite. We write xy for the edge from x to y. Unless stated otherwise, paths and cycles in digraphs are directed, that is, their edges are oriented consistently. A colouring of a digraph is *proper* if no two edges with a common initial vertex, and no two edges with a common final vertex, have the same colour. The *complete digraph* on n vertices has both edges xy and yx for every pair $x \neq y$. As in Section 2.1, a Latin square of order n is the same as a proper colouring of the complete digraph with a loop at each vertex, using n colours, and a transversal is a rainbow 1-factor (Figure 1d).

Directions add a second constraint to every question of the previous chapters. A rainbow path must now also follow the orientation. Both constraints are used in the translations below: a Latin square becomes a coloured complete digraph, a fair-division instance becomes a multipartite digraph, and an arborescence becomes a common basis of two matroids.

6.1 Long rainbow paths and cycles in complete digraphs

Hahn's question from Section 4.1 has a directed version: does every properly coloured complete digraph contain a rainbow Hamilton path? The Maamoun–Meyniel colouring of K_{2^k} answers it negatively. Replace each edge uv of colour f by the two edges uv and vu, both of colour f. The resulting proper colouring of the complete digraph has no rainbow Hamilton path, even if the directions are ignored [51].

For Latin squares the directed question is a question about transversals. Gyárfás and Sárközy conjectured in 2014 that every Latin square of order n has a partial transversal with $n-2$ cells which is *cycle-free*, that is, whose edges in the complete digraph with loops contain no cycle. Such a transversal is a rainbow union of at most two paths. Bowtell, Montgomery, Müyesser and Pokrovskiy proved a stronger statement.

Theorem 6.1 (Bowtell, Montgomery, Müyesser and Pokrovskiy [63])**.** *There is n_0 such that, for $n \geq n_0$, every proper colouring of the complete digraph with loops on n vertices using n colours contains a rainbow path on $n-1$ vertices.*

A loop cannot lie on a path, so the theorem concerns the complete digraph without loops, coloured from a Latin square. It is the directed counterpart of Theorem 4.2, and the cyclic Latin squares of even order show that a rainbow Hamilton cycle need not exist. For almost all Latin squares more is true. Gould and Kelly [142] showed that, as $n \to \infty$, almost all Latin squares of order n have a Hamilton transversal, that is, a rainbow Hamilton cycle in the complete digraph with loops.

The theorem of Bowtell, Montgomery, Müyesser and Pokrovskiy covers more than Latin squares. Their methods apply to every properly coloured digraph whose vertices have in- and outdegree $(1\pm\varepsilon)n$ and whose colours are each used $(1\pm\varepsilon)n$ times, for ε sufficiently small in terms of $1/\log n$. With at least n colours, such a digraph contains a rainbow Hamilton path and a rainbow cycle through all but 100 vertices [63, Theorem 4.1]. For an arbitrary proper colouring of the complete digraph, where colour classes may be small, the best result is weaker.

Theorem 6.2 (Benzing, Pokrovskiy and Sudakov [51])**.** *Every properly coloured complete digraph on n vertices contains a rainbow cycle of length $n - O(n^{4/5})$, and a rainbow union of vertex-disjoint paths with $n - O(n^{2/3})$ edges.*

The difference between the two theorems is in the colourings, not in the host. When each colour class is close to a 1-factor, the error is a constant; when colour classes can have any size up to n, only a polynomial error is known. In undirected complete graphs there is no such gap: every proper colouring of K_n contains a rainbow cycle through all but a constant number of vertices [63, Theorem 1.3]. Benzing, Pokrovskiy and Sudakov asked how far their error terms can be reduced. The natural target is a rainbow cycle of length $n - O(1)$ in every properly coloured complete digraph, which would make the result independent of the sizes of the colour classes.

Theorem 6.1 gives a rainbow path with $n-2$ edges. Gould and Kelly [142] conjectured that one more edge is always possible if a cycle is also allowed.

Conjecture 6.3 (Gould and Kelly, 2021 [142])**.** *Every proper colouring of the complete digraph with loops on n vertices contains a rainbow path or a rainbow cycle with at least $n-1$ edges.*

For a Latin square, such a path or cycle is a partial transversal with $n-1$ cells whose edges form a connected digraph. Gould and Kelly note that, for every n, the conjecture would imply the Ryser–Brualdi–Stein bound $n-1$, Andersen's conjecture and the Gyárfás–Sárközy conjecture. It allows any number of colours. Colour the loops of the doubled Maamoun–Meyniel colouring with new colours. There is still no rainbow Hamilton path, so here the conjecture asks for a rainbow cycle with at least $n-1$ edges.

Benzing, Pokrovskiy and Sudakov [51] also suggested looking for long rainbow paths and cycles in properly coloured tournaments, for example regular ones. Hamilton cycles in regular tournaments have been studied extensively, but their rainbow versions have not.

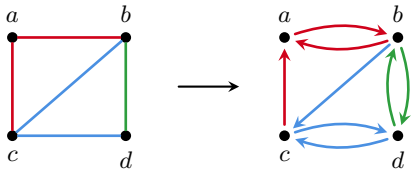
The d-regular analogue, where the host is any d-regular digraph with one incoming and one outgoing edge of each colour at every vertex, is the directed Schrijver problem of Section 4.2.

6.2 Colour degree and outdegree

The colour degree of an edge-coloured graph (Section 5.6) and the outdegree of a digraph are closely related. Czygrinow, Molla, Nagle and Oursler [102] describe two translations.

From a coloured graph to a digraph: at each vertex v, choose one edge of each colour at v and orient it away from v. The outdegree of v in the resulting digraph is exactly its colour degree (Figure 25). The digraph may contain both vw and wv.

From a digraph to a coloured graph: give the edge vw the colour w, that is, colour every edge by its final vertex. The edges leaving v then have distinct colours, so the colour degree of v is at least its outdegree. A cycle $v_1v_2\dots v_\ell v_1$ becomes a rainbow cycle, since its colours $v_2,\dots,v_\ell,v_1$ are distinct. Section 4.3 used the same idea, with colours given by initial vertices, to derive the Caccetta–Häggkvist conjecture from Aharoni's conjecture.



colour degrees $1,3,2,2$ outdegrees $1,3,2,2$

Figure 25: Left: an edge-coloured graph. Right: at each vertex one edge of each colour is chosen and oriented away from the vertex; the arrows keep the colour of their edge. The outdegree of every vertex equals its colour degree. The edges bd and db, for example, are chosen at b and at d respectively.

The second translation shows that a colour-degree condition forcing rainbow cycles of length ℓ would force cycles of length ℓ in oriented graphs, that is, in digraphs without 2-cycles. The thresholds in the two settings are close, but not equal. Li [193] proved that colour degree $(n+1)/2$ forces a rainbow triangle (Section 4.5). For longer even cycles the threshold falls to about $n/3$.

Theorem 6.4 (Czygrinow, Molla, Nagle and Oursler [102])**.** *Let $\ell \geq 4$ be fixed and let n be sufficiently large. If ℓ is even, every edge-coloured graph on n vertices with minimum colour degree at least $(n+5)/3$ contains a rainbow cycle of length ℓ. For every ℓ, every oriented graph on n vertices with minimum outdegree at least $(n+1)/3$ contains a cycle of length ℓ.*

Both bounds are best possible when ℓ is not divisible by 3. For oriented graphs, take a blow-up of a cycle of length 3, with parts V_0, V_1, V_2 and all edges from V_i to V_{i+1}. Every cycle winds around the parts, so its length is divisible by 3. The oriented result is the outdegree version of a theorem of Kelly, Kühn and Osthus, who proved the same bound for the minimum semidegree.

The first translation is the starting point for Lo and Williams [201] in their work on tilings by rainbow triangles (Section 5.6). They first pass to a minimal subgraph with the same colour degrees, in which every colour class is a union of vertex-disjoint stars, and orient each star towards its centre. In the resulting digraph the edges leaving a vertex have distinct colours, and the outdegree is close to the colour degree. This turns their problem into a tiling problem in a digraph.

The constructions behind Theorem 6.4 only exclude cycles whose length is not divisible by 3. For even ℓ divisible by 3, Czygrinow, Molla, Nagle and Oursler show that colour degree $(1/3-\alpha)n$ already suffices for some absolute $\alpha > 0$, so the threshold is lower. They do not determine its value, and their outdegree bound for oriented graphs is likewise sharp only when $3 \nmid \ell$. For $\ell = 3$ the oriented question is the triangle case of the Caccetta–Häggkvist conjecture (Section 4.3).

6.3 Rainbow cycles and envy-free allocations

Rainbow cycles in multipartite digraphs appear in fair division. Suppose that indivisible goods are to be split among n agents, each of whom values a set of goods by the sum of the values of its elements. An allocation is *envy-free up to any good*, or *EFX*, if no agent prefers another agent's bundle after any single good is removed from that bundle. Whether an EFX allocation of all goods always exists is a central open problem of discrete fair division. It is known for three agents, but open for four or more.

One relaxation allows some goods to stay unallocated and asks for an allocation in which every agent values its own bundle at least $1-\varepsilon$ times the value of any other bundle with one good removed. The aim is to leave few goods unallocated. Chaudhury, Garg, Mehlhorn, Mehta and Misra [82] reduced this to a problem about digraphs. For a positive integer d, the *rainbow cycle number* $R(d)$ is the largest k for which there is a k-partite digraph with parts $V_1, \dots, V_k$ of size at most d with two properties. Every vertex has an inneighbour in every other part, but no cycle contains at most one vertex from each part. Colour each edge by the part containing its final vertex. A cycle then uses at most one vertex of each part exactly when its colours are distinct, so the second property says that there is no rainbow cycle (Figure 26). An upper bound on $R(d)$ gives an allocation with fewer unallocated goods.

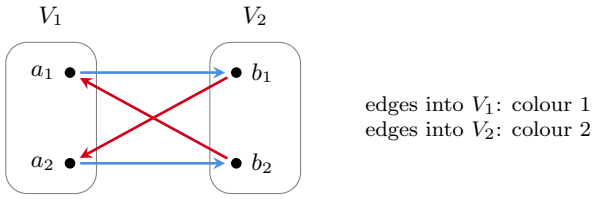


Figure 26: A 2-partite digraph in which every vertex has an inneighbour in the other part. Each edge is coloured by the part containing its final vertex. The only cycle, $a_1b_1a_2b_2a_1$, uses two vertices from each part and two edges of each colour, so it is not rainbow. Hence $R(2) \geq 2$.

Chaudhury, Garg, Mehlhorn, Mehta and Misra proved $d \leq R(d) \leq d^4 + d$. Berendsohn, Boyadzhiyska and Kozma [52] improved this to $d^{2+o(1)}$, and Akrami, Alon, Chaudhury, Garg, Mehlhorn and Mehta [18] and, independently, Chashm Jahan, Seddighin, Seyed-Javadi and Sharifi [162] to $O(d \log d)$. Sivashankar showed that $R(d)$ is linear.

Theorem 6.5 (Sivashankar [251])**.** *For every positive integer d, we have $R(d) < ed$.*

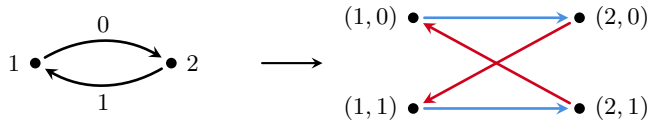
Through the reduction, every instance with n agents has an allocation that is $(1-\varepsilon)$-EFX in the above sense and leaves $O(\sqrt{n/\varepsilon})$ goods unallocated. This is the best guarantee that the reduction can give. The reduction runs in one direction only: full EFX asks for every good to be allocated with $\varepsilon = 0$, and a bound on $R(d)$ does not give that. The exact value of $R(d)$ remains a combinatorial question in its own right.

Problem 6.6 (Berendsohn, Boyadzhiyska and Kozma, 2022 [52])**.** Is $R(d) = d$ for every positive integer d?

This is known for $d \leq 3$ [52], and the case $d = 4$ was verified with a SAT solver (see [251]). Theorem 6.5 leaves a constant factor between d and ed.

The rainbow cycle number also contains a zero-sum problem. For a finite group Γ, let $n(\Gamma)$ be the least n such that every labelling of the edges of the complete digraph on n vertices by elements of Γ has a cycle whose labels multiply, in order, to the identity. Berendsohn, Boyadzhiyska and Kozma [52] observed that this is a special case of the rainbow cycle problem. Take one part $V_i = \{i\} \times \Gamma$ for each vertex i. If the edge ij has label a_{ij}, join (i, x) to (j, xa_{ij}) for every $x \in \Gamma$. Every vertex of V_j then has exactly one inneighbour in V_i. A rainbow cycle through (i_1, x) returns to x only if $x = xa_{i_1 i_2} \cdots a_{i_r i_1}$, so rainbow cycles in the lift are exactly lifts of zero-sum cycles (Figure 27). Hence $n(\Gamma) \leq R(|\Gamma|)+1$, and Theorem 6.5 gives $n(\Gamma) < e|\Gamma| + 1$ [251].

In the lift every pair of parts is joined by a permutation. Restricting to such digraphs defines the permutation rainbow cycle number $R_p(d) \leq R(d)$, for which Berendsohn, Boyadzhiyska and Kozma and, independently, Akrami et al. proved $R_p(d) \leq 2d - 2$. Chashm Jahan et al. improved this to $2d - 4$ for



(a) a $\mathbb{Z}_2$-labelling (b) its lift

Figure 27: (a) A labelling of the complete digraph on two vertices by $\mathbb{Z}_2$; its only cycle has label sum 1. (b) The lift: (i, x) is joined to $(j, x + a_{ij})$, and edges are coloured by the part of the final vertex. The lift is the digraph of Figure 26; its only cycle is not rainbow. A computer check confirms both statements.

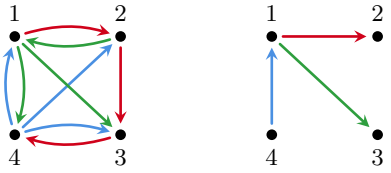
$d \geq 4$ [162]. The cyclic group gives $n(\mathbb{Z}_q) \geq q + 1$, so a positive answer to Problem 6.6 would give the sharp bound $n(\Gamma) \leq |\Gamma| + 1$ for every finite group. According to Sivashankar [251], Campbell et al. have announced this bound and proved it for groups of odd order. The converse fails: a bound on $n(\Gamma)$ says nothing about lifts whose maps between parts are not translations, so Problem 6.6 would remain open even after the zero-sum problem is settled.

6.4 Rainbow arborescences

An *arborescence*, or outbranching, is a spanning tree of a digraph in which every vertex except one, the root, has exactly one incoming edge. As explained in Section 3.6, the arborescences of a digraph are the common bases of two matroids. Aharoni, Kotlar and Ziv conjectured that n disjoint common bases of two matroids of rank n have a rainbow common independent set of size $n - 1$, and this cannot be raised to n in general. Bérczi, Király, Yamaguchi and Yokoi [74] observed that no counterexample of size n is known for arborescences, and conjectured that none exists.

Conjecture 6.7 (Rainbow arborescence conjecture; Bérczi, Király, Yamaguchi and Yokoi, 2024 [74])**.** *If the edges of a digraph on n vertices are coloured with $n - 1$ colours so that each colour class is a spanning arborescence, then there is a spanning arborescence with exactly one edge of each colour.*

Figure 28 shows an example on four vertices. The colour classes may have different roots, and the conjecture does not prescribe the root of the rainbow arborescence. With a prescribed root, deciding whether a rainbow arborescence exists is NP-complete, even when the colour classes have only two distinct roots [74].



three arborescences a rainbow arborescence

Figure 28: Left: three spanning arborescences on four vertices, rooted at 1 (red path), 4 (blue star) and 2 (green). Right: a rainbow arborescence rooted at 4, with one edge of each colour. This example has six rainbow arborescences in total.

Bérczi, Király, Yamaguchi and Yokoi proved the conjecture in several cases. The hardest is when the underlying graph is a cycle [75]. The conjecture also holds when each colour class is a path or a star, when at most two vertices are roots of more than one colour class, and when all colour classes are orientations of one tree. A computer search confirms it for $n \leq 8$. In the other direction, they relaxed the conclusion. Given $2n - 4$ colour classes instead of $n - 1$, there is a rainbow spanning arborescence. From $n - 1$ colour classes one can always find a rainbow arborescence, not necessarily spanning, with $\lfloor n/2 \rfloor$ edges. Closing either gap, to $n - 1$ colour classes or to $n - 1$ edges, is the conjecture itself.

Bérczi, Király, Yamaguchi and Yokoi [74] suggest two further steps. The first strengthens the $\lfloor n/2 \rfloor$ bound in the spirit of a recent theorem of Berger and McGinnis. If $A_1, \dots, A_{n-1}$ are arborescences on n vertices, not necessarily spanning, with $|A_i| \geq \min\{i, \lfloor n/2 \rfloor\}$ edges, is there always a rainbow arborescence with $\lfloor n/2 \rfloor$ edges? The second weakens the conclusion of Conjecture 6.7 from rainbow to properly coloured. Given $n-1$ spanning arborescences on n vertices, does their union contain a spanning arborescence in which the edges leaving each vertex have distinct colours? More strongly, can all edges at each vertex, both entering and leaving, be made to have distinct colours? Each colour may then be used many times, so both questions are weaker than the conjecture.

Random colourings behave as for undirected spanning trees (Theorem 3.22). Bal, Bennett, Cooper, Frieze and Prałat [40] considered the random digraph process, with each edge coloured uniformly from $n(1 + o(1))$ colours. Two conditions are necessary for a rainbow arborescence: at most one vertex has indegree 0, and at least $n-1$ colours have appeared. They showed that with high probability a rainbow arborescence appears as soon as both hold.

6.5 Tournament and digraph collections

Collections of digraphs are defined as for graphs (Section 3.7): a digraph H with m edges is a transversal of $(D_1, \dots, D_m)$ if its edges can be matched bijectively to the D_i so that each edge lies in its digraph.

Every tournament has a Hamilton path. For collections of tournaments this fails for $n = 3$: two oppositely oriented cycles of length 3 have no transversal Hamilton path (Figure 29). Chakraborti, Kim, Lee and Seo [79] showed that small cases are the only exceptions.

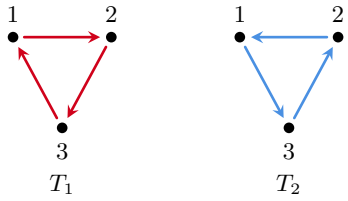


Figure 29: Two tournaments on $\{1, 2, 3\}$. A transversal Hamilton path would consist of an edge of T_1 followed by an edge of T_2, or the other way round. A computer check of all six orderings confirms that none exists.

Theorem 6.8 (Chakraborti, Kim, Lee and Seo [79])**.** *For every sufficiently large n, every collection of $n-1$ tournaments on a common set of n vertices contains a transversal Hamilton path. Every collection of n tournaments on n vertices, all but at most one of which are strongly connected, contains a transversal Hamilton cycle.*

The second statement is sharp: for every $n \geq 3$, a collection of n tournaments in which all but two are strongly connected need not have a transversal Hamilton cycle. Thomason proved that every large tournament contains every orientation of a Hamilton path, and every orientation of a Hamilton cycle except possibly the consistently oriented one. Chakraborti, Kim, Lee and Seo [80] proved the transversal version of this as well.

For general digraphs, the minimum semidegree $\delta^0(D) = \min\{\delta^+(D), \delta^-(D)\}$ plays the role of the minimum degree. Ghouila-Houri proved that every digraph on n vertices with $\delta^0(D) \geq n/2$ has a Hamilton cycle. Chakraborti, Kim, Lee and Seo asked for a transversal version, and Cheng, Li, Sun and Wang proved it.

Theorem 6.9 (Cheng, Li, Sun and Wang [86])**.** *For every sufficiently large n, every collection of n digraphs on a common set of n vertices, each with minimum semidegree at least $n/2$, contains a transversal Hamilton cycle.*

Replacing each edge of a graph by two opposite edges turns a graph of minimum degree $n/2$ into a digraph of minimum semidegree $n/2$. So Theorem 6.9 contains the transversal Dirac theorem of Joos and Kim (Section 5.5) for large n, and Hamilton cycles in digraphs are colour-blind.

The second statement of Theorem 6.8 needs all but one of the tournaments to be strongly connected. Chakraborti, Kim, Lee and Seo suggest a weaker condition. Call a collection of tournaments *strongly rainbow-connected* if for every two vertices x, y there is a transversal path from x to y. A collection with a transversal Hamilton cycle has this property. The converse fails for $n = 3$: take two copies of a cycle of length 3 and one copy of the opposite cycle. Every ordered pair of vertices is joined by an edge of some tournament, but a transversal Hamilton cycle would need three edges of the same orientation from three different tournaments [79].

Conjecture 6.10 (Chakraborti, Kim, Lee and Seo, 2023 [79])**.** *For all sufficiently large n, every strongly rainbow-connected collection of n tournaments on a common set of n vertices contains a transversal Hamilton cycle.*

They also ask whether the assumptions of Theorem 6.8 give transversal cycles of every length from 3 to n.

For digraphs, the next targets are oriented graphs and other orientations of the Hamilton cycle. Keevash, Kühn and Osthus showed that an oriented graph on n vertices with minimum semidegree at least $\lceil (3n-4)/8 \rceil$ has a Hamilton cycle, and that this is best possible (see [86]).

Problem 6.11 (Cheng, Li, Sun and Wang, 2025 [86])**.** Let $D_1, \dots, D_n$ be oriented graphs on a common set of n vertices, each with minimum semidegree at least $\lceil (3n-4)/8 \rceil$. Must the collection contain a transversal Hamilton cycle?

The absorbing structures behind Theorem 6.9 need semidegree about $n/2$, and Cheng, Li, Sun and Wang expect that new ideas are needed at $3n/8$. They also ask for transversal versions of the results of DeBiasio, Kühn, Molla, Osthus and Taylor and of DeBiasio and Molla, which give every orientation of a Hamilton cycle at minimum semidegree about $n/2$.

6.6 Randomly perturbed digraphs

As for trees (Section 3.8), adding a sparse random digraph to a dense digraph gives rainbow spanning structures with an exact palette. Let D_0 be a digraph on n vertices with $\delta^0(D_0) \geq \alpha n$, and add the random digraph $D(n, C/n)$, in which each of the $n(n-1)$ possible edges is present independently with probability C/n. Araujo, Balogh, Krueger, Piga and Treglown showed that the union contains every orientation of every cycle of length between 2 and n, simultaneously, with high probability (see [186]). Krueger and Staudinger proved the rainbow version with exactly n colours.

Theorem 6.12 (Krueger and Staudinger [186])**.** *For every $\alpha > 0$ there is $C > 0$ such that the following holds. Let D_0 be a digraph on n vertices with $\delta^0(D_0) \geq \alpha n$. If $D_0 \cup D(n, C/n)$ is coloured uniformly from $[n]$, then with high probability it contains a rainbow copy of every orientation of every cycle of length between 2 and n, simultaneously.*

The consistently oriented Hamilton cycle was found earlier by Katsamaktsis, Letzter and Sgueglia [170], who asked for every orientation and for universality; Theorem 6.12 answers both. A rainbow Hamilton cycle needs n colours, so the palette is optimal, and the edge probability C/n is optimal up to the constant even without colours. The proof uses Montgomery's distributive absorption.

7 Hypergraphs, designs and set families

A k-uniform hypergraph, or *k-graph*, has edges of size k, and a matching is a set of disjoint edges. Many questions of Section 2 make sense for k-graphs. The case $k = 3$ already appeared in Section 2.1, where a Latin square became a 3-partite 3-graph.

There is also a translation in the opposite direction, from rainbow matchings to ordinary matchings. Given a coloured k-graph, add a new vertex for each colour, and add to every edge the vertex of its colour. Two edges of the same colour now share a vertex. So a rainbow matching of the original k-graph is exactly a matching of the new $(k+1)$-graph (Figure 30). The new hypergraph is $(1, k)$-partite: each edge has one colour vertex and k original vertices. Many results below are proved in this form [202, 106].

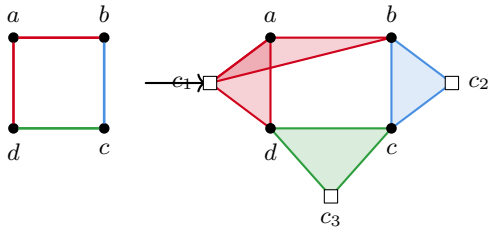


Figure 30: Left: a coloured 4-cycle. Right: each edge receives a new vertex for its colour (squares), giving a 3-graph. The edges ab and da have the same colour, so their triples share the vertex c_1. A computer check of all sets of edges confirms that rainbow matchings on the left are exactly matchings on the right.

Some things change when the uniformity grows. For graphs, n matchings of size n are conjectured to have a rainbow matching of size $n-1$ (Section 2.10). For k-graphs with $k \geq 3$ this fails by a growing amount (Section 7.2), and the number of matchings needed for a full rainbow matching becomes polynomial of degree k (Section 7.1).

7.1 How many matchings force a rainbow matching?

Drisko's theorem (Theorem 2.35) says that $2t-1$ matchings of size t in a bipartite graph have a rainbow matching of size t, and that $2t-2$ are not enough. Aharoni and Berger [2] asked the same question for r-partite r-graphs. They found $2^{r-1}(t-1)$ matchings of size t with no rainbow matching of size t, and conjectured that one more matching always suffices.

Alon [22] disproved this through a problem in additive number theory. Erdős, Ginzburg and Ziv proved that any $2t-1$ integers contain t whose sum is divisible by t. Harborth asked the same for vectors: how long must a sequence in $\mathbb{Z}_t^d$ be to contain t terms summing to zero? Alon turned a sequence in $\mathbb{Z}_t^{r-1}$ with no such t terms into a family of perfect matchings of an r-partite r-graph with parts $A_1, \ldots, A_r$, each a copy of $\mathbb{Z}_t$. The vector s gives the matching with edges $(s_1+i, \ldots, s_{r-1}+i, i)$ for $i \in \mathbb{Z}_t$. A rainbow matching of size t covers every vertex of every part, and summing coordinates shows that the t vectors used would sum to zero. Hence a zero-sum-free sequence of length ℓ gives ℓ matchings with no rainbow matching of size t (Figure 31). Known constructions for Harborth's problem beat $2^{r-1}(t-1)$, for example when $t \geq 3$ is odd and $r \geq 4$.

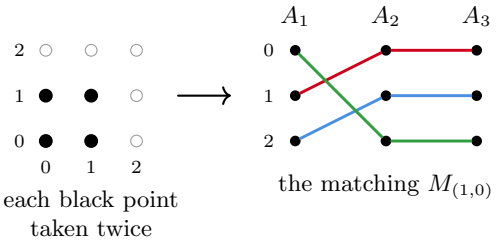


Figure 31: Alon's construction for $r = t = 3$. Left: the four points of $\mathbb{Z}_3^2$ shown in black, each taken twice, form a sequence of length 8 in which no three terms sum to zero. Right: the vector $(1,0)$ gives the perfect matching with edges $(1,0,0)$, $(2,1,1)$, $(0,2,2)$. A computer check confirms that the eight matchings have no rainbow matching of size 3.

Alon asked whether the number of matchings needed is still linear in t for fixed r, as the Erdős–Ginzburg–Ziv constants are; Glebov, Sudakov and Szabó conjectured that it is. Pohoata, Sauermann and Zakharov showed that it is not.

Theorem 7.1 (Pohoata, Sauermann and Zakharov [235]). *For every fixed $r \geq 3$ there are $c_r, C_r > 0$ such that the following holds for every $t \geq 2$. Some family of $c_r t^r$ matchings of size t in an r-partite r-graph has no rainbow matching of size t, while any $C_r t^r$ matchings of size t in any r-graph have a rainbow matching of size t.*

So the answer is of order t^r, far from the linear behaviour of zero-sum problems. The upper bound improves a bound of order t^{r+1} due to Munhá Correia, Sudakov and Tomon. The fractional version behaves differently: Aharoni, Holzman and Jiang [9] showed that any $\lceil rn \rceil$ families of edges of an r-graph, each with a fractional matching of size n, have a rainbow fractional matching of size n.

7.2 Near-full rainbow matchings

The Ryser–Brualdi–Stein question asks how large a rainbow matching n matchings of size n must have. Let $g(r,n)$ be the largest size guaranteed in an r-partite r-graph, and $g'(r,n)$ the same without the partite condition. For $r=2$ the conjecture is $g(2,n) = g'(2,n) = n-1$ (Section 2.10). For larger r the deficit grows with n.

Theorem 7.2 (Bowtell, Freschi, Kronenberg and Yan [60]). *For every fixed $r \geq 3$ and sufficiently large n,*

$$\frac{2n}{r+1} - O_r(1) \leq g'(r,n) \leq g(r,n) \leq n - \Omega_r\big(n^{1-1/r}\big).$$

Previously Aharoni, Charbit and Howard had shown $g(r,n) \leq n - 2^{r-2}$ for even n, a constant deficit. The upper bound shows that no statement of the form $n - O(1)$ holds. The lower bound is far from it, and the authors suspect that the truth is close to the upper bound.

The same question can be asked the other way round, as for graphs in Section 2.10: how large must n matchings be to force a rainbow matching of size n? Let $h(r,n)$ be the smallest s such that any n matchings of size s in an r-partite r-graph have a rainbow matching of size n, and $h'(r,n)$ the same without the partite condition. For $r \geq 3$, Bowtell, Freschi, Kronenberg and Yan showed that $h(r,n) \geq n + \Omega_r(n^{1-1/r})$, so here too the deficit grows, and that $h'(r,n) \leq (r+1)n/2 + o(n)$. They conjectured that all four quantities are asymptotically n.

Conjecture 7.3 (Bowtell, Freschi, Kronenberg and Yan, 2025 [60]). *For every fixed $r \geq 2$, each of $g(r,n)$, $g'(r,n)$, $h(r,n)$ and $h'(r,n)$ is $(1+o(1))n$.*

For g the lower bound $2n/(r+1)$ in Theorem 7.2 refines an argument of Aharoni, Charbit and Howard, and the authors expect that a significantly more complicated strategy is needed to reach $n - o(n)$.

Another way to obtain full rainbow matchings is to bound degrees. Let G be an r-graph with maximum degree Δ whose edges are coloured with n colours. A topological Hall-type theorem of Aharoni, Berger and Meshulam shows that if every colour class has at least $r\Delta$ edges, then there is a full rainbow matching, one edge of each colour [266]. Wdowinski [266] showed that $r\Delta$ cannot be lowered to $r\Delta - 1$, using blow-ups of nets from design theory. If the codegrees of G and the degrees of the colour classes are $o(\Delta)$, then colour classes of size $(1+o(1))\Delta$ suffice, by a theorem of Delcourt and Postle [106]. They conjectured that for properly coloured multigraphs $\Delta + 1$ edges per colour suffice in the bipartite case and $\Delta + 2$ in general, as sparse analogues of the Aharoni–Berger and Gao–Ramadurai–Wanless–Wormald conjectures. Wdowinski disproved both, even when the chromatic index is Δ. The original conjectures, about n colours in Section 2.10, are not affected.

Wdowinski's constructions suggest measuring colour classes against the chromatic index χ' rather than the maximum degree. He asks whether, in a proper colouring of a multigraph with chromatic index χ', colour classes of size $(1+o(1))\chi'$ always give a full rainbow matching, and whether even $\chi' + C$ suffices for a constant C [266]. The theorem of Delcourt and Postle gives the first statement when edge multiplicities are $o(\Delta)$. His constructions rule out $C = 2$ in general and $C = 1$ for bipartite multigraphs, but not larger constants.

7.3 List edge-colouring

Full rainbow matchings also encode list edge-colouring. Let H be a hypergraph in which each edge e has a list $L(e)$ of allowed colours. For every edge e and colour $a \in L(e)$, take the edge $\{(v,a) : v \in e\}$ in a new hypergraph, and give it the colour e. Two of these edges meet exactly when they come from edges of H that share a vertex and receive the same colour a. So a proper colouring of H from the lists is the same as a full rainbow matching of the new hypergraph, which uses one edge of each colour e (Figure 32).

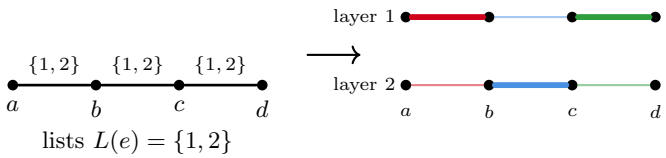


Figure 32: Left: a path *abcd* in which every edge has the list $\{1, 2\}$. Right: each edge e and colour $a \in L(e)$ give a copy of e in layer a, coloured by e. A full rainbow matching takes one copy of each edge, and the copies must be disjoint. The thick copies give the colouring $1, 2, 1$. A computer check confirms that the full rainbow matchings correspond exactly to the two proper colourings $1, 2, 1$ and $2, 1, 2$.

The list edge-colouring conjecture asserts that the list chromatic index of every multigraph equals its chromatic index. Galvin proved it for bipartite multigraphs, which settled Dinitz's problem on Latin squares (Section 3.6). Kahn proved it asymptotically for multigraphs, and showed that a k-graph with maximum degree Δ and codegrees $o(\Delta)$ has list chromatic index $(1+o(1))\Delta$. Through the translation, the theorems of the previous subsection give colour-degree versions [266]. Let the *colour degree* of a colour a be the maximum degree of the edges whose lists contain a, and let Δ be the largest colour degree. If every list has at least $r\Delta$ colours, a proper colouring from the lists exists; if codegrees are small, $(1 + o(1))\Delta$ colours suffice. Galvin's theorem does not extend in this way: Wdowinski found bipartite graphs with maximum colour degree Δ and lists of size Δ that have no proper colouring from the lists.

The chromatic-index version of the previous question has a list form. For a colour a, let H_a be the graph of edges whose lists contain a, and let χ' be the largest chromatic index of the H_a. Wdowinski asks whether lists of size $(1 + o(1))\chi'$, or even $\chi' + C$, always allow a proper colouring from the lists [266]. A positive answer to the matching question would give one here. Every H_a is a subgraph of H, so a positive answer with $\chi' + C$ would give the list edge-colouring conjecture up to an additive constant. His bipartite examples show that $C = 0$ is not enough. He singles out bipartite multigraphs as the case of particular interest.

7.4 Families of sets

The Erdős matching conjecture concerns k-element subsets of $[n]$, where $n \geq km$. It asserts that a family with no m disjoint sets has at most

$$f(n, m, k) = \max\left\{\binom{n}{k} - \binom{n-m+1}{k},\ \binom{km-1}{k}\right\}$$

members. The two terms come from all sets meeting $[m-1]$ and from all k-subsets of $[km-1]$. The case $m = 2$ is the Erdős–Ko–Rado theorem. The rainbow version asks for disjoint sets taken one from each of m families.

Conjecture 7.4 (Huang, Loh and Sudakov, 2012 [159]; Aharoni and Howard, 2016 [10])**.** *Let $n \geq km$. If $\mathcal{F}_1, \ldots, \mathcal{F}_m$ are families of k-element subsets of $[n]$, each with more than $f(n, m, k)$ members, then there are pairwise disjoint sets $F_1 \in \mathcal{F}_1, \ldots, F_m \in \mathcal{F}_m$.*

Taking all families equal shows that the rainbow conjecture contains the Erdős matching conjecture. Huang, Loh and Sudakov proved it for $n > 3k^2m$. Keller and Lifshitz proved it for $n \geq f(m)k$, Frankl and Kupavskii for $n = \Omega(km \log m)$, and Lu, Wang and Yu for $n \geq 2km$ and n sufficiently large (see [129]). For $k = 3$, Gao, Lu, Ma and Yu [129] proved the conjecture for all sufficiently large n. Huang, Li and Wang [158] proved a version for properly coloured hypergraphs. Below $2km$ the conjecture contains the open cases of the Erdős matching conjecture, so a complete answer there would also settle those.

In the r-partite setting, with families of r-tuples from $[n]^r$, the extremal number is $(m-1)n^{r-1}$: take all tuples whose first coordinate lies in $[m-1]$. Aharoni and Howard [10] conjectured that m families of more than $(m-1)n^{r-1}$ tuples always contain a rainbow matching, and proved it for $r \leq 3$. Lu and Yu [206] proved it for $n \geq 3(r-1)(m-1)$, and Kiselev and Kupavskii [179] for every $m \geq 470$. The question has since moved to unequal sizes: for which sequences $f_1, \ldots, f_m$ do the conditions $|\mathcal{F}_i| > f_i$ force

a rainbow matching [188]? Kupavskii and Popova call such sequences *satisfying*. For $r = 1$ a sequence $f_1 \le \cdots \le f_m$ is satisfying exactly when $f_i \ge i - 1$ for every i. Already the case $r = 2$ is open. There the families are sets of edges of $K_{n,n}$, and the question asks which edge counts of m bipartite graphs on the same two parts force a rainbow matching of size m. Kupavskii and Popova propose describing all satisfying sequences for $r = 2$ as the next step. They expect it to give insight for all r and large n, and they also ask for the analogue for k-subsets of $[n]$, which they expect to be much harder.

7.5 Maximin share allocations

Rainbow matchings of set families appear in fair division, in a different way from the rainbow cycles of Section 6.3. Suppose that m indivisible items are to be split among n agents. The *maximin share* of an agent is the largest value it can guarantee by splitting the items into n bundles itself and receiving the worst one. An allocation is an *MMS allocation* if every agent receives a bundle it values at least at its maximin share.

Consider items as vertices and agents as colours. For each agent, the bundles it accepts are edges of its colour; its own optimal split is a perfect matching of its colour. An MMS allocation is then a rainbow matching that uses every colour. For goods with monotone valuations the leftover items can be handed out freely, so the question becomes the following. Every hypergraph on m vertices with n perfect matchings of n edges each, in n different colours, should have a rainbow matching with n edges. For which m does this hold? With $m = 4$ and $n = 2$ it can fail (Figure 33).

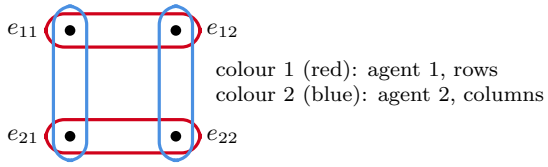


Figure 33: Two agents and four items $e_{11}, e_{12}, e_{21}, e_{22}$. Agent 1 accepts a bundle only if it contains a row, and agent 2 only if it contains a column. Every row meets every column, so the two perfect matchings have no rainbow matching of size 2, and there is no MMS allocation.

Feige determined the answer up to the constant in the error term.

Theorem 7.5 (Feige [122]). *Let $\mu(n)$ be the largest m such that every instance with n agents with monotone valuations and at most m goods has an MMS allocation. Then $\mu(n) = n + \Theta(\log n)$.*

The same holds for chores, where the matching is replaced by a rainbow cover, while for arbitrary valuations $\mu(n) = \log n + O(1)$. In the language of hypergraphs, the theorem says how many vertices beyond n make rainbow matchings unavoidable. With logarithms to base 2, Feige shows that the constant in $\Theta(\log n)$ lies between $0.93 - o(1)$ and $1 + o(1)$ for goods, and asks whether $\mu(n) = n + \log n + o(\log n)$. Pinning it down reduces to a question about hypergraphs on $c + d$ vertices: how many colours of d-matchings, each edge of size at least 2, force a rainbow d-matching for every $d \le c$? The known answers lie between $(2 - o(1))^c$ and $(2.0922 + o(1))^c$.

7.6 Perfect matchings and Hamilton cycles in hypergraph collections

Collections of k-graphs are defined as for graphs (Section 3.7). A *rainbow perfect matching* of k-graphs $H_1, \ldots, H_{n/k}$ on a common set of n vertices consists of disjoint edges $e_i \in H_i$ covering all vertices. For $1 \le \ell < k$, the *minimum ℓ-degree* $\delta_\ell(H)$ is the least number of edges containing an ℓ-set of vertices. Let $\delta(n, k, \ell)$ be the largest minimum ℓ-degree of an n-vertex k-graph with no perfect matching. Taking all H_i equal to such an extremal k-graph gives no rainbow perfect matching, so the uncoloured threshold is a lower bound. For $\ell \ge k/2$ it is exact.

Theorem 7.6 (You [272]). *Let $k \ge 3$ and $k/2 \le \ell \le k - 1$. For sufficiently large n divisible by k, if $H_1, \ldots, H_{n/k}$ are k-graphs on a common set of n vertices with $\delta_\ell(H_i) > \delta(n, k, \ell)$ for every i, then they have a rainbow perfect matching.*

The case $\ell = k-1$ is due to Lu, Wang and Yu [202], whose proof introduced the $(1,k)$-partite form of the colour-vertex translation. For $\ell < k/2$ even the uncoloured thresholds are not known in general. For vertex degrees, rainbow versions of the uncoloured thresholds are known for $k = 3$ and $k = 4$ [207, 203].

For Hamilton cycles in k-graphs, a *tight* cycle has consecutive edges meeting in $k-1$ vertices, and a *loose* cycle has consecutive edges meeting in one vertex. The minimum $(k-2)$-degree threshold for a tight Hamilton cycle is $(5/9+o(1))\binom{n}{2}$, and Tang, Wang, Wang and Yan [258] proved the rainbow version: n k-graphs on a common n-vertex set, each with $\delta_{k-2} \geq (5/9+\gamma)\binom{n}{2}$, have a rainbow tight Hamilton cycle. The general framework of Gupta, Hamann, Müyesser, Parczyk and Sgueglia (Section 5.5) shows that Hamilton ℓ-cycles are colour-blind in several further ranges [149].

Tang, Wang, Wang and Yan [258] conjectured that tight Hamilton cycles are colour-blind for every degree type. Write $h_d(k,n)$ for the minimum d-degree threshold for a tight Hamilton cycle in an n-vertex k-graph.

Conjecture 7.7 (Tang, Wang, Wang and Yan, 2023 [258]). *For all $1 \leq d \leq k-1$ and $\mu > 0$ there is n_0 such that, for $n \geq n_0$, every collection of n k-graphs on a common set of n vertices, each with minimum d-degree at least $h_d(k,n) + \mu n^{k-d}$, has a rainbow tight Hamilton cycle.*

For $d = k-2$ this is their theorem. For most other d the uncoloured threshold $h_d(k,n)$ is itself unknown; Lang and Sanhueza-Matamala conjectured its value, which for $d = k-3$ is $(5/8+o(1))\binom{n}{3}$. The rainbow statement for $d = k-3$ at this value is the first open case Tang, Wang, Wang and Yan single out.

7.7 Bounded colourings

For a single coloured k-graph, the theorems of Section 5.4 have hypergraph analogues. Two bounds are needed for factors. A global bound is needed because a factor has many edges. A local bound is needed as well: colour each edge of the complete k-graph by its $k-1$ smallest vertices. Every colour then has fewer than n edges, but any copy of the complete k-graph $K^{(k)}_{k+1}$ has two edges with the same smallest $k-1$ vertices, so there is no rainbow $K^{(k)}_{k+1}$-factor. Call a colouring of an n-vertex k-graph *μ-bounded* if each colour has at most μn^{k-1} edges, and each $(k-1)$-set lies in at most μn edges of each colour.

Theorem 7.8 (Coulson, Keevash, Perarnau and Yepremyan [99]). *Let $1 \leq \ell < k$, let F be a k-graph on h vertices, and let $\varepsilon > 0$. There is $\mu > 0$ such that for large n divisible by h, every k-graph G on n vertices with $\delta_\ell(G) \geq (\delta^*_\ell(F)+\varepsilon)n^{k-\ell}$ has a rainbow F-factor in every μ-bounded colouring. Here $\delta^*_\ell(F)$ is the asymptotic ℓ-degree threshold for an F-factor.*

The theorem holds even where $\delta^*_\ell(F)$ is not known. Coulson, Keevash, Perarnau and Yepremyan conjectured more generally that above the threshold for any spanning structure, a suitably bounded colouring contains a rainbow copy. Kathapurkar, Morris and Perarnau verified this for loose Hamilton cycles, with only a global bound.

Theorem 7.9 (Kathapurkar, Morris and Perarnau [169]). *Let $1 \leq j < k$ and $\varepsilon > 0$. There is $\mu > 0$ such that for large n divisible by $k-1$ the following holds. If G is an n-vertex k-graph with $\delta_j(G)$ at least εn^{k-j} above the j-degree threshold for a loose Hamilton cycle, then every colouring of G with at most μn^{k-1} edges of each colour has a rainbow loose Hamilton cycle.*

Tight cycles behave differently. Kathapurkar, Morris and Perarnau gave 3-graphs above the codegree threshold for tight Hamilton cycles, with colour classes of only linear size, and with no rainbow tight Hamilton cycle. Split the vertices into three nearly equal parts V_1, V_2, V_3, give every pair inside a part its own colour, and let the edges be the triples with exactly two vertices in one part, coloured by that pair. Each colour has at most n edges. Following a tight Hamilton cycle until it first enters V_3, one finds two consecutive edges sharing a pair inside V_1 or V_2, hence of the same colour (Figure 34).

So for tight Hamilton cycles some local bound is necessary, and the right bounds may depend on the minimum degree. Kathapurkar, Morris and Perarnau propose determining which combinations of local and global bounds force a rainbow tight Hamilton cycle above the degree threshold. For tight cycles no positive result in Dirac hypergraphs is known, and their switching method seems to break down. Even the complete k-graph is not understood. Dudek, Frieze and Ruciński found rainbow tight Hamilton

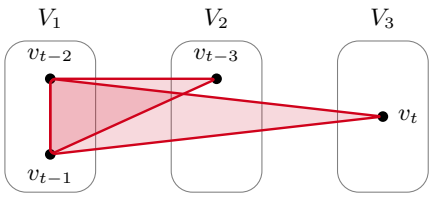


Figure 34: The construction of Kathapurkar, Morris and Perarnau. Triples with two vertices in one part are edges, coloured by that pair. Where a tight cycle first reaches V_3, the two previous vertices lie in one part, so two consecutive triples have the same colour. A computer check for $n = 6$ finds eight tight Hamilton cycles, none of them rainbow.

cycles there when every colour has $o(n)$ edges. They asked whether some colouring with $\Theta(n)$ edges of each colour has none, and believe that it does (see [169]). Alternatively, a global bound of $o(n^{k-1})$ might already suffice, as it does for loose cycles in Theorem 7.9; the construction of Figure 34 shows that this fails in some range of minimum degree. For factors, the local bound in Theorem 7.8 is needed for $F = K^{(k)}_{k+1}$, but the method of Kathapurkar, Morris and Perarnau removes it when F is linear, that is, when two edges of F share at most one vertex. Which F need a local bound is open.

In complete hypergraphs properness is enough for perfect matchings. Li and Xu [196] showed that every proper colouring of the complete r-graph on rn vertices, with $r \geq 2$ and $n \geq 3$, contains a rainbow perfect matching, extending a result of El-Zanati et al. for 1-factorisations.

7.8 Random hypergraphs

Ferber and Krivelevich [123] used a coupling argument of McDiarmid to transfer embedding results to random colourings. In a random k-graph whose edges are coloured uniformly with $(1 + o(1))e(S)$ colours, a rainbow copy of a spanning hypergraph S appears at the threshold for an uncoloured copy, up to a constant factor. With exactly as many colours as edges, Dudek, English and Frieze [113] showed that $p = e^2/n$ is the sharp threshold for a rainbow tight Hamilton cycle in the random k-graph coloured with n colours, for every $k \geq 4$. The rainbow threshold theorem of Section 5.8 gives rainbow Hamilton ℓ-cycles with an exact palette up to a logarithmic factor [152].

7.9 Designs and Latin hypercubes

A *Latin hypercube* of dimension d and order n is an $n \times \cdots \times n$ array of dimension d, filled with n symbols, in which every line obtained by varying one coordinate contains each symbol once. For $d = 2$ this is a Latin square. A *transversal* is a set of n entries, no two agreeing in any coordinate or in their symbol. Translated as in Section 2.1, a Latin hypercube is a $(d + 1)$-partite $(d + 1)$-graph, with a part for each coordinate and one for the symbols, and a transversal is a perfect matching (Figure 35).

layer $k = 0$

0	1	2
1	2	0
2	0	1

layer $k = 1$

1	2	0
2	0	1
0	1	2

layer $k = 2$

2	0	1
0	1	2
1	2	0

Figure 35: The Latin cube of order 3 with entries $i+j+k$ mod 3, shown layer by layer. The shaded entries $(0, 0, 0)$, $(1, 1, 2)$ and $(2, 2, 1)$ have symbols $0, 1, 2$ and no common coordinate, so they form a transversal.

As for Latin squares, the cyclic hypercube with entries $x_1 + \cdots + x_d$ modulo n has no transversal when n and d are both even [88]. The analogue of Ryser's conjecture says that parity is the only obstruction.

Conjecture 7.10 (Wanless, 2011 [264])**.** *If n or d is odd, then every Latin hypercube of dimension d and order n has a transversal.*

For $d = 2$ this is Ryser's conjecture (Conjecture 2.9). It is known for orders $n \leq 5$. For even n and d, there are superexponentially many species of Latin hypercubes without transversals. Child and Wanless constructed, for even d, hypercubes that have transversals but in which every transversal meets one fixed $(d-2)$-dimensional subarray, for every even $n \geq 10$.

Child and Wanless [88] suggest that for $d > 2$ something much stronger than Conjecture 7.10 may hold. Call a Latin hypercube a *confirmed bachelor* if some entry lies in no transversal. Among the Latin hypercubes of dimension $d > 2$ enumerated by computer, confirmed bachelors occur only for $n = d = 4$, and none is known when n or d is odd. Decompositions into transversals appear to be common. These data are consistent with every entry of every Latin hypercube lying in a transversal whenever $d > 2$ and n or d is odd. In the other direction, Ghafari and Wanless found Latin squares with transversals in which asymptotically more than half of the entries lie in no transversal. Child and Wanless ask whether, in higher dimensions, a fixed positive proportion of entries can avoid all transversals; their confirmed bachelors of order 4 achieve only an exponentially small proportion.

Coloured decompositions also arise in design theory. Keevash's general theory of designs covers decompositions of hypergraphs whose edges carry colours. Among his examples is a rainbow triangle decomposition theorem [172]. Let G be a graph whose edges are coloured with $D \geq 4$ colours, all of whose degrees are even and with $3 \mid e(G)$. If G is typical in a quasirandom sense, and every colour class has density below one third of the density of G but not too small, then G decomposes into rainbow triangles. Divisibility is necessary even without colours, and the colours impose no further divisibility condition. Such decompositions encode problems of design theory such as Whist tournaments.

A Main results by chapter

Tables 1–6 list the main line of each section. Each row restates a theorem, conjecture or construction from the text, and the last column says where it is discussed. The column *Obstruction* gives a construction that limits the natural statement; a dash means that the text gives none. Results are stated in the form used in the text, and "for large n" means for all sufficiently large n.

Table 1: Main results on matchings and transversals (Section 2).

Target	Host and colouring	Result	Obstruction	In the text
Decompositions				
decomposition into transversals (orthogonal mate)	some Latin square of order n	exists for every $n \notin \{2, 6\}$ (Bose–Shrikhande–Parker); a random Latin square has one with high probability (Bowtell–Montgomery)	$n = 2, 6$: none (Tarry); even n: squares with no transversal	Conj. 2.1, Thm 2.2
$(1 - o(1))n$ disjoint partial transversals of size $(1 - o(1))n$	every Latin square of order n	holds (Montgomery–Pokrovskiy–Sudakov)	—	Thm 2.3
decomposition into rainbow perfect matchings (Room square)	some optimal colouring of K_{2n}	exists if and only if $n \geq 4$ (Mullin–Wallis)	$n \in \{2, 3\}$	Thm 2.4
perfect 1-factorisation	K_{2n}	known when $2n - 1$ or n is prime; $(1 - o(1))2n$ pairwise Hamiltonian classes (Cheng–Sgueglia)	—	Conj. 2.5
One transversal				
full transversal	Latin square of odd order	open (Ryser); random squares: with high probability (Kwan)	even n: the cyclic square has none	Conj. 2.9, Thm 2.16
full transversal	multiplication table of a finite group G	exists iff the Sylow 2-subgroups are trivial or non-cyclic (Hall–Paige)	nontrivial cyclic Sylow 2-subgroup	Thm 2.11
partial transversal with $n - 1$ cells	every Latin square of order n	for large n (Montgomery); $n - O(\log n / \log\log n)$ for all n (Keevash et al.)	even n: n cells impossible	Conj. 2.19, Thm 2.21
More colours and bounded colourings				
rainbow perfect matching	proper colouring of $K_{n,n}$ with $\geq n^{2-\varepsilon}$ colours	holds for large n (Keevash–Yepremyan; Montgomery–Pokrovskiy–Sudakov)	n colours: cyclic square of even order	Conj. 2.24

Target	Host and colouring	Result	Obstruction	In the text
full transversal	$n \times n$ array, each symbol at most $(n-1)/16$ times	holds (Erdős–Spencer); bound $(1-o(1))n$ open	each symbol n times: see next row	Thm 2.27, Prob. 2.30
partial transversal with $n-1$ cells	equi-n-square (each symbol exactly n times)	$n - n^{1-\Omega(1)}$ cells (Chakraborti et al.)	some squares have none with more than $n - \Omega(\sqrt{n})$ cells	Conj. 2.26, Thm 2.29
Families of matchings				
rainbow matching of size n	$2n-1$ matchings of size n, bipartite graph	holds (Drisko)	$2n-2$ matchings from a $2n$-cycle	Thm 2.35
full rainbow matching	n matchings of size $n+1$, bipartite multigraph	size $n + o(n)$ suffices (Pokrovskiy)	size n: $n-1$ copies of M_1 and one of M_2 in a $2n$-cycle	Conj. 2.33
full rainbow matching	n matchings of size $n+2$, multigraph	size $n + 20n^{1-1/16}$ suffices (Correia–Pokrovskiy–Sudakov)	size $n+1$ is not enough	Conj. 2.37
Steiner triple systems				
matching with $\lfloor n/3 \rfloor - 1$ edges	Steiner triple system of order n	for large n (Montgomery)	—	Conj. 2.41

Table 2: Main results on spanning trees and bases (Section 3).

Target	Host and colouring	Result	Obstruction	In the text
Decomposing into rainbow spanning trees				
n rainbow spanning trees partitioning the edges	optimal colouring of K_{2n}	for large n, even with isomorphic trees (Glock–Kühn–Montgomery–Osthus)	—	Conj. 3.3, Thm 3.7
$\lfloor n/2 \rfloor$ edge-disjoint rainbow spanning trees	proper colouring of K_n	$(1-o(1))n/2$ trees (Montgomery–Pokrovskiy–Sudakov)	—	Conj. 3.5, Thm 3.6
$(1-\varepsilon)m$ disjoint rainbow forests with k edges	proper colouring, km edges, colour classes of size $\leq m$	holds for $1 \leq k \leq 2m$ (Xu)	$k > 2m$: possibly no rainbow forest with k edges	Thm 3.8
Rainbow copies of a given tree				
every tree on $(1-o(1))n/k$ vertices	locally k-bounded colouring of K_n	holds (Montgomery–Pokrovskiy–Sudakov)	proper colourings: no rainbow Hamilton path when $n = 2^k$	Thm 3.9, Conj. 3.10
every tree with at most n edges	proper colouring of the hypercube Q_n	holds (Crawford–Sankar–Schildkraut–Spiro)	colouring by coordinates: n colours, no rainbow cycle	§3.4

Target	Host and colouring	Result	Obstruction	In the text
Labellings and Ringel's conjecture				
decomposition of K_{2n+1} into copies of a tree with n edges	near-distance colouring of K_{2n+1} (rainbow copy suffices)	for large n (Montgomery–Pokrovskiy–Sudakov; Keevash–Staden)	—	Conj. 3.13, Thm 3.14
graceful labelling of every tree	difference colouring of K_{n+1} on $\{0,\dots,n\}$	open; approximate versions known	—	Conj. 3.12
every tree on n vertices, maximum degree $\leq \Delta$	$K_{\mathbb{Z}_n}$, edge ab coloured $a+b$	for large n (Müyesser–Pokrovskiy)	the same colouring on $\mathbb{Z}_2^k$: no rainbow Hamilton path	Thm 3.15
Matroid bases				
n disjoint rainbow bases (Rota)	n bases of a rank-n vector space or matroid	$(1-o(1))n$ disjoint; $(1+o(1))n$ cover all elements (Montgomery–Sauermann)	—	Conj. 3.16, Thm 3.18
common independent set of size $n-1$, one element per F_i	two matroids, n disjoint common independent sets of size n	open	Latin squares without transversals: n is impossible	Conj. 3.20
Collections and random colourings				
every tree with maximum degree $\leq cn/\log n$	$n-1$ graphs, each with minimum degree $\geq (1/2+\alpha)n$	transversal copy exists (Montgomery–Müyesser–Pehova)	—	Thm 3.21
rainbow spanning tree	random graph process, $\geq n-1$ random colours	appears when the graph is connected and has $n-1$ colours (Frieze–McKay)	—	Thm 3.22

Table 3: Main results on paths and cycles (Section 4).

Target	Host and colouring	Result	Obstruction	In the text
Long paths and cycles in proper colourings				
path on $n-1$ vertices	proper colouring of K_n	holds for large n (BMMP)	$n=2^k$: no rainbow Hamilton path (Maamoun–Meyniel)	Thm 4.2, Fig. 14
cycle through all but C vertices	proper colouring of K_n	some constant C (BMMP); value of C unknown	1-factorisations have $n-1$ colours: no rainbow Hamilton cycle	Prob. 4.3

Target	Host and colouring	Result	Obstruction	In the text
path with $(1-o(1))d$ edges	proper colouring of a d-regular graph	holds (Bucić et al.)	—	Thm 4.4
Hamilton cycle	K_n, each colour used at most αn times	$\alpha = 1/38$ suffices	classes of size $< n/2$: every rainbow path misses $(\ln n)/42$ vertices (Pokrovskiy–Sudakov)	§4.1
Orderings of groups				
ordering of S with distinct partial sums	$S \subseteq \mathbb{Z}_p \setminus \{0\}$, p prime	holds for large p (Pham–Sauermann)	—	Conj. 4.6, Thm 4.7
ordering of S with distinct partial sums	$S \subseteq G \setminus \{0\}$, G finite abelian	holds for $\lvert S\rvert \geq n_0$ (Kuang–Wang)	—	Thm 4.8
directed path with $d - o(d)$ edges	d-regular digraph, one in- and one out-edge of each of d colours at every vertex	holds (Wilson); $d-1$ edges open	—	Prob. 4.9, Thm 4.10
Short cycles				
cycle of length $\leq \lceil n/r \rceil$	n vertices, n colours, each used $\geq r$ times	$r \leq 2$; length $\leq n/r + \alpha_r$ for fixed r (Hompe–Huynh)	—	Conj. 4.13
some rainbow cycle	proper colouring, average degree d	$d \geq C \log n \log\log n$ suffices (Alon et al.)	hypercube: $d = \log_2 n$, no rainbow cycle	Conj. 4.16
Dense hosts				
triangle	colour degree $\geq (n+1)/2$	holds (Li; Li, Ning, Xu and Zhang)	proper colouring of $K_{n/2,n/2}$	§4.5
Hamilton cycle	$\delta(G) \geq n/2$, each colour used $\leq \mu n$ times	some $\mu > 0$ (Coulson–Perarnau)	fails for $\mu > 1/8$; for proper colourings 1/8 conjectured exact	Thm 4.17, Conj. 4.18

Table 4: Main results on general spanning structures (Section 5).

Target	Host and colouring	Result	Obstruction	In the text
Proper and locally bounded colourings				
tiling by rainbow copies of F	proper colouring, $\delta(G) \geq (1-1/\chi(F))n + C$	holds (Yuster)	rainbow F-factor impossible: a 1-factorisation has $n-1$ colours	Thm 5.1
rainbow triangles covering all but C vertices	proper colouring of K_n with n colours	all but $n^{1-\varepsilon}$ vertices (Rödl nibble)	no example rules out $C = 2$	Conj. 5.2

Target	Host and colouring	Result	Obstruction	In the text
copy of H, $\Delta(H) \le \Delta$, spanning	quasirandom host, locally Λ-bounded and globally $(1-\gamma)e(G)/e(H)$-bounded	holds (Ehard–Glock–Joos)	global bound $e(G)/e(H)$ is forced by counting colours	Thm 5.3
cyclic decomposition of K_{2n+1} into copies of H	H with n edges and $\Delta(H) \le \Delta$	cyclic packing with $(1-\varepsilon)n$ edges; trees: for large n	—	Conj. 5.4
Globally bounded colourings				
copy of any H with $\Delta(H) \le \Delta$	K_n, each colour at most $n/(51\Delta^2)$ times	holds (Böttcher–Kohayakawa–Procacci)	classes of size $16n/\Delta^2$ can fail (Sudakov–Volec)	Thm 5.5
H with at most r cherries	K_n, each colour at most $n/(1512\, r^{3/4})$ times	holds (Sudakov–Volec)	9-bounded colourings and trees with $\Theta(n^{4/3})$ cherries	Thm 5.6
F-factor	$\delta(G) \ge (\delta_F + \varepsilon)n$, each colour at most μn times	holds; same threshold as uncoloured (Coulson–Keevash–Perarnau–Yepremyan)	—	Thm 5.7
r-colourable H of bounded degree and bandwidth $\le \beta n$	$\delta(G) \ge (1-1/r+\varepsilon)n$, each colour at most μn times	holds (Glock–Joos)	—	Thm 5.8
Graph collections				
transversal Hamilton cycle	n graphs of minimum degree $\ge n/2$	holds; colour-blind (Joos–Kim)	—	§5.5
transversal F-factor	minimum degree $\ge (\delta_F^{\mathrm{T}} + \varepsilon)rn$ in each graph	holds (Montgomery–Müyesser–Pehova)	bridgeless F: threshold rises to $1/2$ (one bipartite layer)	Thm 5.9, Fig. 23
transversal r-th power of a Hamilton cycle	minimum degree $\ge (r/(r+1)+\varepsilon)n$ in each graph	holds (Gupta et al.); exact threshold open	—	Thm 5.10
Colour degree, decompositions and random colourings				
tiling by rainbow triangles	colour degree $\ge cn$	$c = 5/6 + \varepsilon$ suffices (Lo–Williams); $5/7$ conjectured	colour degree $5n/7 - 1$ can fail (Lo–Williams)	Thm 5.11, Conj. 5.12
α-decomposition into rainbow F-factors	K_n, globally $(1-\alpha)v(F)n/(2e(F))$- and locally $\eta n/\log^{2a(F)} n$-bounded	holds (Kim–Kühn–Kupavskii–Osthus)	global bound is forced by counting	Thm 5.13
rainbow copy of a minimal member of $\mathcal{F}$	random subset of X, uniformly coloured with $k \ge \ell$ colours	threshold $\le Kq_f(\mathcal{F}) \log \ell$ (Han–Yuan)	—	Thm 5.14

Table 5: Main results on edge-coloured digraphs (Section 6).

Target	Host and colouring	Result	Obstruction	In the text
Complete digraphs				
path on $n-1$ vertices	complete digraph with loops, proper colouring with n colours (a Latin square)	for large n (Bowtell–Montgomery–Müyesser–Pokrovskiy)	cyclic squares of even order: no rainbow Hamilton cycle	Thm 6.1
path or cycle with $n-1$ edges	complete digraph with loops, any proper colouring	open	doubled Maamoun–Meyniel colouring: no rainbow Hamilton path	Conj. 6.3
cycle of length $n-O(1)$	complete digraph, any proper colouring	length $n-O(n^{4/5})$ (Benzing–Pokrovskiy–Sudakov)	—	Thm 6.2
Colour degree and outdegree				
rainbow cycle of even length $\ell \geq 4$	colour degree $\geq (n+5)/3$	holds (Czygrinow–Molla–Nagle–Oursler)	$3 \nmid \ell$: a coloured complete 3-partite graph	Thm 6.4
directed cycle of length $\ell \geq 4$	oriented graph, outdegree $\geq (n+1)/3$	holds (Czygrinow–Molla–Nagle–Oursler)	$3 \nmid \ell$: blow-up of a directed triangle	Thm 6.4
Fair division and arborescences				
rainbow cycle number $R(d)$	multipartite digraphs, parts of size $\leq d$	$d \leq R(d) < ed$ (Sivashankar)	$R(d) \geq d$	Thm 6.5, Prob. 6.6
rainbow spanning arborescence	$n-1$ colour classes, each a spanning arborescence	special cases (e.g. underlying cycle) and $n \leq 8$; holds with $2n-4$ classes	—	Conj. 6.7
Collections and random digraphs				
transversal Hamilton path / cycle	$n-1$ tournaments / n tournaments, all but one strongly connected	for large n (Chakraborti–Kim–Lee–Seo)	$n=3$: two opposite directed triangles	Thm 6.8, Fig. 29
transversal Hamilton cycle	n digraphs, minimum semidegree $\geq n/2$	for large n (Cheng–Li–Sun–Wang)	—	Thm 6.9, Prob. 6.11
every orientation of every cycle, simultaneously	$\delta^0(D_0) \geq \alpha n$ plus $D(n, C/n)$, coloured uniformly from $[n]$	with high probability (Krueger–Staudinger)	—	Thm 6.12

Table 6: Main results on hypergraphs, designs and set families (Section 7).

Target	Host and colouring	Result	Obstruction	In the text
Families of matchings				
rainbow matching of size t	N matchings of size t in an r-graph, $r \geq 3$ fixed	N of order t^r is necessary and sufficient (Pohoata–Sauermann–Zakharov)	zero-sum-free sequences beat $2^{r-1}(t-1)$ (Alon)	Thm 7.1, Fig. 31
rainbow matching of size $g(r,n)$	n matchings of size n, r-partite r-graph, $r \geq 3$	$g(r,n) \geq 2n/(r+1) - O_r(1)$; conjectured $n - o(n)$	$g(r,n) \leq n - \Omega_r(n^{1-1/r})$ (Bowtell–Freschi–Kronenberg–Yan)	Thm 7.2, Conj. 7.3
full rainbow matching	r-graph of maximum degree Δ, each colour class $\geq r\Delta$ edges	holds (Aharoni–Berger–Meshulam)	$r\Delta - 1$ is not enough (Wdowinski)	§7.2
Set families and fair division				
pairwise disjoint $F_i \in \mathcal{F}_i$ (rainbow Erdős matching)	m families of k-subsets of $[n]$, each larger than $f(n,m,k)$	$n \geq 2km$ and n large; all large n for $k = 3$	the two Erdős constructions	Conj. 7.4
MMS allocation (rainbow matching using every colour)	n agents with monotone valuations, at most $\mu(n)$ goods	$\mu(n) = n + \Theta(\log n)$ (Feige)	4 items, 2 agents (rows versus columns)	Thm 7.5, Fig. 33
Collections and bounded colourings				
rainbow perfect matching	n/k k-graphs with $\delta_\ell > \delta(n,k,\ell)$, $k/2 \leq \ell < k$	holds for large n; colour-blind (You)	the uncoloured extremal k-graphs	Thm 7.6
rainbow tight Hamilton cycle	n k-graphs with $\delta_{k-2} \geq (5/9+\gamma)\binom{n}{2}$	holds (Tang–Wang–Wang–Yan); other degree types conjectured	—	Conj. 7.7
rainbow F-factor	$\delta_\ell(G) \geq (\delta_\ell^*(F)+\varepsilon)n^{k-\ell}$, μ-bounded colouring	holds (Coulson–Keevash–Perarnau–Yepremyan)	a local bound is needed for $F = K_{k+1}^{(k)}$	Thm 7.8
rainbow loose Hamilton cycle	above the loose-cycle degree threshold, each colour $\leq \mu n^{k-1}$ times	holds (Kathapurkar–Morris–Perarnau)	tight cycles: linear colour classes can fail	Thm 7.9, Fig. 34
Latin hypercubes				
transversal	Latin hypercube of dimension d, order n, n or d odd	open; known for $n \leq 5$	n and d both even: the cyclic hypercube has none	Conj. 7.10

B History of the major conjectures

For each conjecture, Table 7 gives the year in which it was posed, the main steps towards it with the years of the cited sources, and its present status.

Table 7: History of the major conjectures.

Conjecture	Statement	Milestones	Status	In the text
Euler's officers (1782)	no Latin square of order $n \equiv 2 \bmod 4$ decomposes into transversals	1900: true for $n = 6$ (Tarry); 1959–60: false for every other such $n > 2$ (Bose, Shrikhande, Parker); 2025: random Latin squares decompose (Bowtell–Montgomery)	disproved	Conj. 2.1
Hall–Paige (1955)	a group table has a transversal iff the Sylow 2-subgroups are trivial or non-cyclic	1947: abelian groups (Paige); 2009–2018: reduction and proof (Wilcox; Evans; Bray et al.)	proved	Thm 2.11
Ryser (1967)	every Latin square of odd order has a transversal	2016: random Latin squares (Kwan); 2022: their number of transversals (Eberhard–Manners–Mrazović)	open	Conj. 2.9
Ryser–Brualdi–Stein (1967–1991)	every Latin square has a partial transversal with $n - 1$ cells	1969: $2n/3$; 1977: $3n/4$; 1978: $n - \sqrt{n}$; 1982/2008: $n - O(\log^2 n)$ (Shor; Hatami–Shor); 2020: $n - O(\log n / \log\log n)$ (Keevash et al.); 2023: $n - 1$ (Montgomery)	proved for large n	Conj. 2.19, Thm 2.21
Stein (1975)	every equi-n-square has a partial transversal with $n - 1$ cells	1991: full transversal if each symbol appears $\leq (n-1)/16$ times (Erdős–Spencer); 2017: counterexamples missing $(\ln n)/42$ cells (Pokrovskiy–Sudakov); 2024: $n - n^{1-\Omega(1)}$ cells, and $n - \Omega(\sqrt{n})$ is best possible (Chakraborti et al.)	disproved	Conj. 2.26
Kotzig (1963)	K_{2n} has a perfect 1-factorisation	true when $2n - 1$ or n is prime; 2026: $(1 - o(1))2n$ pairwise Hamiltonian classes (Cheng–Sgueglia)	open	Conj. 2.5
Aharoni–Berger (bipartite) (2009)	n matchings of size $n + 1$ in a bipartite multigraph have a full rainbow matching	size $2n$ by a greedy argument; then $7n/4$, $5n/3$, $3n/2 + 1$; 2016: $n + o(n)$ (Pokrovskiy); 2021: $n + O(n^{3/4})$ (Correia–Pokrovskiy–Sudakov)	open	Conj. 2.33
Aharoni–Berger (r-partite) (2009)	$2^{r-1}(t-1)+1$ matchings of size t in an r-partite r-graph have a rainbow matching of size t	2011: disproved via zero-sum sequences (Alon); 2022: the right number has order t^r (Pohoata–Sauermann–Zakharov)	disproved	Thm 7.1
Ringel (1963)	K_{2n+1} decomposes into copies of any tree with n edges	1967: graceful labellings (Rosa); 2020: proof for large n (Montgomery–Pokrovskiy–Sudakov; Keevash–Staden)	proved for large n	Conj. 3.13, Thm 3.14
Graceful tree (1960s)	every tree is graceful	2016: almost all trees almost graceful (Adamaszek et al.); 2025: bijective labelling with few repeated differences (Letzter–Pokrovskiy–Williams)	open	Conj. 3.12

Conjecture	Statement	Milestones	Status	In the text
Brualdi–Hollingsworth (1996)	every optimal colouring of K_{2n} decomposes into rainbow spanning trees	constant numbers of trees; 2017: linearly many (Balogh–Liu–Montgomery; Pokrovskiy–Sudakov); 2018: $(1-o(1))n$ (Montgomery–Pokrovskiy–Sudakov); 2019: exact, isomorphic trees (Glock–Kühn–Montgomery–Osthus)	proved for large n	Conj. 3.3, Thm 3.7
Rota's basis conjecture (1989)	n bases of a rank-n space split into n rainbow bases	1994: the Alon–Tarsi conjecture for n implies it (Huang–Rota), so it holds in characteristic 0 for $n = p \pm 1$; 2025: $(1-o(1))n$ disjoint rainbow bases (Montgomery–Sauermann)	asymptotic form proved	Conj. 3.16, Thm 3.18
Hahn / Andersen (1980 / 1989)	proper colourings of K_n have a rainbow Hamilton path (Hahn), or a path on $n-1$ vertices (Andersen)	1984: Hahn's form fails for $n = 2^k$ (Maamoun–Meyniel); 2016: $n - O(n^{3/4})$ (Alon–Pokrovskiy–Sudakov); 2017: $n - O(\sqrt{n}\log n)$ (Balogh–Molla); 2026: $n-1$ vertices (Bowtell–Montgomery–Müyesser–Pokrovskiy)	proved for large n	Conj. 4.1, Thm 4.2
Hahn–Thomassen (1986)	colour classes of size αn force a rainbow Hamilton cycle in K_n	$O(n^{1/3})$ (Hahn–Thomassen); $n/64$ (Albert–Frieze–Reed); 2012: $n/38$ (Böttcher–Kohayakawa–Procacci); classes just below $n/2$ can fail (Pokrovskiy–Sudakov)	proved	§4.1
Frieze–Krivelevich (2008)	colour classes of size αn force every bounded-degree spanning tree	2012: every graph of maximum degree Δ with classes $n/(51\Delta^2)$ (Böttcher–Kohayakawa–Procacci); 2015: the order n/Δ^2 is optimal (Sudakov–Volec)	proved	Thm 5.5
Graham (1971)	every $S \subseteq \mathbb{Z}_p \setminus \{0\}$ has an ordering with distinct partial sums	small sets by the polynomial method and rectification; 2024: Bedert–Kravitz; 2026: large p (Pham–Sauermann)	proved for large p	Conj. 4.6, Thm 4.7
Caccetta–Häggkvist / Aharoni (1978 / 2017)	outdegree r (resp. colour classes of size r) forces a (rainbow) cycle of length $\leq \lceil n/r \rceil$	Aharoni's form implies Caccetta–Häggkvist; 2018: Aharoni's case $r = 2$ (DeVos et al.); 2022: length $\leq n/r + \alpha_r$ (Hompe–Huynh)	open	Conj. 4.12, Conj. 4.13
Keevash–Mubayi–Sudakov–Verstraëte (2007)	$Cn\log n$ edges in a proper colouring force a rainbow cycle	2023: average degree $(\log n)^{1+o(1)}$ (Alon–Bucić–Sauermann–Zakharov–Zamir); applications to locally decodable codes	open	Conj. 4.16
Brouwer (1981)	every Steiner triple system of order n has a matching with $(n-4)/3$ edges	2023: $\lfloor n/3 \rfloor - 1$ edges for large n (Montgomery)	proved for large n	Conj. 2.41
Rainbow arborescence (2024)	$n-1$ spanning arborescences have a rainbow spanning arborescence	2025: underlying cycles (Bérczi–Király–Yamaguchi–Yokoi)	open	Conj. 6.7

C Index of conjectures and problems

Table 8 lists every conjecture and problem stated in the text, in order of appearance.

Table 8: Index of the conjectures and problems.

	Source	Statement	Status
Conj. 2.1	Euler, 1782	no Latin square of order $n \equiv 2 \bmod 4$ decomposes into transversals	disproved
Conj. 2.5	Kotzig, 1963	K_{2n} has a perfect 1-factorisation	open
Conj. 2.8	Wanless, 1999	$K_{n,n}$ has a perfect 1-factorisation iff $n = 2$ or n is odd	open
Conj. 2.9	Ryser, 1967	every Latin square of odd order has a transversal	open
Conj. 2.13	Snevily, 1999	$k \times k$ subsquares of odd abelian group tables have transversals	proved
Conj. 2.14	Friedlander, Gordon and Tannenbaum, 1981	orthomorphisms with all cycles of length k, for $k \mid n-1$	proved for large groups
Conj. 2.19	Ryser–Brualdi–Stein; sources from 1967, 1975, 1991	partial transversal with $n-1$ cells; full one for odd n	$n-1$ part proved for large n; odd part open
Conj. 2.22	Anastos, Fabian, Müyesser and Szabó, 2022	matching with prescribed numbers a_i of edges from each M_i	open; known for 3 matchings
Conj. 2.24	Akbari and Alipour, 2004	proper colourings of $K_{n,n}$ with $n^2/2$ colours have rainbow perfect matchings	proved
Conj. 2.26	Stein, 1975	equi-n-squares have partial transversals with $n-1$ cells	disproved
Prob. 2.30	Pokrovskiy and Sudakov, 2017	colour classes of size $(1-o(1))n$ force a rainbow perfect matching of $K_{n,n}$?	open
Conj. 2.31	Chakraborti, Christoph, Hunter, Montgomery and Petrov, 2024	equi-n-squares have partial transversals with $n - C\sqrt{n}$ cells	open
Conj. 2.32	Stein, 1975	colourings of $K_{n,n-1}$ proper on the larger side have rainbow matchings of size $n-1$	open
Conj. 2.33	Aharoni and Berger, 2009	n matchings of size $n+1$ in a bipartite multigraph give a full rainbow matching	open
Conj. 2.36	Aharoni, Berger, Chudnovsky, Howard and Seymour, 2016	$2n$ matchings of size n in a graph give a rainbow matching of size n	open; $3n-3$ known
Conj. 2.37	Gao, Ramadurai, Wanless and Wormald, 2017	n matchings of size $n+2$ in a multigraph give a full rainbow matching	open
Conj. 2.41	Brouwer, 1981	Steiner triple systems have matchings with $(n-4)/3$ edges	proved for large n
Conj. 2.43	Ferber and Kwan, 2019	almost every Steiner triple system is resolvable	open
Conj. 3.3	Brualdi and Hollingsworth, 1996	optimal colourings of K_{2n} decompose into rainbow spanning trees	proved for large n
Conj. 3.4	Constantine, 2002	the same with isomorphic trees	proved for large n
Conj. 3.5	Kaneko, Kano and Suzuki, 2002	proper colourings of K_n have $\lfloor n/2 \rfloor$ disjoint rainbow spanning trees	open; $(1-o(1))n/2$ known
Conj. 3.10	Montgomery, Pokrovskiy and Sudakov, 2018	proper colourings of K_n contain every tree on $n-C$ vertices	open; $n-o(n)$ known
Conj. 3.12	Graceful tree conjecture; Ringel, Kotzig and Rosa, 1960s	every tree is graceful	open

	Source	Statement	Status
Conj. 3.13	Ringel, 1963	K_{2n+1} decomposes into copies of any tree with n edges	proved for large n
Conj. 3.16	Rota, 1989; see Huang and Rota	n bases of an n-dimensional space split into n rainbow bases	open; asymptotic form proved
Conj. 3.17	Alon and Tarsi, 1992	even and odd Latin squares of even order n differ in number	open; known for $n = p \pm 1$
Conj. 3.19	Kahn, 1994	an $n \times n$ array of bases has a choice that is a basis in every row and column	open
Conj. 3.20	Aharoni and Berger, 2009	n common independent n-sets of two matroids have a rainbow one of size $n-1$	open
Conj. 4.1	Andersen, 1989	proper colourings of K_n have rainbow paths on $n-1$ vertices	proved for large n
Prob. 4.3	Akbari, Etesami, Mahini and Mahmoody, 2007	do 1-factorisations of K_n have rainbow cycles through all but two vertices?	open; all but C known
Conj. 4.5	Chen and Li, 2005	colour degree d forces a rainbow path with $d-1$ edges	open; $2d/3+1$ known
Conj. 4.6	Graham, 1971	subsets of $\mathbb{Z}_p \setminus \{0\}$ have orderings with distinct partial sums	proved for large p
Prob. 4.9	Directed Schrijver problem; Bucić, Frederickson, Müyesser, Pokrovskiy and Yepremyan, 2025	properly d-coloured d-regular digraphs have rainbow paths with $d-1$ edges?	open; $d - o(d)$ known
Prob. 4.11	Bedert, Bucić, Kravitz, Montgomery and Müyesser, 2025	do subsets of all finite groups have orderings with distinct partial products?	open
Conj. 4.12	Caccetta and Häggkvist, 1978	outdegree r forces a directed cycle of length $\leq \lceil n/r \rceil$	open
Conj. 4.13	Aharoni, 2017	n colour classes of size r force a rainbow cycle of length $\leq \lceil n/r \rceil$	open; $r \leq 2$ known
Conj. 4.16	Keevash, Mubayi, Sudakov and Verstraëte, 2007	$Cn \log n$ edges in a proper colouring force a rainbow cycle	open; $n(\log n)^{1+o(1)}$ known
Conj. 4.18	Peng and Yan, 2024	proper colourings with $\delta \geq n/2$ and classes $\leq n/8$ have rainbow Hamilton cycles	open
Conj. 5.2	Müyesser, 2023	proper n-colourings of K_n have rainbow triangle packings missing C vertices	open
Conj. 5.4	Ehard, Glock and Joos, 2019	cyclic decompositions of K_{2n+1} into any bounded-degree H with n edges	open; known for trees
Conj. 5.12	Lo and Williams, 2024	colour degree $(1-(r-1)/(r^2-2))n$ forces a tiling by rainbow K_r	open
Conj. 6.3	Gould and Kelly, 2021	proper colourings of the complete digraph with loops have rainbow paths or cycles with $n-1$ edges	open
Prob. 6.6	Berendsohn, Boyadzhiyska and Kozma, 2022	is $R(d) = d$?	open; known for $d \leq 4$
Conj. 6.7	Rainbow arborescence conjecture; Bérczi, Király, Yamaguchi and Yokoi, 2024	$n-1$ spanning arborescences have a rainbow spanning arborescence	open
Conj. 6.10	Chakraborti, Kim, Lee and Seo, 2023	strongly rainbow-connected tournament collections have transversal Hamilton cycles	open
Prob. 6.11	Cheng, Li, Sun and Wang, 2025	oriented collections with semidegree $\lceil (3n-4)/8 \rceil$: transversal Hamilton cycle?	open
Conj. 7.3	Bowtell, Freschi, Kronenberg and Yan, 2025	g, g', h, h' are all $(1+o(1))n$	open

	Source	Statement	Status
Conj. 7.4	Huang, Loh and Sudakov, 2012; Aharoni and Howard, 2016	rainbow version of the Erdős matching conjecture	open; proved for $n \geq 2km$, n large
Conj. 7.7	Tang, Wang, Wang and Yan, 2023	tight Hamilton cycles are colour-blind for every degree type	open; $d = k-2$ known
Conj. 7.10	Wanless, 2011	Latin hypercubes with n or d odd have transversals	open; known for $n \leq 5$

D Threshold gaps

Figure 36 shows six questions in which a proved bound and an obstruction are separated by a gap.

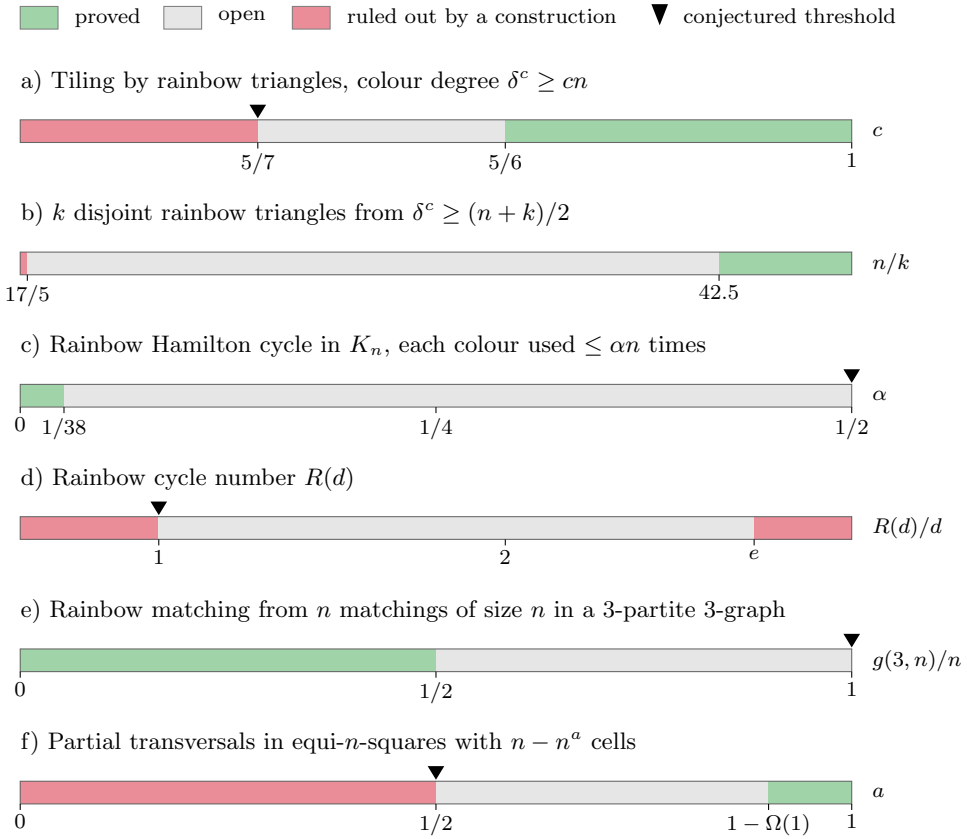


Figure 36: Ranges in which the natural statement is proved, open or ruled out. Bars are drawn to scale on the axis at the right. Sources in the text: (a) Thm 5.11, Conj. 5.12 (uncoloured threshold 2/3); (b) §5.6 (holds for $n \geq 42.5k + 48$); (c) §4.1 (classes of size just below $n/2$ can fail); (d) Thm 6.5, Prob. 6.6; (e) Thm 7.2, Conj. 7.3 ($n - O(1)$ is ruled out); (f) Thm 2.29, Conj. 2.31 (the proved exponent is some constant below 1; 0.9 is schematic).

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